36. Omitted. [8]
36A. Climate change and all other matters related thereto.[9]
37. Omitted. [10]
38. Omitted. [11]
39. Omitted. [12]
40. The Water (Prevention and Control of Pollution) Act, 1974 (6 of 1974).
41. The Water (Prevention and Control of Pollution) Cess Act, 1977 (36 of 1977).
42. The Air (Prevention and Control of Pollution) Act, 1981 (14 of 1981).
43. The Indian Forest Act, 1927 (16 of 1927).
44. The Wild Life (Protection) Act, 1972 (53 of 1972)
45. The Forest (Conservation) Act, 1980 (69 of 1980).
46. The Environment (Protection) Act, 1986 (29 of 1986).
47. The Public Liability Insurance Act, 1991 (6 of 1991).
48. The National Green Tribunal Act, 2010 (19 of 2010).*

NOTE:- The Ministry of Environment, Forest and Climate Change will be responsible for overall policy in relation to forests, except all matters, including legislation, relating to the rights of forest dwelling Schedule Tribes on forest lands.**

[8] Omitted vide Amendment Series no. 349 dated 04.04.2019 (transferred to D/o Animal Husbandry and Dairying).
[9] Inserted vide Amendment series no. 306 dated 31.07.2014.
[10] Omitted vide Amendment Series no. 349 dated 04.04.2019 (transferred to D/o Animal Husbandry and Dairying).
[11] Omitted vide Amendment Series no. 334 dated 06.07.2017.
[12] Omitted vide Amendment Series no. 334 dated 06.07.2017.
* Inserted vide Amendment series no. 334 dated 06.07.2017.
** Inserted vide Amendment series no. 285 dated 17.03.2006.

2.1. Control of Pollution

The 'Control of Pollution' Division administers a Central Sector Scheme –'Control of Pollution' which started from the year 2018-19 and it is continued in the FY 2025-26. The outlay of the scheme for the year 2021-22 to 2025-26 is Rs.3,802.5 crore. The CoP scheme comprises of the following components:

(1) National Clean Air Programme (NCAP)

- The National Clean Air Programme (NCAP) is a flagship initiative launched to improve air quality across 130 cities (82 non-attainment cities + 40 non-attainment cities and Million-plus population cities +8 Million-plus population cities) in 24 States / UTs through a comprehensive, integrated and collaborative approach.
- The programme focuses on reducing concentrations of Particulate Matter (PM10)—fine airborne particles primarily generated from sources such as road dust, vehicular emissions, construction activities, open waste burning, and industrial operations.
- NCAP aims to achieve a reduction of up to 40% in PM10 levels compared to the 2019–20 baseline, or alternatively, ensure cities meet the National Ambient Air Quality Standard (NAAQS) of 60 µg/m³ for PM10 by the year 2025–26.
- Each city under the NCAP has developed a Clean Air Action Plan (CAP) with defined annual targets to reduce PM10 concentrations. These plans focus on sectoral interventions across road dust, solid waste management, vehicular emissions, construction activities, and industrial pollution. Under Clean Air Action Plan (CAP) cities are mandated to identify key pollution sources, prioritize them based on impact, and deploy targeted mitigation measures to achieve quantifiable improvements in ambient air quality.
- These city-level Clean Air Action Plans are aligned with broader State-level and National-level frameworks to ensure consistency in approach, avoid duplication of efforts, and enable resource pooling
- Funding for the implementation of City Action Plans (CAPs) is leveraged through convergence with various Central Government schemes, including:
  - Ministry of Housing and Urban Affairs (MoHUA):
  - Swachh Bharat Mission – Urban (SBM-U)
  - Atal Mission for Rejuvenation and Urban Transformation (AMRUT)
  - Smart Cities Mission
  - PM E-Bus Sewa Scheme
  - Urban Transport
  - Ministry of Petroleum and Natural Gas (MoPNG):
  - Sustainable Alternative Towards Affordable Transportation (SATAT)
  - Ministry of Road Transport and Highways (MoRTH):
  - Vehicle Scrapping Policy
  - Ministry of Heavy Industries (MHI):
  - PM E-DRIVE Scheme
  - Ministry of Power (MoP):
  - SAMARTH – National Biomass Mission (Sustainable Agrarian Mission on Use of Agri-Residue in Thermal Power Plants)
  - Ministry of New and Renewable Energy (MNRE):
  - National Bioenergy Programme
- To address critical funding gaps, performance-based incentive grant is provided under NCAP.
- The XV Finance Commission through its Million-Plus Cities Challenge Fund (MPCCF) made provision of Rs. 16,539 crores for targeted actions in 48 Million-Plus Cities / Urban Agglomeration (42 MPCs + 6 UAs). Besides, critical gap funding of Rs. 3591 crores was provided in 82 cities under Central sectoral Scheme (CSS) 'Control of Pollution(CP)' of MoEF&CC. Overall, Rs. 13,852.22 crore (Rs. 11,457.34 crores for targeted actions in 48 Mil-

lion-Plus Cities / Urban Agglomeration and Rs. 2,394.88 crore to the 82 cities though Control of Pollution, Central Sector Scheme) have been distributed to these cities so far.

- As of date, a cumulative amount of ₹13,852.22 crore has been released to 130 cities under NCAP, including:
  - **₹2,394.88** crore to 82 non-attainment cities under the CP scheme of MoEF&CC
  - **₹11,457.34** crore to 48 Million Plus Cities/ Urban Agglomerations under the MPCCF of the XV Finance Commission
- Cities receive performance-linked grants to implement their action plans. Better-performing cities are incentivized through additional funding and recognition.
- The programme is anchored with multi-sectoral initiative involving the coordinated efforts of the Central and State Governments, Urban Local Bodies (ULBs), and other stakeholders.
- The National Level Apex Committee, headed by the Hon'ble Minister EF&CC, provides overall guidance to the program. Till date 43 meetings (04 Apex Committee, 08 Steering Committee, 14 Monitoring Committee and 17 Implementation Committee meetings) were held at National level.
- During FY 2024–25, a total of 534 meetings were conducted at the state and city levels, comprising 27 State Steering Committee meetings, 48 Air Quality Monitoring Committee (AQMC) or State-Level Monitoring and Implementation Committee (SLMIC) meetings, and 459 District-Level Monitoring and Implementation Committee (DLMIC) meetings.
- In addition, CPCB officers undertook field visits in all 130 NCAP cities to provide ground-level feedback and technical support.
- NCAP has also been brought under the purview of State and District-level DISHA Committees, enhancing accountability and convergence. These committees are chaired by the Chief Minister or Lieutenant Governor in the case of Union Territories at the state level, and by the elected Member of Parliament (Lok Sabha) from the respective district at the district level.
- To strengthen implementation, the Ministry has established Project Management Units (PMUs) at SPCBs / PCCs and in 130 cities, staffed with trained technical personnel. These units provide hands-on support to States and UTs. Additionally, more than 60 Institutes of Repute have been engaged to offer technical assistance, research support, and capacity augmentation.
- A structured Capacity Building Program has been designed to deliver periodic training to stakeholders across all levels, ensuring institutional readiness and sustained programmatic effectiveness.
- The programme implementation is tracked through the PRANA portal- through the PRANA portal—a centralized, web-based dashboard developed by MoEF&CC. It captures both physical progress (such as clean air action plan proposed, number of activities completed, meetings held) and financial progress (funds allocation, released and utilized) across all 130 cities.
- Air quality monitoring is conducted at 855 stations (500 manual stations and 355 CAAQMS stations) in 130 NCAP cities and the data is transmitted to CPCB through concerned SPCB / PCCs throughout the year. Performance of cities is assessed based on annual average of PM10 concentrations and Air Quality Index (AQI) computed from the air quality data collected annually through these stations.
- The Swachh Vayu Survekshan initiative ranks cities based on CAP implementation and air pollution mitigation efforts. Top-performing cities are felicitated annually on National Swachh Vayu Diwas (7th September).
- Swachh Vayu Survekshan Framework released in 2022-23 on measures taken by cities on road dust, SWM, vehicular emissions, industrial pollution, C&D waste management, other

emissions (DG sets and commercial establishments), IEC activities

- Participation and ranking of all 130 cities in 3 groups (Population below 3 Lakh; 3-10 Lakh; & above 10 Lakh)
- Self-assessment by cities; Approval by State-level Monitoring Committee; Evaluation by CPCB; Field verification by Independent Committees
- 4 Survekshans carried out (2021-22, 2022-23, 2023-24,2025-26)
- Cash award conferred on Swachh Vayu Diwas (7th September) to top 3 performing cities in each group
- Mission LiFE has been integrated into State and City Action Plans to promote environmentally responsible citizen behaviour.
- Implementation of Key Air Quality Improvement Activities under NCAP as on date is as below:
  - A total of 16,424.14 km of roads have been paved end-to-end to reduce dust emissions from open surfaces.
  - 802 Mechanical Road Sweepers (MRS) have been procured to support routine dust removal operations.
  - 9,367.31 acres of green areas have been developed to enhance urban vegetation and improve ambient air quality.
  - 8,232 tonnes per day (TPD) of capacity has been created through the establishment of Construction and Demolition (C&D) waste processing sites.
  - 3,234 electric buses have been purchased and deployed to reduce vehicular emissions in urban transport systems.
  - 689 EV charging stations have been installed to support the transition to cleaner mobility.
  - 111 PNG / electric crematoria have been established to replace traditional biomass-based cremation practices.
  - 1,125.21 lakh metric tonnes of legacy waste have been remediated, contributing to improved waste management and reduced open burning.
- The focused actions by 130 cities under NCAP have shown positive results with 103 cities showing reduction in PM10 concentration in 2024-25 with respect to 2017-18, 64 cities have shown reduction in PM10 levels by more than 20% with respect to base year 2017-18 and 25 of these cities have achieved a reduction of more than 40%. A total of 22 Cities have met NAAQS and have PM10 Concentrations less than 60 µg/m3.
- The key initiatives taken under NCAP in FY 2025-26 are as follows:
  - Introduction of new Air Quality Challenge Method to access funds for FY 2025-26 for 82 cities funded under CP Scheme of MoEF&CC.
  - Cities mandated to prepare saturation-based plans (85 cities prepared)
  - Cities to achieve saturation in following 5 key activities by 2025-26

  (i) Road improvement works for dust control including end to end paving, greenery development and mechanical road sweeping etc.

  (ii) Greening of open areas, traffic corridors and mini forest including Miyawaki plantation

  (iii)Improvement of traffic junctions for decongestion

  (iv) Air pollution control measures in crematoriums:

  - Road improvement works for dust control including end to end paving, greenery development and mechanical road sweeping etc.
  - Greening of open areas, traffic corridors and mini forest including Miyawaki plantation

  - Improvement of traffic junctions for decongestion
  - Air pollution control measures in crematoriums:
    - Conversion to cleaner fuels, or

Investigators, Dr. S. K. Goyal, Chief Scientist & Head, CSIR-NEERI and Dr. Neeraj Sharma, Chief Scientist, TPE Division, CRRI, New Delhi.

(ii) The study under the project entitled "Decision Support System for Air Quality Management in Delhi and its Bordering Districts in the NCR" was conducted under the supervision of Principal Investigator, Dr. Gaurav Govardhan, Scientist 'D' IITM, Pune.

(iii) The study under the project entitled "Evaluation of Retro-fitment of 2 & 3 Wheeler Auto-rickshaws with Electric Drive as an option for improvement in air quality in NCR" has been completed under the supervision of Principal Investigator. Dr. Ravindra Kumar, GM, ARAI, Kothrud, Pune.

14) Grievance redressal and facilitation

The Commission has set up a grievance redressal mechanism for dealing with the complaints/ grievances received.

15) Budget Allocation for the Financial Year 2025-26

	(Rs. In crore)
(a) Grant in Aid General	30.88
(b) Grant for Creation of Capital Assets	00.10
(c) Grant in Aid Salaries	08.00
Total	38.98

2.2. Central Pollution Control Board

a. Introduction

Under the provisions of the Water (Prevention & Control of Pollution) Act, 1974, the Central Government constituted the 'Central Board for the Prevention and Control of Water Pollution' on September 23, 1974. The name of the Board was amended to Central Pollution Control Board (CPCB) under the Water (Prevention & Control of Pollution) Amendment Act, 1988 (No. 53 of 1988). The CPCB has been entrusted with the added responsibilities of Air Pollution Control since May, 1981 under the provisions of the Air (Prevention and Control of Pollution) Act, 1981. The enactment of the Environment (Protection) Act, 1986, which is the umbrella legislation for enforcement of measures for protection of environment, and several notifications of Rules under the Act widened the scope of activities of the CPCB.

CPCB is a statuary body and receives 100 % Grand in Aid from the MoEF&CC for executing its functions and responsibilities, with a total budget allocation of Rs.116.20 Crores as Grant-in-Aid in the Budget Estimate for FY 2025-26 (Under BE).

The newly developed Regional Directorate Laboratories at Shillong and Pune were inaugurated and laid the foundation stone for a new building of the CPCB, Delhi on 50th CPCB Foundation Day, by the Hon'ble Minister, EF&CC.

b. Important Activities

1. Air Quality Monitoring Network

The following steps have been taken to monitor and improvement of air quality:

- Under Sub-section 2 (h) of Section 16 of the Air (Prevention and Control of Pollution)

ity monitoring at these locations is carried as per the Guidelines for Water Quality Monitoring, 2017 circulated by MoEF & CC.

- The network comprises of monitoring locations in 28 States and 6 Union Territories (UTs). Monitoring network covers 2260 locations on rivers, 587 on Lakes, 143 on Ponds, 102 on Tanks, 1271 Ground Water locations and 559 monitoring locations on other water bodies under National Water Quality Monitoring Programme (NWMP).
- Real Time Water Quality Monitoring of River Ganga is carried out at 40 stations (Main stem -17; Tributaries-23) for 12 parameters including pH, BOD, DO, Temperature, EC, Chloride, COD, Turbidity, Nitrate, TOC, water level and water depth.
- Manual water quality monitoring of river Ganga is being carried out by CPCB at 112 locations in 5 main stem States through the concerned SPCBs namely Uttarakhand (UK), Uttar Pradesh (UP), Bihar, Jharkhand and West Bengal (WB). River water quality is assessed for primary water quality criteria notified for outdoor bathing in terms of pH (6.5-8.5), Dissolved Oxygen (DO) (≥5mg/L), Bio-chemical Oxygen Demand (BOD) (≤3mg/L), Faecal Coliform (FC) (≤2500 MPN/100ml) and Faecal Streptococci (FS) (≤500 MPN/100ml). Sampling was carried out on fortnightly basis and the data of year 2024 (January to December) was analysed using statistical tool median which is simply a measure of central tendency and it is observed that:
- pH & DO (median) are meeting the primary water quality criteria for bathing at all the monitoring locations along the entire stretch.

3. Polluted River Stretches

- River Location not complying to BOD criteria (> 3 mg/L) is identified as Polluted Location. Two or more polluted locations in a continuous sequence is called Polluted River Stretch (PRS). Accordingly, PRS are categorized under five Priority Classes based on maximum BOD level observed. Priority I: exceeding 30 mg/L; Priority II: between 20-30 mg/L; Priority III: between 10-20 mg/L; Priority IV: between 6-10 mg/L; and Priority V: between 3-6 mg/L
- During 2018, 351 PRS on 323 Rivers were identified in 31 States/ UTs based on water quality data of 521 rivers (2016 & 2017). Comprehensive Action Plans were formulated by States/ UTs and Action plans are under implementation & are reviewed at State & Central Level. At State level, action plans are reviewed by four membered River Rejuvenation Committee (RRC) of concerned State/ UT comprising Director (Environment), Director (Urban Development), Director, (Industries) and Member Secretary, SPCB/PCC under supervision & coordination of Principal Secretary, (Environment). Overall responsibility lies with Chief Secretaries of the States/UTs.
- During 2022, 311 PRS on 279 rivers in 30 States/UTs were identified, based on water quality data of 603 rivers in 2019 & 2021. They are categorized under five Priority Classes: I (> 30 mg/L) – 46; II (20-30 mg/L) – 16, III (10-20 mg/L) – 39, IV (6-10 mg/L) – 65, V (3-6 mg/L) – 145.
- Comparative assessment of PRS during 2018 vs 2022 revealed that total number of PRS have been reduced from 351 (year 2018) to 311 (year 2022) despite increase in number of river monitoring locations. The number of PRS has reduced in 13 States and number of PRS in 5 States/UTs has remained same.
- Water quality of 180 PRS out of 351 identified in 2018 has improved. 106 river stretches have come out of the list (BOD <3mg/L) and 74 have shifted to lower priority class. Priority class of 108 PRS has re-

mained same in both years 2018 and 2022, indicating that there was no further impact of pollution despite increased activities in the catchment area.

- Further, during 2025, based on water Quality of 2022 & 2023, 296 PRS were identified on 271 rivers in 32 States / UTs (out of 623 rivers). The number of PRS were reduced from 351 in year 2018 to 296 in year 2025. Total 149 no. of river stretches identified as polluted during 2018 are not identified as polluted based on assessment of water quality data for the years 2022 and 2023.

4. National Mission for Clean Ganga (NMCG) – Namami Gange Programme

Three Projects were sanctioned to CPCB under Namami Gange Programme, with a primary emphasis on enhancing the water quality of the Ganga River and improving overall public health. The details are:

1. Pollution Inventorization, Assessment and Surveillance (PIAS) sanctioned in year 2011 and subsequently extended till 2023. The project re-sanctioned on 24.05.2023 with an amount of Rs 113.85 Crore for three years w.e.f. 01.04.2023.
2. Water Quality Monitoring (WQM) Project for River Ganga was sanctioned in year 2013 and extended on 20.06.2020 for six years till 19.06.2026 with an amount of Rs 126.17 Crore.
3. Strengthening of Environmental Regulators (SER) was first sanctioned on 19.06.2013 with an amount of Rs 69.26 Crore and next phase of SER project was sanctioned on 18.02.2022 for 3 years with an amount of Rs 66.21 Crore. The third phase of the project was sanctioned on 07.05.2025 with an amount of Rs. 56.54 Crore for three years w.e.f. 07.05.2025.

- CPCB has undertaken various sector specific initiatives to enhance compliance. This includes implementation of sector specific charters by involving State Govt. departments and expert institutes like CPPRI, VSI, NSI and Industry Association.
- Charters have been developed with the objective to facilitate industries to adopt cleaner technology waste minimization practices as well as upgradation of effluent treatment facility in the five major industrial sectors like Pulp & paper, Distillery, Sugar, Textile and Tannery which together contribute effluent discharge of about 70% and BOD load of 90% of total Industrial pollution discharge by GPIs in Ganga Basin.
- The exercise of annual inspection of Grossly Polluted Industries (GPIs) and implementation of Charter in major water intensive industrial sectors has resulted in an increase of 82% GPI compliance in 2023, compared to just 39% in 2017. Also, 47.2% reduction of BOD load was observed from 26 TPD in 2017 to 13.73 TPD in 2023 from industrial effluent.
- Charter implementation has yielded remarkable reductions in specific freshwater consumption and effluent discharge across various sectors in the Ganga Basin:
- Pulp & Paper sector: The specific freshwater consumption decreased from 16.91 KL/MT in 2017 to 10.11 KL/MT in 2024, marking a 40% reduction.
- Sugar sector: The specific freshwater consumption decreased from 299 KL/MT in 2017 to 68.03 KL/MT in 2024, marking a 77.2% reduction.
- Textile sector: The specific freshwater consumption decreased from 141 KL/MT in 2019 to 30.46 KL/MT in 2024, marking a 78.3% reduction.
- Distillery sector: The specific freshwater consumption decreased from 15 KL/KL in 2017 to 4.65 KL/KL in 2024, marking a 69% reduction.

5. National Ambient Noise Monitoring Net-

work

The following steps have been taken to monitor and improvement of noise pollution:

- The National Ambient Air Quality in respect of Noise for day & night has been notified in Schedule III under Rule 3 of The Environment (Protection) Rules, 1986.
- The CPCB in association with State Pollution Control Boards (SPCBs) has established National Ambient Noise Monitoring Network (NANMN) in 10 Metro-Cities (Bengaluru, Chennai, Delhi, Hyderabad, Kolkata, Lucknow, Bhopal, Vishakhapatnam, Vijayawada and Mumbai) having 82 Noise Monitoring Stations.

6. **Development of Portal**

Development of Common Consent Portal for the Unified Consent & Authorization Management System, Development of Environmental Relief Fund Portal, Common EPR Portal and Eco-Mark Portal.

7. **Important steps taken for industrial pollution control**

- CPCB has initiated Online Continuous Emission & Effluent Monitoring System (OCEMS) to promote self-regulation among highly polluting industries (17 categories), Grossly Polluting Industries (GPIs) and Common facilities i.e. Incinerators of Common Hazardous Waste Treatment Storage and Disposal Facilities (TSDFs), Incinerators of Common Bio-medical Waste Treatment and Disposal Facilities (CBWTFs), Common Effluent Treatment Plants (ETPs) and Sewage Treatment Plants (STPs). Currently, about 7,649 Industries have installed OCEMS & providing online data through about 44,000 devices on 15-minute basis to CPCB, SPCBs & PCCs
- CPCB has revised the methodology for classifying sectors based on their pollution potential. Based on the Pollution Index (PI), industries have been categorized into Red, Orange, Green & White Category. PI is a function of water pollution, air pollution, hazardous waste generation, fuel consumption and amount of waste water generation. So far, 419 industrial / non-industrial sectors (Red: 124, Orange: 135, Green: 94, White: 54 and Blue:12) have been categorized based on revised criteria of Pollution Index. A new category "blue category" was introduced to separate out Essential Environmental Services (ESS) required for management of waste generated from domestic / household activities.

CPCB has issued directions on 12.02.2025 to all SPCBs/PCCs under section 18(1)(b) of the Water Act, 1974 and the Air Act, 1981 for implementation of Classification-2025.

- Environmental Standards have been notified under Schedule-I: Aluminium Industries, Pesticide Industries and Pulp & Paper Industries, Amendment in the Notification (07.12.2015) of coal or lignite based thermal power plants w.r.t. specific water consumption limits vide dated 03.07.2025 and applicability of SO2 emission standards for the Category A,B,C Thermal Power Plants vide dated 11.07.2025. by MoEF&CC. Draft Standards notified for Water quality standards for Coastal water and Marine and Distillery So far, environmental standards have been notified under Schedule–I of EPA, 1986 for 79 sectors. For remaining sectors, general standards notified under Schedule-VI of EPA are applicable.
- Based on Polluter pay Principle, CPCB, SPCBs and PCCs started imposing Environmental Compensation (EC) against the industries, common facilities which are violating the environmental norms / standards for strict compliances.

dance with EPR guidelines notified by MoEF&CC in February, 2022 through 4th Amendment to PWM Rules.

CPCB has framed the Standard Operating Procedure (SOP) and developed the online portal for certification of compostable Manufacturers/Sellers. There are 56,355 Brand-Owners, 2,926 Producers and 55 Importers have been issued registration. As per provision 4 (h) of PWM Rules, 2016, certificates have been issued to 299 compostable Manufacturers / Sellers. Further, CPCB has prepared "Compendium of Eco-Alternatives to Single Use Plastics (SUP)".

CPCB initiated the Phase-I of the audit of registered PWPs generating EPR certificates >200 tons on the EPR portal. Further, letter has been issued to all CPCB RDs and SPBC/ PCCs for initiating the phase II of the audit with 325 certificate generating PWP units. Phase II of the audit is scheduled to be completed on December 31, 2025.

e. E-Waste Management:

E-Waste (Management) Rules, 2022 has been notified on 02nd November, 2022 and implemented since 01st April, 2023 and have superseded E-Waste (Management) Rules 2016. CPCB has developed an online EPR Portal for the management of E-Waste with the provisions of registration of stakeholders, transection of EPR credits between Producers, Recycler & Refurbishes for the fulfillment of EPR obligations.

As on date, 10,326 Producers, 381 Recyclers, 99 Manufacturers & 128 Refurbishers are registered under E-Waste Portal. The annual processing capacity of 381 registered recyclers is 33,17,787.576 TPA. In FY 2023-24, registered recyclers transferred a total of 2,61,220 MT of EPR certificates to registered producers, against the EPR obligation of 2,66,616 MT.

f. Battery Waste Management:

Battery Waste Management (Amendment) Rules, 2022 as amended has been notified on 22.08.2022. Accordingly, 3,937 Producers and 473 Recyclers were registered on the Battery Waste Portal till date.

g. Waste Tyre Management:

CPCB has developed Registration Modules for Producers & Recyclers of waste tyres. Accordingly, 266 Producers and 558 Recyclers were registered on the Waste Tyre EPR Portal till date. The annual processing capacity of 559 registered recyclers is 17,077,430.4 TPA. In FY 2023-24, 198 registered recyclers transferred a total of 19,12,849.503 MT of EPR certificates to registered producers, against the EPR obligation of 20,38,502.515 MT.

11. Industrial CAAQMS Portal:

Industrial Air Quality Management Portal has been developed to collect, store, and analyze data from Continuous Ambient Air Quality Monitoring System installed in industrial premises. The portal supports visualization of ambient air quality data in graphical as well as tabular formats. At present, there are 562 stations integrated to the network.

12. Sameer App:

Sameer App was created by CPCB to create public awareness about air quality and provide a platform for grievances redressal. This App provides hourly updated information on AQI for more than 289 cities using real time data. It can be readily accessed using the interactive map format. It provides real-time AQI, pollutant-wise sub-index and monthly calendar de-

picting overview of AQI for each monitoring location. Daily AQI bulletin published by CPCB is uploaded on the App at 4 PM. The App also issues advisories through push notifications to the public as per prevailing air quality levels. This CAAQMS data is made available to public and various agencies through the CPCB CCR Portal (https://airquality.cpcb.gov.in). Since its launch, it has been downloaded by 2.5 lakhs users on Android and iOS platforms.

Also, Sameer App provides a facility for lodging grievances related to air pollution. The complainant can upload photographs along with complaint, and geo-coordinates are automatically captured for pinpointing exact location of source of air pollution for facilitating prompt action by concerned agencies. The complaints are automatically forwarded to implementing agencies depending on the location of the complaint. For Delhi-NCR, more than 40 agencies responsible for mitigation of air pollution are configured on the App which is well integrated with complaint management system of implementing agencies. Since January, 2025 till date, more than 3855 complaints are received on Sameer App. Out of these around 2152 complaints (56%) were resolved through coordination with more than 40 implementing agencies.

13. Social Media Platform

CPCB has created Social Media Platforms to use them for dissemination of information on pollution related issues, creating awareness on functions, schemes and activities of CPCB and giving citizens the facility to reach out to CPCB for raising air pollution related queries. Presently, CPCB is reachable at Twitter, Facebook, Instagram and YouTube. In current year, more than 400 creatives have been posted on Social Media Platforms by CPCB.

Social Media Accounts are also being used to redress public complaints on air pollution in Delhi NCR, wherein public grievances received on CPCB account are being forwarded to the handles of concerned local agencies.

Annexure - R&D projects granted financial assistance by the Commission

S. No.	Name of project along with Detailed Project Report (DPR)	Cost of project and year wise Budget Allocated	Originally targeted date of completion	Name of PI and co-PI allotted	Current status (ongoing/ completed)
1.	Use of Artificial Intelligence (AI)/ Machine Learning (ML) tool for vehicle counting by uploading the CCTV footage on Cloud Platform.	Rs. 50.0 lakhs + GST @ 18%)	15 months (30.06.2024)	Dr. K.V. George, Chief Scientist & Head, Air Pollution Control Division, NEERI, Nagpur	Field study completed. Report awaited.

S. No.	Name of project along with Detailed Project Report (DPR)	Cost of project and year wise Budget Allocated	Originally targeted date of completion	Name of PI and co-PI allotted	Current status (ongoing/ completed)
2.	Addressing Vehicular Traffic Induced Road Dust re-suspension with S&T based Action Plans for Air Quality Improvement in Delhi NCR	Rs. 48.5 lakhs + GST @ 18%)	12 months (31.03.2024)	Dr. S. K. Goyal, Chief Scientist & Head, CSIR-NEERI, Dr. Neeraj Sharma, Chief Scientist, TPE Division, CRRI	Completed
3.	An Autonomous Drone Swarm Framework for Real Time Air Quality Monitoring and Pollutant Quantification in NCR hotspots.	Rs. 34.2792 lakhs	24 months (18.01.2025)	Dr. Ravi Kumar, Associate Professor, Thapar Institute of Engineering and Technology, Patiala, Punjab	Ongoing
4.	Decision Support System for Air Quality Management in Delhi and its Bordering Districts in the NCR	Rs. 31.98 lakhs	24 months (01.03.2025)	Dr. Gaurav Govardhan, Scientist 'C' IITM, Pune	Completed
5.	"Trial of Ambient Air Purification System (APS) in a Market Place in Delhi NCR" Instead of market place , trial study will conducted at Anand Vihar, ISBT, Delhi	Rs. 59.3154 lakhs including Tax	06 months (18.01.2024) extended upto March, 31, 2024	Dr. K.S. Rajan, Professor, Dept. of Chemical Eng. SASTRA Deemed University, Thanjavur	Completed

S. No.	Name of project along with Detailed Project Report (DPR)	Cost of project and year wise Budget Allocated	Originally targeted date of completion	Name of PI and co-PI allotted	Current status (ongoing/ completed)
6.	Under actuated Filterless Air Cleaner Retrofit for Rolling Stock and Vehicles	Rs. 31.35 lakhs	24 months (13.02.2025)	Dr. Bahni Ray, Assistant Professor, Department of Mech. Eng., IIT Delhi , Shri Karan Rao, Founder, CEO, Swachh. io	Ongoing
7.	Evaluation of Retro-fitment of 2 & 3 Wheeler Auto-rickshaws with Electric Drive as an option for improvement in air quality in NCR	Rs. 105.0 lakhs	12 months (20th February 2024) extended upto 31.07.2024)	Shri Ravindra Kumar, GM, ARAI, Kothrud, Pune	Completed

2.3. Hazardous Substances Management

a. Introduction

Hazardous Substances Management (HSM) Division is the nodal point in the Ministry for management and safe use of chemicals and wastes. The HSM Division is involved in formulation and administration of rules pertaining to wastes and chemicals. The Division is also the nodal point for multilateral conventions related to wastes and chemicals representing Government of India.

b. Major programmes/activities

i. **Solid Waste Management Rules, 2016:**
Solid Waste Management Rules, 2016 were notified on 8th April, 2016 and its applicability has been extended beyond municipal areas to urban agglomerations, census towns, notified industrial townships, etc. The source segregation, door to door collection of waste has been mandated to channelize the waste for useful purposes such as recovery, reuse and recycle.

ii. **E-Waste (Management) Rules, 2022:**
E-Waste (Management) Rules, 2022 notified on 2nd November, 2022 and is in force since 1st April, 2023. New rules covers 106 Electrical and Electronic Equipment (EEE) including Solar PV waste. These Rules are a transformative step towards implementation of the announcement made by Hon'ble Prime Minister to promote Circular Economy. New rules provide Extended Producer Responsibility (EPR) regime for e-waste recycling according to the needs of the current scenario.

iii. **Hazardous and Other Wastes (Management**

and Transboundary Movement) Rules, 2016: The Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 were notified on 4th April, 2016 incorporating waste management hierarchy in the sequence of priority i.e. prevention, minimization, reuse, recycling, recovery, co-processing and safe disposal. These rules shall apply to every occupier of the facility who is engaged in handling, generation, collection, storage, packaging, transportation, use, treatment, processing, recycling, recovery, pre-processing, utilization, offering for sale, transfer or disposal of hazardous and other wastes. The import/export of hazardous and other waste is also being regulated thorough the said Rules.

iv. Bio-Medical Waste Management Rules, 2016:

Bio-Medical Waste Management Rules, 2016 were notified with the objectives to improve segregation, collection, processing, treatment and disposal of infectious bio-medical waste in an environmentally sound and safe manner. These rules are applicable to all persons who generate, collect, receive, store, transport, treat, dispose, or handle bio medical waste in any form including hospitals, nursing homes, clinics, dispensaries, veterinary institutions, animal houses, pathological laboratories, blood banks, AYUSH hospitals, clinical establishments, research or educational institutions, health camps, medical or surgical camps, vaccination camps, blood donation camps, first aid rooms of schools, forensic laboratories and research labs.

v. Plastic Waste Management Rules, 2016:

Plastic waste generated in the country, based upon information provided by State Pollution Control Boards (SPCBs) / Pollution Control Committees (PCCs) to Central Pollution Control Board, the quantity of plastic waste generation during the period 2018-19 to 2022-23 is given below:

Financial Year	Plastic waste Generation (TPA)
2018-19	3360043.45
2019-20	3469781.73
2020-21	4126808.44
2021-22	3901802.06
2022-23	4136188.83

The Plastic Waste Management Rules, 2016, as amended, provide the statutory framework for plastic waste management in an environmentally sound manner throughout the country. The Rules prohibits manufacture, import, stocking, distribution, sale and use of plastic sheets less than fifty microns in thickness in the country. There is complete ban on sachets using plastic material used for storing, packing or selling gutkha, tobacco and pan masala. The Rules also provide measures for recycling of plastic waste and its end of life disposal in an environmentally sound manner.

In order to address pollution caused by littered and unmanaged plastic waste the Ministry of Environment, Forest and Climate Change has banned identified single use plastic items, which have high littering potential and low utility with effect from 1st July, 2022 under Plastic Waste Management Rules. Plastic carry bags having

thickness less than one hundred and twenty microns have also been prohibited with effect from the 31st December, 2022. Further, over and above the Plastic Waste Management Rules, 2016, as per Annual Report of Central Pollution Control Board, thirty five States/UTs have issued notifications/orders to introduce regulations pertaining to complete or partial ban on plastic carry bags and/or identified single-use plastic items.

The Ministry of Environment, Forest and Climate Change notified the Guidelines on Extended Producer Responsibility (EPR) for plastic packaging vide Plastic Waste Management (Amendment) Rules, 2022, on 16th February, 2022. The Guidelines stipulate mandatory targets on EPR, recycling of plastic packaging waste, reuse of rigid plastic packaging and use of recycled plastic content. The guidelines provide for moving towards sustainable plastic packaging and reducing the plastic foot print. The implementation of ban on identified single use plastic items coupled with extended producer responsibility on plastic packaging will reduce pollution caused by littered and unmanaged plastic waste.

The following steps have been taken to strengthen implementation of Plastic Waste Management Rules, 2016 and to implement ban on identified Single Use Plastic items:

(i) All thirty-six States/UTs have constituted the Special Task Force under the chairpersonship of the Chief Secretary / Administrator for elimination of identified single use Plastic items and effective plastic waste management. A National Level Taskforce has also been constituted by the Ministry for taking coordinated efforts to eliminate identified single use plastic items and effective implementation of Plastic Waste Management Rules, 2016.

(ii) Directions have been issued under Section 5 of the Environment (Protection) Act, 1986 for setting up of institutional mechanism for enforcement of provisions of Plastic Waste Management Rules, 2016 to all State Pollution Control Boards / Pollution Control Committees. Directions have also been issued to E-commerce companies, leading Single Use Plastic sellers/users, and plastic raw material manufacturers with respect to phasing out of identified single use plastic items. Separately, custom authorities have been asked to stop the import of banned SUP items.

(iii) For effective monitoring of ban on identified Single Use Plastic items and plastic waste management in the country, the following online platforms are in operation: (a) National Dashboard for monitoring of comprehensive action plan implementation, (b) CPCB Monitoring Module for Compliance on Elimination of Single Use Plastic, and (c) CPCB Grievance Redressal App.

(iv) Regular enforcement drives have been undertaken by CPCB, SPCBs/PCCs along with local authorities to implement ban on identified single use plastic items and on plastic carry bags having thickness less than one hundred twenty microns.

The total Extended Producer Responsibility (EPR) of registered Producers, Importers and Brand owners (PIBOs) for plastic packaging is around 8.3 million tonnes for the year 2024-25, as per national dashboard of the Centralized Online Portal on Plastic Packaging, under EPR Guidelines. Already, around 57746 PIBOs and 2969 Plastic Waste Processors are registered on the Centralized EPR portal for plastic packaging. The quantum of EPR certificates generated by registered plastic waste processors after processing of plastic packaging waste including recycling amounts to 18 million tonnes.

vi. Environment (Construction and Demolition) Waste Management Rules, 2025:

For the first time the Government notified Construction & Demolition Waste Management Rules, 2016. The rules apply to everyone who generates construction and demolition waste such as building materials, debris, and rubble waste resulting from construction, re-modelling, repair and demolition of any civil structure of individual or organization or authority. The rules mandate for collection, transportation, storage and re-processing of C&D waste. Emphasis was given on recycling/ recovery of valuable materials and C&D waste as 'RESOURCE'. Scope of the rules includes construction & demolition/ renovation of buildings, excavation, road/ flyover/ underpass/ bridge construction, laying utility pipelines etc. Timeline were prescribed for creation of physical reprocessing facility at all places for recycling C&D waste.

To further strengthen the Construction and Demolition waste management in the country. The Ministry of Environment, Forest and Climate Change has notified the Environment (Construction and Demolition) Waste Management Rules, 2025, vide G.S.R. 219(E), dated 2nd April 2025. These rules supersede the earlier Construction and Demolition (C&D) Waste Management Rules, 2016, and will come into effect from 1st April 2026. The revised framework marks a significant policy shift, aiming to promote environmentally sustainable management of construction and demolition waste while fostering a circular economy. The rules introduced the concept of Extended Producer Responsibility (EPR) for construction and demolition waste. The Rules set specific targets for the use of recycled materials in construction, reconstruction, and road infrastructure projects, encouraging the reuse of materials that would otherwise contribute to environmental degradation.

vii. Battery Waste Management Rules, 2022:

Ministry of Environment, Forest and Climate Change, Government of India published the Battery Waste Management Rules, 2022 on 24th August, 2022 to ensure environmentally sound management of waste batteries. New rules replace Batteries (Management and Handling) Rules, 2001. The rules cover all types of batteries, viz. Electric Vehicle batteries, portable batteries, automotive batteries and industrial batteries. The rules function based on the concept of Extended Producer Responsibility (EPR) where the producers (including importers) of batteries are responsible for collection and recycling/refurbishment of waste batteries and use of recycled materials from wastes into new batteries.

viii. Ash Utilisation notification, 2021:

The Ministry of Environment, Forest and Climate Change has reviewed the existing fly ash notification, 1999 and issued new notification on 31st December, 2021 with an aim to achieve 100% ash utilisation generated from coal and lignite based thermal power plants. The notification mandates thermal power plants to achieve 100% utilisation of current generation of ash in 3-5 year cycle and legacy ash in 10 years. The eco-friendly purposes for utilisation of ash have been prescribed. Environmental compensation is to be imposed on thermal power plants and other user agencies for non-compliance of provisions. Users agencies involved in road construction, roads and flyover embankments laying, construction of dams, mining, manufacture of ash based products, building construction have been mandated to use ash or ash based products located within 300 km radius of thermal power plants.

ix. Environment Protection (End-of-Life Vehi-

cles) Rules, 2025:

The Ministry of Environment, Forest and Climate Change notified the Environment Protection (End-of-Life Vehicles) Rules, 2025 on 06.01.2025 for environmentally sound management of end-of-life vehicles. The rules apply to all types transport and non-transport vehicles, except agricultural tractor, agricultural trailer, combine harvester, power tiller. Under the rules, the producers of transport and non-transport vehicles have been given mandatory Extended Producer Responsibility (EPR) targets for scrapping of end-of-life vehicles, in the range 8% to 18% for the vehicles placed in the market 20 years back for non-transport vehicles and 15 years back for transport vehicles. Registered Vehicle Scrapping Facilities (RVSFs) have been obligated to generate EPR certificates based on the weight of the steel recovered by scrapping of end-of-life vehicles or any other steel scrap materials from the automobile sector processed at RVSF.

x. Environment Protection (Management of Contaminated Sites) Rules, 2025:

The Ministry of Environment, Forest and Climate Change notified the Environment Protection (Management of Contaminated Sites) Rules, 2025 on 24.07.2025 for identification and management of contaminated sites in the country. The process of identification, determination and remediation of contaminated sites has been defined under the said Rules. Under the Rules, the responsible person has been mandated to prepare the remediation plan, carry out the remediation activities of the contaminated sites as per the remediation plan, and submit the progress reports periodically to CPCB and the concerned SPCB/ PCC. In case of orphan contaminated sites and sites under temporary possession of banks or judicial proceedings, SPCB/ PCC has been given the responsibility to prepare the remediation plan and conduct the remediation activities. The expenditure for remediation activities in this regard is to be met by the Central and State Government in the ratio and manner prescribed in the Rules.

xi. Chemical Safety:

The Ministry of Environment, Forest and Climate Change notified the Manufacture, Storage and Import of Hazardous Chemicals (MSIHC) Rules, 1989 and the Chemical Accidents (Emergency Planning, Preparedness and Response) (CAEPPR) Rules, 1996 for ensuring chemical safety in the country. Objective of these rules is to prevent chemical accidents from industrial activities and mitigate impacts of chemical accidents. Occupier of the hazardous chemicals is required to declare major hazards of industrial activity to public and prepare on-site emergency plan & safety reports, conduct regular safety audits and mock-drills, etc.

xii. Public Liability Insurance Act, 1991:

The Public Liability Insurance Act, 1991 enacted with a purpose to provide immediate relief to the persons affected by accident occurring while handling hazardous substance and for matters connected therewith or incidental thereto. PLI Act, 1991 covers death or injury to any person other than workman (General Public) or damage to property resulting from an accident while handling hazardous substance. The Act covers 179 chemicals and flammable substances. As per Section 7A of the Amended Act, 1992, the Major Accident Hazard (MAH) units shall contribute an amount equal to the amount of premium of the insurance policy to the Environment Relief Fund (ERF) created by the Central Government.

xiii. Circular Economy:

- 'Circular Economy' is a focus of Government of India. India had brought in market based EPR regulation in 2022 which has led to the growth of recycling sector across five waste streams viz. plastic packaging, E-waste, battery waste, waste tyre and used oil.
- The EPR regulation has become a tool for operationalization of "Waste to Wealth" and Circular Economy.
- Under the EPR concept the producer, importer or brand owner (PIBOs) who introduces the product in the market becomes obligated under law for its environmentally sound management. Regulations on market based Extended Producer Responsibility (EPR) principle have been notified in respect of plastic packaging waste, battery waste, e-waste, waste tyre, used oil, end-of-life vehicles, Scrap of Non- Ferrous Metals and construction & demolition waste.
- Introduction of market based EPR regulation by MoEFCC since 2022 in different waste streams has not only led to rapid growth of recycling sector but has also led to environmental protection and resource conservation through environmentally sound management of wastes and reduction in use of virgin material.
- Extended Producers Responsibility (EPR) is a policy tools being implemented as an environmental protection strategy to achieve objectives of Circular economy by making the manufacturer or Importer or Producer or Brand Owner of the product to take-up responsibility for the life-cycle of their products placed on market, after its end of use.
- As per the mandate given under the above EPR framework, till date following 05 EPR Portals have been developed by CPCB and are operational for implementation of EPR Frameworks:
  - · EPR Portal for Plastic Packaging
  - · EPR Waste Tyre Management Portal
  - · EPR Portal for Battery Waste Management
  - · E-waste Management System
  - · EPR Portal for Used Oil Management.
- Common EPR portal, Uniform Consent Portal and electronic trading platform for exchange of EPR certificates between recyclers and producers are under development.
- Across all EPR regulations, Producers/ OEMs are registered with CPCB and recyclers/refurbishers are registered with concerned SPCB/PCCs.
- Verification of EPR certificates of recyclers is done by CPCB/SPCB through periodic audits.
- Status on Implementation of Extended Producer Responsibility Frameworks (as on 20/12/2025):

S.No	Waste Type	Registered Producers Nos.	Registered Recyclers (Nos.)	EPR Certificate generated (in tonnes)	Total EPR certificate transferred (in tonnes)
1.	Plastic Packaging	57,428	2950	179.07 lakh	144.42 lakh
2.	E-Waste	10,709	382	10.51 lakh	7.69 lakh
3.	Battery Waste	4072	488	13.48 lakh	10.28 lakh

Change has notified the Ecomark Rules on 26th September 2024 which have replaced the Ecomark scheme of 1991.

The objective of the scheme is to encourage the demand for environment-friendly products, promote lower energy consumption, resource efficiency and circular economy. The scheme seeks to ensure accurate labelling and prevent misleading information on green products.

Ecomark will enable consumers to make informed purchase decision as well as encourage manufacturers to transition to production of environment friendly products leading to promotion of green industries.

Governance structure of Ecomark comprises of Steering Committee chaired by the Secretary, MoEFCC, Central Pollution Control Board (as implementing agency) and Technical Committees. Steering Committee has been constituted on 23rd October 2024 and has members from concerned Ministries/Departments, experts and institutions, and industry associations.

Ecomark is eligible to be availed by those products holding a valid license or a certificate of conformity of Indian Standards under the BIS Act, 2016 and/or falls under the scope of a mandated quality control order issued by the Central Government.

Products accredited under the Ecomark Scheme will adhere to specific environmental criteria, ensuring minimal environmental impact.

At present, the scope of the scheme covers 17 different product categories comprising of 80 types of products such as food additives, soaps, detergents, batteries etc.

2.4. Policy and Law

a. Introduction and Objective

The Policy and Law (P&L) Division deals with the administration/ establishment matters of the NGT; framing of NGT Recruitment Rules/ Amendments in the NGT Act and NGT Recruitment Rules; coordination with the Ministry of Law & Justice and handling of court cases in the Supreme Court of India, various High Courts and NGT (all its Benches).

The P&L Division handles the administrative issues of NGT including their budget alongwith portion of budget related to the ESZ and CP Divisions, which are administrative-in-charge of Dahanu Taluka Environment Protection Authority and Loss of Ecology Authority, respectively. The Division is allocated funds under the Budget Head "National Green Tribunal" and the budget under this is further allocated to –

- National Green Tribunal;
- Grants-in-Aid to Loss of Ecology Authority, Chennai, and
- Grants-in-Aid to Dahanu Taluka Environment Protection Authority, Ahmedabad.

Legal Monitoring Cell (LMC)

A Legal Monitoring Cell set up in the Policy & Law Division monitors the progress of court cases and also acts as an effective coordinator with various Divisions of the Ministry regarding court cases. The total number of court cases, as on 30.12.2025, pending before the Supreme Court of India, High Courts, National Green Tribunal and other subordinate courts is 3115.

2.5. National Green Tribunal (NGT)

The National Green Tribunal (NGT) was set up on 18th October, 2010 under the NGT Act, 2010 for the purpose of effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources including enforcement of any legal right relating to environment and giving relief and compensation for damages to persons and property and for matters

connected therewith or incidental thereto.

The Tribunal has five places of sitting i.e., the Principal Bench at New Delhi and Zonal Benches at Pune, Kolkata, Bhopal and Chennai. As per Section 4(1) of NGT Act, 2010 the Tribunal shall consist of (a) one full time Chairperson, (b) not less than ten but subject to maximum of twenty full time Judicial Members as the Central Government may, from time to time, notify and (c) not less than ten but subject to maximum of twenty full time Expert Members, as the Central Government may, from time to time, notify. MoEF&CC, in June 2024, advertised the five (05) vacant posts of Judicial Members and five (05) vacant posts of Expert Members in the NGT. Upon completing the process of selection , 02 Judicial and 04 Expert Members of NGT have been appointed in August 2025. Out of these, four Expert Members have assumed the charge in NGT.

3.1. Environmental Impact Assessment (EIA)

1. Environmental Impact Assessment (EIA) serves as a planning instrument designed to incorporate environmental considerations into the development process from the very beginning. In India, EIA was first applied in 1978 for river valley projects and was subsequently extended to major public sector projects that required approval from the Public Investment Board (PIB). These practices were formally institutionalized through the EIA Notification of 1994, which came into force on 27 January 1994. Initially, the notification required environmental clearance for 29 categories of projects, a number that was later increased to 32 based on investment criteria. In 2006, a revised EIA Notification was introduced with the objective of creating a more transparent, efficient, and decentralized system. This revision integrated environmental safeguards at the planning stage and placed greater emphasis on public participation. It also marked a shift from investment-based criteria to impact-based assessment, making environmental clearance mandatory for 39 categories of projects.
2. The EIA Notification, 2006 and the CRZ Notification, 2011 have been amended periodically to further streamline the environmental clearance process. These amendments have also sought to address emerging issues by integrating environmental considerations into the development process, in line with the objective of sustainable development. While granting Environmental Clearance (EC) to development projects, specific conditions, environmental safeguards, and mitigation measures are prescribed to ensure their effective implementation during both the construction and operational phases. These safeguards are aimed at minimizing adverse impacts, inter alia, on (i) air quality, (ii) water quality, (iii) land degradation, (iv) biodiversity, and (v) wildlife habitats. In addition, projects are required to implement measures such as rainwater harvesting, water conservation, greenbelt development and plantation, wildlife conservation plans, and other related initiatives.
3. Environmental Clearance to Developmental Projects

   In accordance with the provisions of the Environment Impact Assessment (EIA) Notification, 2006, meetings of the various sector-specific Expert Appraisal Committees (EACs) were convened during the year for appraisal of Category 'A' projects pertaining to sectors such as Industry, Thermal Power, River Valley and Hydroelectric Projects, Coal Mining, Non-Coal Mining, Infrastructure and Coastal Regulation Zone (CRZ), as well as Nuclear, Defence and allied projects.

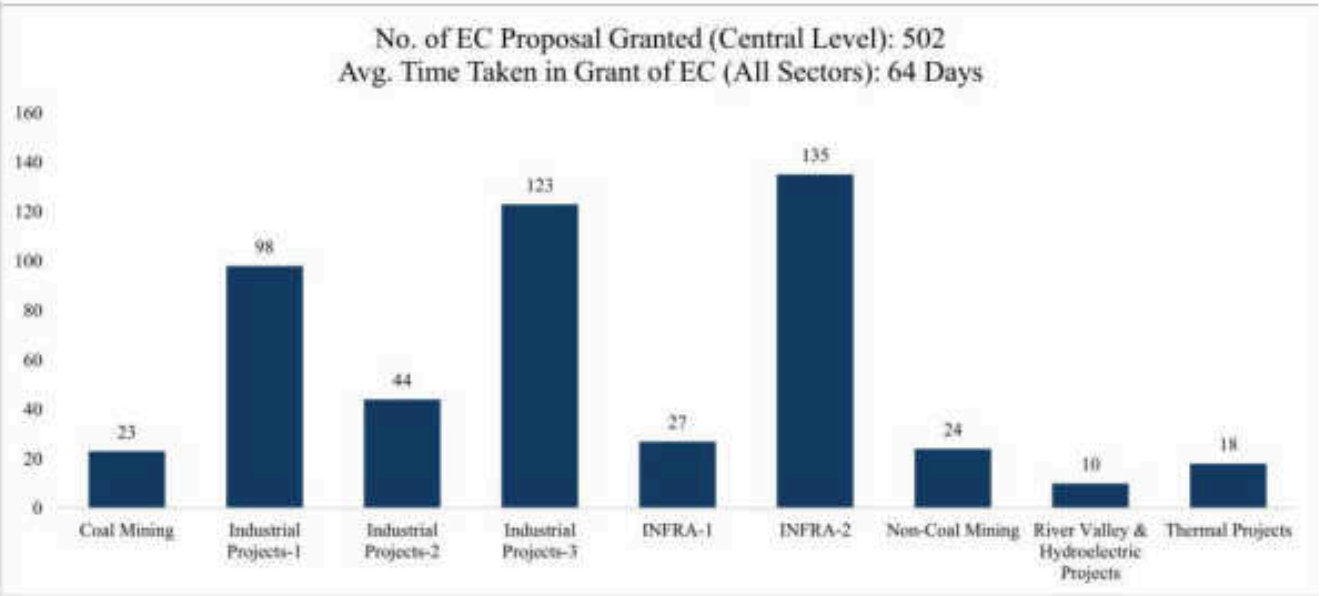
   During the period from 1 January 2025 to 31 December 2025, Environmental Clearance (EC) was accorded to a total of 502 projects. The sector-wise distribution of Environmental Clearances granted by the Ministry during the year 2025 (up to 31 December 2025) is depicted in Figure 1. The average time taken for grant of Environmental Clearance across all sectors at the Central level was 64 days, as against the stipulated timeline of 105 days.

4. Implementation of Single-Window Integrated Environmental Management System.

   With a view to strengthening e-Governance and enhancing ease of doing business, the PARIVESH portal was launched in 2018 as a single-window platform for Environment, Forest, Wildlife and CRZ clearances and was extended to all States and Union Territories in 2019. The portal is currently operational across 36 States / UTs, 19 Regional Offices and the Ministry.

   The PARIVESH system has been progressively

Fig. 1 - No. of EC proposals Granted (Central Level)



upgraded in line with statutory requirements and has ensured uninterrupted service delivery. To further improve efficiency and user experience, PARIVESH 2.0 was developed by leveraging GIS and advanced data analytics for end-to-end online appraisal, grant of clearances and compliance monitoring.

PARIVESH 2.0 includes modules such as Know Your Approval (KYA), Know Your Customer (KYC) and a GIS-enabled Decision Support System (DSS), and enables complete online processing of Category 'A' and Category 'B' Environmental Clearance proposals at the Central and SEIAA levels, respectively. The platform has also been integrated with national digital systems including GatiShakti, the Ministry of Coal systems and the PMG Portal.

In pursuance of G.S.R. 702(E) and 703(E) dated 12.11.2024, the Ministry issued an Office Memorandum dated 14.01.2025 prescribing the procedure for processing projects requiring prior Environmental Clearance but exempted from obtaining Consent to Establish. The entire workflow, including seeking comments from SPCBs / PCCs, has been automated through the PARIVESH portal.

The paperless clearance management system under PARIVESH 2.0 has strengthened e-Governance, improved ease of doing business and contributed to reduction of carbon footprint. Key outcomes during 2025 are as follows:

a. Prior Planning: About 5,500 users utilised the Know Your Approval (KYA) module for early-stage assessment of environmental sensitivity of proposed projects.
b. Informed Decision-Making: Over 1,300 ToR and EC proposals were appraised and granted using the GIS-enabled Decision Support System (DSS).
c. Expeditious Processing: Systemic reforms and automation reduced the average time for grant of EC at the Central level to 64 days, against the stipulated timeline of 105 days.

5. Compliance and Monitoring Division (CMD)
Post Project Monitoring of Environment Clearance Condition: Clearances under EIA Notifi-

cation, 2006 are granted by MoEFCC for Category 'A' projects and by State Level Impact Assessment Authorities (SEIAAs) / UTIAAs for Category 'B' projects. Similarly, clearances under CRZ Notification, 2011 are also granted by MoEFCC or the respective SEIAAs as the case may be after the project has been recommended by the concerned Coastal Zone Management Authority. While granting environmental clearances under both these Notifications, various conditions and environmental safeguards are stipulated which are required to be implemented by the project proponent during various stages of project cycle.

The objectives of Post-Project Clearance Monitoring are:

- to ensure that actions have been taken to incorporate the environmental safeguards during the project cycle in accordance with the conditions stipulated in the Environmental Clearance letter; and
- to take appropriate corrective measures to check adverse impact on environment during operation of the respective projects.

Monitoring of projects with respect to conditions stipulated in the environmental clearance issued under EIA Notification, 2006 and Coastal Regulation Zone (CRZ), 2011 is carried out through the Eleven (11) Regional Offices (ROs) and Nine (9) Sub Regional Office as also through the Consent mechanism of the State Pollution Control Boards and UT Pollution Control Committees.

As a step towards strengthening the Environmental Compliance and Monitoring for effective implementation of provisions of Environment (Protection) Act, 1986; the Ministry has created a dedicated and separate "Compliance and Monitoring Division" (CMD) vide Office Order dated 13.07.2023.

The major responsibilities of CMD are:

- Monitoring of EC granted under the EIA Notification, 2006,
- Functional auditing of SEIAA/CZMA and post EC monitoring in relation to Court cases.
- Parliament Matter, VIP reference, PG, RTI etc.

C&M Division examines the monitoring reports submitted by Regional Offices. In cases of major non-compliance, further follow up action is taken for effecting compliance including issue of show cause notice followed by directions under Environment (Protection) Act, 1986 on case to case basis.

Environment Audit Rules, 2025

In line with the Government's commitment to Ease of Doing Business (EoDB), principles of trust-based governance and to expand the resource pool for compliance monitoring by involving professionals, the Ministry of Environment, Forest and Climate Change (MoEFCC) notified the Environment Audit Rules, 2025 on 29th August 2025 with the goal of attaining minimum government and maximum governance for enforcing compliance responsibilities on one hand and ensuring EoDB on the other. The key objectives of the rules are to institutionalize environmental audits to bridge manpower and infrastructural gaps in compliance monitoring, promote transparency and encourage self-compliance.

Accreditation of the EIA consultants with Quality Council of India (QCI / National Accreditation Board of Education and Training (NABET) :

The environmental appraisal of development projects is undertaken as per the provisions of EIA Notification 2006 based on EIA/EMP re-

4.1 Conservation and Survey- III (Biodiversity) Division

a. Introduction, objectives and functions

The areas of work dealt within CS III (Biodiversity) division can be broadly categorized under following categories:

- **International Commitments:**

I. Convention on Biological Diversity (CBD).
II. Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization.

- **National Legislation:**

Biological Diversity Act, 2002 (amendment in 2023)

- CBD is one of the key agreements adopted during the Earth Summit held in Rio de Janeiro in 1992. The three objectives of the CBD are: conservation of biological diversity, sustainable use of its components, and fair and equitable sharing of benefits arising out of the use of genetic resources. Pursuant to ratification of the CBD by India on 18 February 1994, several steps were initiated to meet the commitment s under the Convention, and to realize the opportunities offered by the Convention. India enacted the Biological Diversity Act in 2002 to give effect to the provision of this Convention. India has also prepared a National Biodiversity Action Plan (NBAP) in 2008, and an Addendum to NBAP was prepared in 2014 with 12 National Biodiversity Targets, developed in line with the 20 global Aichi biodiversity targets. India also submit a National Report to CBD providing an update on progress in National Biodiversity Targets. Recently India submitted its 23 National Targets in line with the Kunming Montreal Global Biodiversity Framework (KMGBF) on 10th September 2024 to the CBD portal. Further India submitted its updated National Biodiversity Strategy and Action Plan on 31st October 2024 to the CBD portal.
- The Nagoya Protocol on Access to Genetic Resources (A BS) and the Fair and Equitable Sharing of Benefits Arising from their Utilization, adopted under the aegis of CBD in 2010. India signed the Protocol on 11 May 2011, and ratified it on 9 October 2012. The Nagoya Protocol on ABS entered into force on 12 October 2014. The Nagoya Protocol is being implemented at national level inter alia through the B D Act.
- Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES) was established by the Governments of various countries in April 2012 as the biodiversity counterpart of Intergovernmental Panel on Climate Change, for strengthening the science-policy interface on biodiversity so as to provide policy makers with reliable, independent and credible information on biodiversity, thus promoting human well-being and sustainable development through sustainable use of biodiversity.
- In pursuance to CBD, India enacted the Biological Diversity Act, 2002 (amendment 2023), and notified the Biological Diversity Rules in 2004, which were superseded by the Biological Diversity Rules 2024.
- The Act provides for conservation of biological diversity, sustainable use of its components and fair and equitable sharing of the benefits arising out of the use of biological resources, knowledge and for matters connected therewith or incidental thereto. It primarily aims at giving effect to the provisions of the CBD suiting to our national needs and requirements. The provisions of the Nagoya Protocol on Access and Benefit Sharing to which India is a Party, are also being implemented through the Biological Diversity Act.
- The Biological Diversity Act, 2002 is implemented through a three-tier institutional mech-

anism: NBA, State Biodiversity Boards (SBBs) and Biodiversity Management Committee s (BMCs). NBA is a statutory body that performs facilitative, regulatory and advisory functions for the Government on biodiversity related issues.

b. Important activities undertaken

- The Ministry celebrated the "International Day on Biological Diversity 2025" on 22 May 2025. The theme of the event is "Harmony with Nature and Sustainable Development". The event was well attended by Senior officials from the Central / State Government, college student, NGOs and Biodiversity Management Committees members and made commitments towards biodiversity conservation.
- During the event, the booklet on National Biodiversity Strategy and Action Plan 2024-2030 was released along with following booklets and brochures:
  i. Summary for Stakeholders: India's Updated National Biodiversity Strategy and Action Plan (NBSAP 2024-2030).
  ii. Brochure on the preparation of India's Seventh National Report (NR 7) to the Convention on Biological Diversity (CBD)
  iii. Curriculum and Prospectus: MoEFCC-IICA-NBA-UNDP Certificate Programme in Business & Biodiversity Conservation.
  iv. Compendium of Biodiversity Heritage Sites in India.
  v. Brochure on The Biological Diversity (Access to Biological Resources and Knowledge Associated thereto and Fair and Equitable Sharing of Benefits Regulations, 2025)

c. Progress/Innovations

Internationally Recognized Certificate of Compliance (IRCC) issued under Nagoya Protocol: Nagoya Protocol on ABS requires Parties to issue a permit or its equivalent at the time of access as evidence that access to genetic resources was based on prior informed con sent and that mutually agreed terms were established. The Protocol further requires that Parties make information on the permit or its equivalent available to the ABS Clearing House for it to constitute an Internationally Recognized Certificate of Compliance (IRCC). Since October 2015, 3556 (6 0.24%) approval granted to users of biological resources in the ABS CH website for generating IRCC out of 5903.

d. Important cumulative achievements

i. 2.76 lakhs BMCs have been established in 28 States and 8 Union Territories across India.
ii. 2.72 lakhs PBRs have been documented by the BMCs with the support of SBBs and NBA.
iii. 54 Biodiversity Heritage sites have been declared 19 States
iv. So far, NBA distributed Access Benefit Sharing (ABS) to the tune of Rs. 70/- crore to all SBBs / UTBCs, research institutions, Forest departments etc.
v. So far, India has generated 3,556 Internationally Recognized Certificate of Compliance (IRCC). India accounts for 57.35% of the total 6,200 IRCCs published by other Parties to the convention on biological diversity portal.

e. Promulgation of new Acts / Rules

- Vide Gazette notification no. 332 dated 29th April, 2025, and in supersession of the Access and Benefit Sharing Regulation, 2014, notified the Biological Diversity (Access to Biological Resources and Knowledge Associated thereto and Fair and Equitable Sharing of Benefits) Regulations, 2025.
- Amendment of Rule 19 and associated Forms 11 and 12 of the Biological Diversity Rules, 2024, by The Biological Diversity (Amendment) Rules, 2025, vide Gazette Notification G.S.R. 295(E) dated 6th May 2025 and it came into force on 1st November 2025

f. Important Conferences

Workshops organized in India:

- Consultation for reporting on seventh National Report on19 September 2025 in New Delhi.
- National cum State-level Inception Workshop on the Gol- UNDP-GEF7 BD Project entitled "Strengthening Institutional Capacities for Securing Biodiversity Conservation Commitments", from 14-15 November, 2025 in Shillong, Meghalaya.
- National-Level Inception Workshop for the Preparation of India's Seventh National Report (NR-7) to the Convention on Biological Diversity (CBD) on 19.09.2025
- Validation workshop for the GBFF project on "Conservation of Biodiversity, its Sustainable Use, Fair and Equitable Sharing of Benefits in India" (CONSERVE) on 05-06 July 2025, Faridabad.
- BIOFIN Annual Planning and Review Meeting on 24 January 2025, Chennai.
- Strengthening Institutional Capacities for Securing Biodiversity Conservation Commitments; Project orientation workshop cum planning meeting on 23 January 2025, Chennai.

International conferences:

- Second resumed session of the sixteenth meeting of the Conference of the Parties to the Convention on Biological Diversity; the fifth meeting of the Conference of the Parties to the Convention on Biological Diversity serving as the meeting of the Parties to the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from Their Utilization; the eleventh meeting of the Conference of the Parties to the Convention on Biological Diversity serving as the meeting of the Parties to the Cartagena Protocol on Biosafety from 25–27 February 2025, Rome, Italy.
- Regional workshop on knowledge management for biodiversity for countries in Asia 11–13 March 2025, Bangkok, Thailand.
- Sub-regional dialogue on biodiversity monitoring and reporting with a focus on ecosystem restoration (Target 2 of the Kunming-Montreal Global Biodiversity Framework), 2–5 September 2025, Bangkok, Thailand.
- Twenty-seventh meeting of the Subsidiary Body on Scientific, Technical and Technological Advice, 20–24 October 2025, Panama City, Panama.
- First meeting of the Subsidiary Body on Article 8(j) and Other Provisions of the Convention on Biological Diversity Related to Indigenous Peoples and Local Communities, 27–3 0 October 2025, Panama City, Panama.

g. Major publications

Compendium of Biodiversity Heritage Sites in India released during the International Day for Biological Diversity, May 22, 2025

h. Autonomous bodies

- The National Biodiversity Authority is a body corporate established in accordance with the provisions of Section 8 of the Biological Diversity Act, 2002, at Chennai w.e.f. 1st October 2003. It is an autonomous, statutory and regulatory organization which is intended to implement the provisions of Biological Diversity Act, 2002. The Biological Diversity Act 2002 came into force in 2003. The Act extents to the whole of India. The objectives of the Act are conservation, sustainable utilization and fair and equitable sharing of benefits arising out of the use of biological resources and associate d knowledge. The Act is being implemented through a three-tiered institutional structures (NBA at National level, Stat e Biodiversity Boards at State level and Biodiversity Management Committees at local level)
- The main objectives of NBA are:

i. To regulate access to biological resources of the country to conserve and sustainable use of biological diversity;

cember 12–15, 2024) and BSI-NRC, Dehradun (August 23–29, 2025).

- Environmental Awareness Initiatives:
  - A special lecture on 'The Changing Paradigm of our Environment' was organized at BSI-ANRC, Port Blair, in association with SURYA, India, as part of the 'Mission LiFE' (Lifestyle for Environment) initiative.

c. Progress/innovations

New Discoveries: BSI published Plant Discoveries 2024, a significant enumeration adding 1 family, 10 genera, 410 species, and 23 infraspecific taxa to the Indian flora. Of these discoveries, 245 were entirely new to science, while 188 represented new distributional records within the country, with seed plants and fungi each contributing the maximum share (36%) of novelties. These findings update the total recorded plant diversity in India to 56,177 taxa, encompassing 22,368 angiosperms, 15,986 fungi, 9,117 algae, 3,151 lichens, 2,850 bryophytes, 1,325 pteridophytes, 1,297 microbes, and 83 gymnosperms.

Digitization, BSI maintains the online digital platforms like "Indian Virtual Herbarium" (https://ivh.bsi.gov.in/), e-Plant Checklist of India and e-Flora of India (https://efloraindia.bsi.gov.in/), e-Archives (https://archive.bsi.gov.in/), Rare Books (https://bsi.gov.in/page/en/rare-books), Journal Nelumbo (http://nelumbo-bsi.org), Medicinal Plant Database (https://bsi.gov.in/page/en/medicinal-plant- database) etc. which serve as important online resources for the Botanical fraternity and promoting taxonomic research in the country.

During the reporting period, a total of 40,503 herbarium sheets were successfully digitized, and 88,056 associated metadata records were meticulously prepared and integrated into the online archives.

d. Important Cumulative achievements

- Floral Survey, collection and identification: BSI undertook 88 floral surveys and local tours across diverse phytogeographical regions of the country as part of 80 distinct research initiatives. Three Herbarium Consultation Tours (HCTs) were also executed at various herbaria within India. These combined efforts during the reference period resulted in the collection and precise identification of 4,922 floral specimens across different plant categories.
- Ex-situ Conservation: BSI actively collected, multiplied, and introduced approximately 673 plant species into its network of botanic gardens. At the Acharya Jagadish Chandra Bose Indian Botanic Garden, new Medicinal Plants and Wild Edible Fruit sections established and a taxonomic garden and a sensory garden have been initiated. New facilities as Geographical Information System (GIS) laboratory and a Mushroom laboratory established at the Central National Herbarium (CNH) in Howrah.

e. Continuing Committees / Commissions

- The 14th meeting of BSI Research Advisory & Monitoring Committee (RAMC) was convened at Kasauli, Himachal Pradesh, running from April 18th to April 20th, 2025.

f. Important conferences organised in India or international conferences

BSI was represented at key international conservation forums by senior scientists who underscored India's commitment to biodiversity documentation and policy.

- Dr. S. S. Dash, Scientist-F & In-charge, Technical Section, BSI-Headquarters, participated in the IUCN World Conservation Congress 2025 in Abu Dhabi as part of the Indian delegation. During the congress, India launched its National Red List Roadmap and Vision 2025–

2030, a landmark initiative emphasizing science-based frameworks for comprehensive assessment and conservation of the nation's rich biodiversity.

- Dr. Avishek Bhattacharjee, Scientist-E, BSI-Central National Herbarium, Howrah, attended the 20th Meeting of the CITES Conference of Parties (CoP-20) held in Samarkand, Uzbekistan, from November 24 to December 5, 2025.

g. Multilateral Environmental Agreements (MEAs) with Countries and International Organizations

BSI expanded its collaborative research network by signing three new Memoranda of Understanding (MoUs). These agreements were established with Berhampur University in Odisha, the Green For Life Foundation in Kolkata, and the esteemed Royal Botanic Garden in Edinburgh, UK, formalizing partnerships for future scientific cooperation and joint research initiatives.

h. List of major publications

During the reference period, BSI published 11 volumes of the Flora of India (Volumes 8, 11, 14, 16, 19, 21, 27, 28, 30, 31, 32). Completion of the state flora publications, with three volumes each published for Gujarat and Chhattisgarh. BSI also produced a consolidated identification resource titled 400 Wild Mushrooms of India, an Illustrated Field Guide, designed to assist researchers and students for differentiation between edible and non-edible mushrooms based on appearance.

New titles slated for imminent publication:

- Bharat Vriksh-Trees of India (Two volumes), which will depict India's tree wealth.
- Botanical Paintings, a coffee table book showcasing the work of Indian botanical artists from the post-independence period.
- Amazing Antarctica (Cynoprokaryotes and Algae), a book offering the first insights into the microbial life of the Larsemann Hills following the First Indian Scientific Expedition to the region.



***Aeschynanthus chayangtajoensis* Chowlu from A N islands.**



***Centranthera grandiflora* Benth Salisb. Scouting**



***Desmos chinensis* Lour**



Training session for school students demonstrating propagation methods



Prepared saplings of *Bombax insigne*



Quick Response Code

4.4 Biosphere Reserves

Conservation & Survey - I (Biosphere Reserve) Division

a. Introduction, Objective and Functions

Biosphere Reserves are large landscapes, which represents terrestrial and coastal/marine ecosystems or a combination thereof. These are designated within the framework of UNESCO's Man and Biosphere Programme. The Biosphere Reserves represent characteristic ecosystems in different biogeographic zones and local communities as one of the integral component.

Biosphere Reserves are demarcated into following three inter-related zones viz:

i. Core Zone- strictly protected for conservation of landscapes, ecosystems, species, and genetic variation.
ii. Buffer Zone- surrounds the core; used for research, education, and sustainable activities.
iii. Transition Zone - outermost part; fosters economic and human development that is socio-culturally and ecologically sustainable.

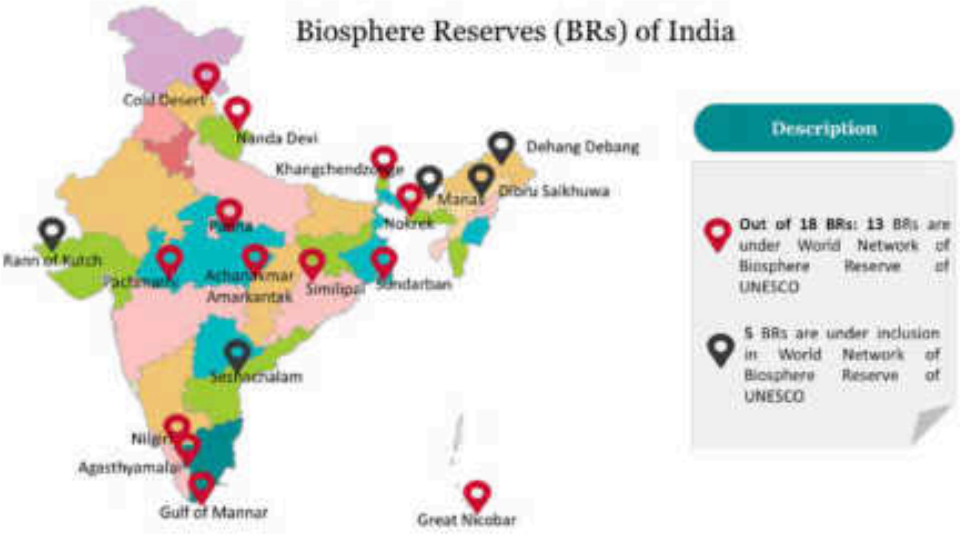
Man and the Biosphere (MAB) Programme is a flagship intergovernmental scientific initiative of UNESCO launched in 1971 to improve the relationship between people and their environment through interdisciplinary research, capacity building, and policy advice. The MAB Programme contributes to achieving the Sustainable Development Goals (SDGs) by promoting conservation of biodiversity, sustainable use of natural resources, and socio-economic development of local communities.

CS I (Biosphere Reserve) Division of the Ministry of Environment, Forest & Climate Change administers a Centrally Sponsored Scheme namely 'Biodiversity Conservation' scheme, which is a

List of Biosphere Reserves in India

S.No.	Name of the BR	Year of Designation	Year of inclusion in the WNBR	States	Area (sq.km)
1	Nilgiri*	1986	2000	Tamil Nadu, Kerala and Karnataka	5520
2	Nanda Devi*	1988	2004	Uttarakhand	6407.03
3	Nokrek*	1988	2009	Meghalaya	820
4	Gulf of Mannar*	1989	2001	Tamil Nadu	10500
5	Sunderbans*	1989	2001	West Bengal	9630
6	Manas	1989		Assam	2837
7	Great Nicobar*	1989	2013	Andaman and Nicobar Islands	885
8	Simlipal*	1994	2009	Odisha	5569
9	Dibru-Saikhowa	1997		Assam	765
10	Dehang-Dibang	1998		Arunachal Pradesh	5111.50
11	Pachmarhi *	1999	2009	Madhya Pradesh	4981.72
12	Khangchendzonga*	2000	2018	Sikkim	2931.12
13	Agasthyamalai*	2001	2016	Kerala, Tamil Nadu	3500
14	Achanakmar-Amarkantak*	2005	2012	Madhya Pradesh, Chhattisgarh	3835.46
15	Great Rann of Kutch	2008		Gujarat	12454
16	Cold Desert*	2009	2025	Himachal Pradesh	7770
17	Seshaschalam	2010		Andhra Pradesh	4755.58
18	Panna*	2011	2020	Madhya Pradesh	2998.98

included in the World Network of Biosphere Reserves



sub-scheme under umbrella scheme 'Conservation of Natural Resources and Eco-systems'. Under the scheme, the Division supports Biosphere Reserve (designated under framework of UNESCO's Man and the Biosphere Programme) by providing grants-in-aid to the States where the Biosphere Reserves are located, for conservation and development activities as specified in their Management Plan. The implementation of this scheme is undertaken by the State Forest Departments. Thus, the Ministry provides grants-in-aid to the states on cost sharing basis i.e. 60:40 (Central: State) for other States and 90:10 (Central: State) to North Eastern and Himalayan States.

b. Important activities undertaken

- The Ministry is implementing integrated Management Plan (IMP) for Biosphere Reserve for the Period of Five Year since 2022-23 in four identified Biosphere Reserves namely Nanda Devi (Uttarakhnad), Pachmarhi (Madhya Pradesh), Dehang-Dibang (Arunachal Pradesh) and Agasthyamala (Kerala & Tamil Nadu). The Ministry provides fund to said five States in the said period for implementation of IMP for Biosphere Reserve.
- As a nodal point, the Ministry has submitted a ten-year periodic review report for eight Biosphere Reserves Nanda Devi, Similipal, Pachmarhi, Nokrek, Nilgiri, Gulf of Mannar, Achanakmar-Amarkantak, and Sundarbans to the MAB Secretariat, UNESCO, Paris for their consideration.
- Additionally, two proposals for the inclusion in the World Network of Biosphere Reserves viz. Dehang-Dibang and Cold Desert, have also been submitted to the MAB Secretariat, UNESCO for consideration. Cold Desert has been included in the UNESCO's World Network of Biosphere Reserve.
- States Governments have been asked for identification of new areas for designation as Biosphere Reserves.
- Participation in ICC-MAB meeting at China during September 26-28, 2025.

c. Progress / Innovation

Three new proposals for designation as Biosphere Reserves from the state of Madhya Pradesh, Viz. Kanha, Pench, and Bandhavgarh areas, received from the State of Madhya Pradesh was submitted to the MAB Secretariat, UNESCO for consideration in the next session of ICC-MAB.

d. Important Cumulative achievements

- During the ICC-MAB meeting in China in 2025, the Cold Desert Biosphere Reserve was officially included in the UNESCO World Network of Biosphere Reserves, marking a new addition from India. India now has 13 Bio-

sphere Reserves in the UNESCO World Network of Biosphere Reserves, the highest number in South Asia.

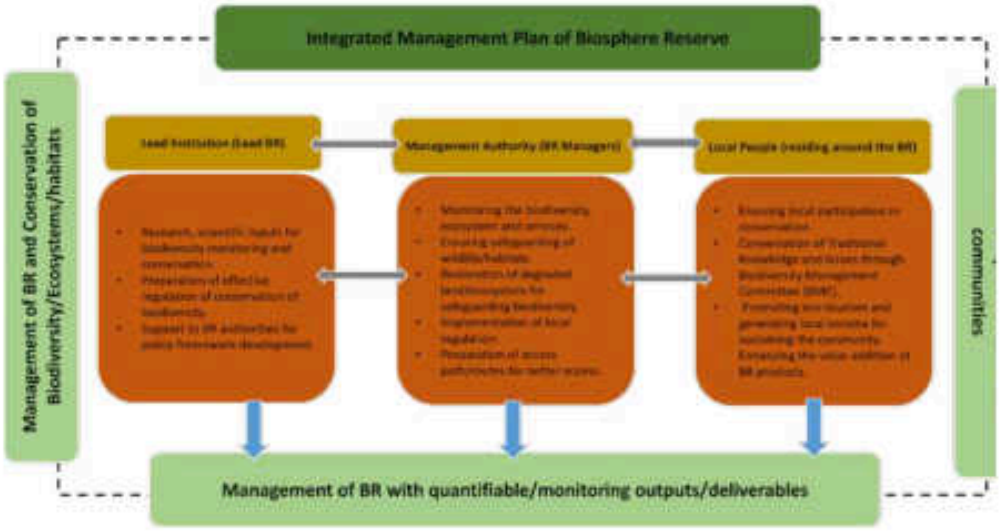


Cold Desert Biosphere Reserve

Scheme/Programme under the Division-

a. Name of the scheme: Biodiversity Conservation scheme

- 'Biodiversity Conservation' scheme is a sub-scheme under Centrally Sponsored Umbrella Scheme namely Conservation of Natural Resources and Ecosystem of the Ministry of Environment, Forest and Climate Change. Under the scheme, the financial supports are provided to the States for Management Plan of Biosphere Reserve (designated under framework of UNESCO's Man and the Biosphere Programme). The targeted beneficiaries of the Scheme are primarily the population dependent on the Biosphere Reserves. The scheme is different from other conservation related schemes and has the focus on the welfare of local inhabitants through provision of supplementary and alternate livelihood support to the people in the buffer and transition zones in order to reduce biotic pressure on biodiversity of the natural reserves of core zone.
- Since FY 2022-23, Integrated Management Plan for Five Years period in four idenified Biosphere Reserves namely Nanda Devi (Uttarakhand), Pachmarhi (Madhya Pradesh), Dehang-Dibang (Arunachal Pradesh) and Agasthyamala (Tamil Nadu and Kerala) is being implemented on pilot basis. IMPs provide a detailed roadmap covering biodiversity protection, promotion of sustainable livelihoods, eco-tourism development, capacity building, and awareness programmes. IMPs also set out mechanisms for research, monitoring and community participation, thereby ensuring a holistic approach that combines ecological integrity with socio-economic development.



During the period of 01.01.2025 to 31.12.2025 the following amount for implementation of IMP for Biosphere Reserve has been released to five said States:

Period	Amount (Rs. in lakhs)
01.01.2025 to 31.03.2025	60.21
01.04.2025 to 31.12.2025	216.89

b. Progress/Achievement

- Integrated Management Plan for Biosphere Reserve for the period of five years is in progress (in fourth year) in four Biosphere Reserves.
- Cold Desert Biosphere Reserve was included in the UNESCO's World Network of Biosphere Reserves, marking a new addition from India. India now has 13 Biosphere Reserves in the UNESCO World Network of Biosphere Reserves, the highest number in South Asia.
- The Ministry is actively considering for designating new Biosphere Reserve. Three new proposals for designation as Biosphere Reserves from the State of Madhya Pradesh, Viz. Kahan, Pench, and Bandhavgarh areas, received from the State of Madhya Pradesh was submitted to the MAB Secretariat, UNESCO for consideration in the next session of ICC-MAB.

4.5 Eco-Sensitive Zone

- The purpose of declaring Eco-sensitive Zone (ESZ) is to create a kind of "Shock Absorber" for specialized ecosystems like Protected Areas such as National Parks and Wildlife Sanctuaries. The ESZs act as a transition zone from areas requiring high protection to areas of lower protection where development may be permitted. There are a total of 688 National Parks and Wildlife Sanctuaries (Protected Areas) declared in the country.
- Notifying or declaration of an Eco-Sensitive Zones around the Protected Areas is an important objective of the Ministry of Environment, Forest and Climate Change for sustainable development and conservation of Wildlife and Biodiversity around the Protected Areas. The Ministry of Environment, Forest and Climate Change declare or notifies an area as Eco-Sensitive Zone (ESZ) around a Protected Area based on its need for special protection owing to because of its landscape, wildlife, biodiversity and historical value, etc. The purpose is to regulate the developmental activities in order to maintain the carrying capacity of that area and to ensure sustainable development taking into consideration in consonance with the needs and aspiration of the local people.
- For declaration of an ESZ around a Protected Area, proposals received from the State Government as per Guidelines formulated for the purpose and shared with the State Governments and Union territory administrations. As of now, out of total 688 Protected Areas across the country, the Ministry has published 355 final ESZ notifications covering 498 Areas an ESZ area of 73602.71 sq. km as on 31st December, 2025.
- A graphical representation on the progress of declaration of ESZs around Protected Areas since 2014 onwards and the areas in square Kilometers (sq. km) covered under ESZ regulations are given below:

5.1 National Plan for Conservation of Aquatic Eco-System (NPCA)

1. Wetlands Division

a. Introduction, objectives, and functions

The Wetlands Division, MoEF&CC works towards sustainable management and conservation of wetlands in the country. The Division's objective of conservation and management of wetlands is divided into four broad categories i.e., Implementation of National Plan for Conservation of Aquatic Ecosystems (NPCA), Ramsar Convention, Regulatory framework and Capacity building.

b. Important activities undertaken

(i) The Ministry of Environment, Forest, and Climate Change (MoEFCC) celebrated World Wetlands Day on 2nd February 2025 at the Parvati Arga Bird Sanctuary, a Ramsar Site in Gonda, Uttar Pradesh. This event was organised in collaboration with the Uttar Pradesh Department of Forest and the State Wetland Authority. The celebrations began with a bird-watching event, followed by the inauguration of an exhibition by Hon'ble Chief Minister Shri. Yogi Adityanath and Hon'ble Minister of State for Environment, Forest and Climate Change, Govt. of India Shri. Kirti Vardhan Singh.

(ii) Three regional workshops were organised under the Sahbhagita Mission at Ayodhya for Northern States on 1st February 2025, at Udaipur for Western States on 3–4 July 2025, and at Shimla for Northern States on 30–31 October 2025.

(iii) Indian delegation led by Hon'ble Minister of Environment, Forest and Climate Change (Head of the Delegation), Government of India, Shri Bhupender Yadav, participated in 15th meeting of the Conference of Contracting Parties (COP-15) -2025 of the Ramsar Convention on Wetlands held during 23rd to 31st July, 2025 in Victoria Falls (Zimbabwe). The theme of COP15 was "**Protecting Wetlands for our Common Future**".

c. Progress / Innovations

(i) **Increase in number of Ramsar sites** - The Wetlands division of the Ministry facilitated the designation of 11 more Ramsar sites (Annexure-1) to take the total number of Ramsar sites in the country from 85 to 96 as on December 2025. Two more have been added in January 2026 with total 98 Ramsar sites in the country. Currently, India stands first in Asia and third in the world in terms of the number of Ramsar Sites.

(ii) **Wetland City accreditation for the first time in India**- Two cities namely, Indore in Madhya Pradesh and Udaipur in Rajasthan accredited as 'Wetland Cities' under the Ramsar Convention during COP-15 for their outstanding commitment to safeguarding urban wetlands for the benefit of people and nature. This further cemented India's leadership in integrating wetlands into urban planning and community stewardship.

d. Important cumulative achievements

(i) **Adoption of India's Draft Resolution**: India's Draft Resolution 23.20 on "Promoting Sustainable Lifestyles for the Wise Use of Wetlands" was formally adopted by the COP. This resolution, aligned with India's Mission LiFE, reinforces the global call for responsible consumption and participatory conservation practices.

(ii) Elected as Standing Committee Member: India was elected to the Ramsar Standing Committee for the 2026–2028 triennium, marking its third tenure on the Committee since the 1999–2002 period.

(iii) Membership in Key Ramsar Bodies: India secured membership in two critical working structures of the Ramsar Convention. (a) Work-

ing Group on RIS updating – focusing on enhancing the quality and consistency of Ramsar Information Sheets. (b) CEPA Oversight Panel – overseeing communication, education, participation, and awareness initiatives under the Convention.

(iv) Ground truthing and boundary demarcation of Wetlands is a continuous process and under progress.

e. Promulgation of new Acts/Rules

The Wetlands (Conservation and Management) Rules, 2017 will be amended and notified based on the suggestions received / to be received from the States and UTs.

f. Continuing Committees / commissions

During the CoP15 of Ramsar Convention held from July 20-31st, 2025 at Zimbabwe, India was elected to the Ramsar Standing Committee for the 2026–2028 triennium, marking its third tenure on the Committee since the 1999–2002 period

g. Important Conferences / Workshops organized

(i) **Regional Workshop on Conservation and Wise Use of Wetlands for Northern States** - MoEF&CC organised a regional workshop for Northern states on 1st February 2025 at Ayodhya, Uttar Pradesh. The key points covered were, updates on the progress of ground-truthing and boundary demarcation of wetlands based on the SAC Atlas, 2021, major achievements such as the designation of Ramsar Sites and the preparation of health cards for wetland areas, identification of key challenges faced in the conservation and management of wetlands within each region conservation and provided a comprehensive platform for sharing challenges and best practices.

(ii) **Regional Workshop on Conservation and Wise Use of Wetlands for Central and Western States** – MoEF&CC organised a regional workshop for Western and Central States on 03-04 July 2025 at Udaipur, Rajasthan. The workshop served as a platform to share best practices, deliberate on policy and governance frameworks, and explore solutions to challenges in wetland conservation.

As part of the event, two significant publications were launched: The ANUP Newsletter (January–June 2025), which highlights recent wetland conservation developments across the country. The "Wetland Conservation: Do's and Don'ts" booklet, which offers easy to-understand guidelines for communities and decision-makers engaged in wetland stewardship. The workshop served as a platform to share best practices, deliberate on policy and governance frameworks, and explore solutions to challenges in wetland conservation. Technical sessions covered a range of themes including Wetland City Accreditation under Ramsar Convention, participatory wetlands conservation, integrating wetlands with local development planning, interaction with Wetland Mitras to develop effective engagement strategies.

(iii) **Regional Workshop on Conservation and Wise Use of Wetlands for Northern States, Shimla**

MoEF&CC organised a regional Sahbhagita Workshop on Conservation and Wise Use of Wetlands for Northern Region States in Shimla, Himachal Pradesh during 30-31st October 2025. During the workshop, representative from Tata motors briefed about the efforts / ways the company is doing for the conservation of wetlands within their company jurisdiction apart from CSR initiatives, which could be taken by the other company for conservation of wetland. During this workshop, 3 publications were released (i) Sustainable Lifestyles for the Wise Use of Wetland (Postcard), (ii) Peatlands

ecosystem-based and community-centric approaches into a comprehensive coastal risk framework. The assessments account for the high susceptibility of India's coastline to climate change impacts and evaluate physical, ecological, and social vulnerabilities to derive Cumulative Coastal Vulnerability indices. This integrated analysis supports evidence-based planning, targeted climate adaptation measures, and ecosystem-enhancing interventions aimed at reducing long-term risks for coastal communities.

(xi) **State Action Plan on Climate Change of Lakshadweep Islands**

The draft Lakshadweep State Action Plan on Climate Change (LAKSAPCC) has been prepared with a focus on advancing sustainable development pathways for the Union Territory under multiple projected climate futures, including IPCC Representative Concentration Pathway (RCP) scenarios. Most importantly, as a small island system, climate change scenarios were developed under all RCP pathways, with predictions modelled up to the year 2100 to capture long-term climatic shifts and their implications for the archipelago's fragile ecological and socio-economic systems. The plan integrates climate vulnerability assessments, sectoral impact analyses, and ecosystem-based adaptation strategies tailored to the unique geomorphological and ecological sensitivities of Lakshadweep. Emphasis is placed on enhancing freshwater security, safeguarding coastal and marine ecosystems, strengthening disaster risk reduction, and promoting climate-resilient livelihoods, thereby providing a comprehensive framework to guide long-term climate action in these fragile islands.

(xii) **Development of Sustainable Beaches**

Scientific, environmental and safety evaluations along with detailed planning exercises were carried out to facilitate the development of environmentally sustainable and internationally certified beaches. NCSCM prepared 10 Detailed Project Reports (DPRs) and conducted site-specific assessments for priority beaches, including Marina Beach (Marina Beach in Chennai is considered the world's second longest natural urban beach), Thiruvanmiyur, Palavakkam, Uthandi, among other beaches in Tamil Nadu. The work involved comprehensive analysis of the distance from the High Tide Line, beach morphology, water quality, bathymetry and coastal processes, safety and service infrastructure, and ecosystem management requirements in alignment with Blue Flag certification criteria. These technically robust DPRs and assessments provided coastal authorities and State Government with a strong scientific basis for implementing sustainable beach management practices that enhance environmental quality, visitor safety, and long-term coastal resilience.

c. NIRANTAR Vertical IV on Climate Change Support

NCSCM is one of the institutions under Vertical IV – Climate Change Support of the NIRANTAR programme, working in close partnership with the GB Pant National Institute of Himalayan Environment (GBPNIHE), Almora. Together, the institutions are jointly implementing a project on high-altitude wetlands, undertaking coordinated field campaigns, shared scientific assessments, and a series of wetland awareness programmes aimed at fostering a deeper mutual understanding of both coastal and Himalayan ecosystems, while emphasising how climate change acts as a central driver affecting these extreme and highly sensitive environments.

In 2025, NCSCM advanced its contribution to this national initiative with a focused assessment of carbon dynamics in fragile Himalayan wetland systems, particularly the peatlands of the North and Northeast. The institute conducted detailed field

6.1 Forest Conservation

1. Name of the Division: Forest Conservation (FC) Division

a. Introduction, objectives and functions

The Forest Conservation Division of the Ministry of Environment, Forest and Climate Change (MoEF&CC) is entrusted with the implementation of the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980. The Adhniyam came into force on 25th October, 1980 and has since been amended in 1988 and 2023.

In accordance with the provisions of the Adhiniyam, prior approval of the Central Government is mandatory for : (i) de-reservation of forest land; (ii) use of forest land for non-forest purposes; (iii) assignment of forest land by way of lease or otherwise; and (iv) clear felling of naturally grown trees for the purpose of reforestation.

The Adhiniyam is regulatory in nature and is aimed at strengthening ecological security, ensuring conservation of forests, and promoting the sustainable management of forest resources in the country. The scope of the Act was further strengthened by the Forest (Conservation) Amendment Act, 2023, enacted on 04.08.2023, with the objective of enhancing forest management alongside conservation, clarifying its applicability across different categories of land, encouraging private plantations, and aligning its provisions with the evolving ecological, social, and environmental changes witnessed over the past four decades.

To implement the provisions of the Adhiniyam, Rules and Guidelines have been issued by the Central Government. The Van (Sanrakshan Evam Samvardhan) Rules, 2023 have been notified by the Central Government to operationalize the provisions of the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980. These Rules lay down the procedure for approval of diversion of forest land for non-forest purposes, implementation of compensatory afforestation, monitoring, and enforcement mechanisms in accordance with the provisions of the Adhiniyam.

Proposals received from the States and Union Territories for diversion of forest land are scrutinized by the Forest Conservation Division and thereafter examined by the Advisory Committee constituted under Section 3 of the Adhiniyam. Each proposal is analyzed with due diligence, and diversion of forest land is permitted only in unavoidable circumstances, and then only to the barest minimum extent required for non-forestry purposes.

Based on the recommendations of the Advisory Committee, the Central Government accords prior approval subject to conditions including payment of Net Present Value (NPV), undertaking compensatory afforestation over non-forest land or degraded forest land, and implementation of need-based mitigation measures. Such measures may include Catchment Area Treatment (CAT) Plans, Wildlife Management Plans, Soil and Moisture Conservation works, and maintenance of safety zones, all of which are stipulated in the approvals granted by the Central Government.

During the year of 2025 (From 01.01.2025 to 28.12.2025), 104 proposals involving diversion of 12,649.57 ha of forest land were granted approval by the Ministry under the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980. Against this approval, compensatory afforestation has been stipulated over an 13,768.35 ha of land comprising of 8,223.13 ha of degraded forest land and 5545.22 ha of non-forest land.

b. Key legislative and Policy Reforms carried out in 2025 under the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980

The Van Sanrakshan Evam Samvardhan) Amend-

b. Important activities undertaken

Important activities undertaken during the period 01.01.2025 to till date with respect to SU Division is as under:

i. The division has played a key role in giving the final shape to India State of Forest Report (ISFR) 2023. The details activities undertaken by the Forest Survey of India area as under:

A. Forest cover mapping and Tree cover assessment:

Forest Survey of India (FSI) carries out in wall-to-wall forest cover assessment of the country on biennial basis through interpretation of satellite data and publishes the information in the 'India State of Forest Report'. So far, eighteen cycles of forest cover assessment have been completed from 1987 to 2023. The results of 18th cycle Forest Cover Mapping (FCM) were published in ISFR 2023, and currently the work of 19th Cycle FCM, for ISFR 2025 is in progress.

In addition to forest cover mapping, assessment of tree cover of the country is also being carried out using the Trees Outside Forests (TOF) inventory data. As per ISFR 2023, there is an increase of 156.41 km2 (0.02%) in the forest cover and 1,289.40 km2(1.16%) in the tree cover as compared to the last assessment, ISFR 2021. Thus, there is an overall increase of 1,445.81 km2 (1.18%) of forest cover and tree cover put together at national level. The details of different types of forests are shown in Fig. 1 below.

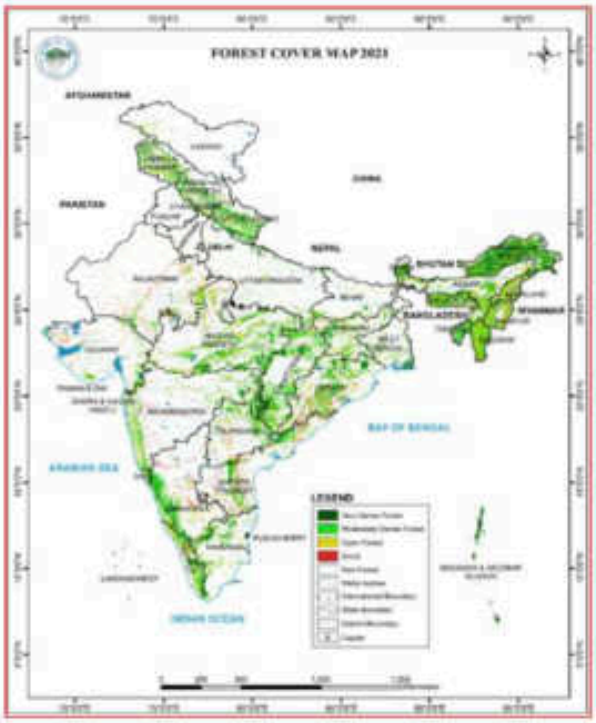


Fig.1 Forest Cover Mapping

B. Monitoring of Forest fires and forest fire risk zonation mapping

(i) Near Real Time Monitoring of Forest Fires: FSI has been alerting State Forest Departments of forest fire locations detected by the MODIS (Moderate Resolution Imaging Spectro-radiometer) sensor on- board Aqua and Terra Satellites of NASA since 2004. Since then, there has been continuous upgradation in the forest fire alert systems.

(ii) The FSI Fire Alerts System 3.0 has the features like improved Custom Filter to boost accuracy level, Large Forest Fire monitoring programme to automatically identify and track large forest fires, Map Server based Van Agni Geo-portal to view dynamic display of alerts along with other thematic layers, availability of Web Map Service for integration with State Forest Departments geo-portal, Customized alerts to the State Forest Depart-

6.8 National Afforestation and Eco-development Board

a. Introduction

The National Afforestation and Eco-Development Board (NAEB), is responsible for promoting afforestation, tree planting, ecological restoration and eco-development activities in the country, with special attention to the degraded forest areas and lands adjoining the forest areas, national parks, sanctuaries and other protected areas as well as the ecologically fragile areas like the Western Himalayas, Aravallis, Western Ghats etc. The Desertification Cell (DC) of the Ministry has been merged with the NAEB and works related to Land Degradation Neutrality, desertification, drought and matters related to the United Nations Convention to Combat Desertification (UNCCD) are being handled in the NAEB.

Further, the erstwhile scheme of National Afforestation programme (NAP) has been merged with the National Mission for a Green India (GIM). Notes on works under GIM Directorate are given in a separate communication.

The main schemes and programmes undertaken by the NAEB are the Eco-Development Forces (EDF) scheme jointly operated with the DGTA which is supported from the Ministry's budget, National CAMPA funded Nagar Van Yojana (NVY) and School Nursery Yojana (SNY) schemes and the "Ek Ped Maa Ke Naam" (EPMKN) campaign, a unique initiative aimed at promoting afforestation along with nurturing the bond between people and mother nature, on a "Whole of Government" and "Whole of Society" approach.

b. Important Activities undertaken

Under the EPMKN programme, more than 130 crore seedlings have been planted in the period under report i.e. from 01.01.2025 to 31.12.2025 as reported on MeriLiFE Portal by public, institutions, agencies, government and non-government alike.

A total of 620 Nagar Vans have been sanctioned in 28 States and 3 Union Territories of the country under NVY at a total cost of Rs. 654.30 Crores since the launch of the scheme and a total of 75 Nagar Vans have been sanctioned during the reference period.

The NAEB / GIM / DC is monitoring and guiding the works being done on a convergence mode in the concerned States under the Aravalli Landscape Restoration / Aravlli Green Wall (AGW) project and the implementation of the DPRs on 13 Rivers' Rejuvenation through Forestry Interventions.

c. Progress / Innovation

- Local Indigenous species have been stressed upon for plantation and Microforests have been promoting with due handholding of the States and Institutions such as Jawahar Navodaya Vidyalayas, Kendriya Vidyalayas etc. making use of expertise of the scientists from ICFRE etc.

d. Important / cumulative achievement

- More than 260 crore seedlings planted under the campaign since 2024 till as reported on MeriLiFE Portal.
- A total of 620 Nagar Vans have been sanctioned in 28 States and 3 Union Territories of the country under NVY since the launch of the scheme.





The Governing Body Meeting of the Central Zoo Authority held under the chairpersonship of Shri Bhupender Yadav, Hon'ble Minister for Environment, Forest and Climate Change, Government of India on 22nd April 2025, at Indira Paryavaran Bhawan, Jorbagh Road, Aliganj, New Delhi.

7.4 Project elephant

1. Project Elephant Division

a. Introduction, Objectives and Functions

India has considerable biological and cultural diversity, along with a history of conservation initiatives. Of the 13 Asian countries where elephants are present, India accounts for more than 60% of the global wild Asian elephant population. Asian elephants in India are mainly distributed across the southern, north-eastern, east central, and northern regions. The Asian elephant is listed as Endangered on the IUCN Red List, indicating a significant risk of extinction in the wild. In India, elephants are protected under Schedule I of the Wildlife Protection Act, 1972. To support habitat restoration and mitigate human-elephant conflict, the Government of India launched Project Elephant in 1992 as a Centrally Sponsored Scheme by the Ministry of Environment, Forests & Climate Change. The project was established with the following objectives:

- To protect elephants, their habitats, and corridors.
- To address the challenges of human-elephant conflict.
- To ensure the welfare of captive elephants.

The implementation of Project Elephant is through the Centrally Sponsored Scheme-Project Tiger & Elephant with the grant to State Governments with a funding ratio of Central: State share of 60:40 and 90:10 for Northeast and Himalayan states. Presently the Project is being implemented in 23 States/UTs.

b. Important Activities Undertaken

- 21st Steering Committee Meeting of Project Elephant, chaired by Hon'ble Minister Shri Bhupendra Yadav at IGNFA, Dehradun, reviewed progress on corridors, conflict mitigation, and elephant reserve management.
- The World Elephant Day 2025 was organized jointly by the Ministry of Environment, Forest & Climate Change (MoEF&CC) and the Government of Tamil Nadu in Coimbatore to celebrate elephants as India's National Heritage Animals and promote their conservation. The world elephant day celebration was presided over by Hon'ble Union Minister of State for Environment, Forest and Climate Change, Shri Kirti Vardhan Singh in the esteemed presence of senior officials, conservationists, students, and representatives from civil society.
- Mass Awareness Campaign: NMNH and Project Elephant conducted outreach among 15 lakh students(approx) across 5,518 schools in southern India, administering a multilingual pledge for elephant conservation.
- The Gaj Gaurav Awards (2025) were presented during the World Elephant Day event for exemplary efforts in elephant conservation & management. The Award was conferred:

- Jointly to Shri Ganesh Tamang (Mahout), Shri Sumit Gogoi (Attendant) from Arunachal Pradesh

- Jointly to Shri Kesu Singh Walke (Asst. Mahout), Shri Sahadan Ram Lakada (Asst. Mahout) from Madhya Pradesh
- Jointly to Shri M. Murali (Anti-Poaching Watcher), Shri S. Karthikeyan (Forest Guard) from Tamil Nadu

Shri Irshad Ali (Mahout) from Uttar Pradesh.

- Railway Collision Mitigation: MoEF&CC–WII–Railway jointly surveyed 127 (110 in elephant range and 17 in two tiger range states) vulnerable stretches spanning 3,452 km. Of these, 77 stretches (1,965 km across 14 States) prioritized for 705 mitigation structures (ramps, crossings, fencing, and overpasses).
- Elephant-Human Conflict in the State of Assam, Chhattisgarh and Jharkhand, India (2000-2023)- Trends, Challenges & Insights: The report, covering 2000–2023, analyses conflict trends using spatial data, field surveys, and forest department records to identify hotspots and assess the influence of land use changes on conflict severity and elephant mortality. It proposes mitigation strategies to promote coexistence between local communities and elephants and was released by the Hon'ble Minister, EFCC during the 21st Steering Committee Meeting of Project Elephant at IGNFA, Dehradun.
- Regional Action Plans (RAPs): The Regional Action Plan (RAP) aims to promote the long-term conservation of elephants and their habitats through a coordinated, landscape-based approach. Recognizing the transboundary movement of elephants, the RAP focuses on maintaining ecological connectivity, mitigating human-elephant conflicts (HEC), and fostering community participation. Drafting Committees have been constituted for Southern and North Eastern regions to prepare a draft plan.
- Advisory on Tusk Trimming in Captive Elephants: Elephant tusks, made of dentine (ivory), serve vital roles in foraging, defence, and mating but are prone to fractures and diseases, especially in captivity. Proper tusk care, including safe trimming, is essential for their welfare, while improper methods can cause severe or fatal injuries. The advisory provides ethical and scientific guidelines for safe tusk trimming in captive elephants, ensuring compliance with animal welfare and legal standards. It was released by the Hon'ble Minister, EFCC during the 21st Steering Committee Meeting of Project Elephant at IGNFA, Dehradun.
- Healthy Feet, Healthy Elephants: Foot ailments such as abscesses, cracks, rot, arthritis, and infections are major causes of morbidity in captive elephants, often resulting from poor hygiene, improper trimming, unsuitable flooring, and lack of routine inspection. To address these issues, Project Elephant, in collaboration with veterinarians and technical experts, developed "Healthy Feet, Healthy Elephants: A Comprehensive Guide to Foot Care in Captive Asian Elephants"-the first structured, science-based manual offering practical guidance for mahouts, veterinarians, and caretakers on effective foot health management and welfare of captive elephants.

c. Important Cumulative Achievements

- DNA Profiling of Captive Elephants: Over 2,000 biological samples (Approx. 70% of the captive elephant population) from 25 states have been collected and analysed via the Gajah Suchana App in collaboration with WII.
- All-India Synchronized Elephant Population Estimation (2022) completed nationwide.

d. Continuing Committees/ Commissions

- A sub-committee to prepare a draft Regional Action Plan for Southern India Elephant Range States was constituted on 19.02.2025.
- Captive Elephant Healthcare and Welfare

Committee (CEHWC) – 5th meeting held on 07.03.2025.

- Central Project Elephant Monitoring Committee (CPEMC) – 8th meeting held on 17.03.2025.
- A sub-committee to prepare a draft Regional Action Plan for North- Eastern India Elephant Range States was constituted on 06.08.2025.
- A sub-committee to identify train-hit accidents hotspots from the last 10 years was constituted on 07.08.2025.
- A sub-committee to study overlapping areas of Tiger and Elephant Reserves was constituted on 24.09.2025.

e. Important Conferences/ Meetings

- Regional Consultation on HEC Management in Northeastern India held on 21 January 2025 at Guwahati.
- 21st Steering Committee Meeting of Project Elephant held on 26 June 2025 at IGNFA, Dehradun.
- Experience Sharing Workshop on Good Practices in Elephant Habitat Mnagemnt, Corridor management and Human Elephant Conflict mitigation was held on August 12th 2025 at Coimbatore
- 22nd Steering Committee Meeting of Project Elephant held on 21st Decemeber 2025 at Sundarbans Tiger Reserve, West Bengal.

f. MEAs/ International Cooperation

- 4th Asian Elephant Range State meeting organized by IUCN from 5th to 7th February 2025, the meeting was attended by Shri Ramesh Kumar Pandey (IGF) and Dr. Dheeraj Mittal (AIGF).
- A two-member Indian delegation, led by Shri Ramesh Kumar Pandey, Additional Director General of Forests (Wildlife/Project Elephant), MoEFCC, along with Dr. Parag Nigam, Scientist 'G', Wildlife Institute of India, participated in the 12th Meeting of the IUCN Species Survival Commission (SSC) Asian Elephant Specialist Group (AsESG) held in Bien Hoa, Vietnam, from 17th to 21st September 2025.
- Reported data of all MIKE sites in India for the year 2024 to CITES secretariat.

g. Major Publications

i. Report on Suggested Measures to Mitigate Elephant & Other Wildlife Train Collisions on Vulnerable Railway Stretches: a consolidated report for 14 States.
ii. State-wise report on understanding elephant conflict issues for suggesting conflict reduction measures for the States of Assam, Chhattisgarh and Jharkhand
iii. Trumpet Newsletter (Vol. IV, Issue 3 & Vol. V, Issue 1) – quarterly updates.
iv. Trumpet Special Edition – Elephants and Tribes of India – highlighting tribal participation in conservation.
v. Healthy Feet, Healthy Elephants: A Guide to Foot Care in Captive Asian Elephants.
vi. Elephant Care and Management: A Guide for Elephant Handlers
vii. Trumpet Special Edition – Elephant Population Estimation and Use of Technology in mitigating Human Elephant conflict

2. Scheme / Programme under Project Elephant Division

Centrally Sponsored Scheme – Project Elephant

a. Progress / Achievements

- Strengthened habitat protection and conflict mitigation across elephant range States.
- DNA profiling and digital documentation of captive elephants through Gajah Suchana App.
- Support extended to 23 States/UTs for protection, conservation, habitat improvement, human-elephant conflict mitigation and captive elephant welfare activities.
- Capacity building through regional workshops and training.
- Public awareness through World Elephant Day

heritage for the benefit, education and enjoyment of the people, the NTCA:

1. Provides statutory authority to Project Tiger so that compliance of its directives becomes legal.
2. Fosters accountability of Center and States in management of Tiger Reserves within our federal Structure, by providing for a tripartite MoU with States, Reserve Management and NTCA.
3. Provides for an oversight by Parliament.
4. Addresses livelihood interests of local people in areas surrounding Tiger Reserves.

Functions of the NTCA

Powers and functions of the National Tiger Conservation Authority as prescribed under section 38 O of the Wildlife (Protection) Act, 1972, as amended in 2006 are as under: -

a. to approve the tiger conservation plan prepared by the State Government under sub section (3) of Section 38 V of the Act
b. evaluate and assess various aspects of sustainable ecology and disallow any ecologically unsustainable land use such as, mining, industry and other projects within the tiger reserves
c. lay down normative standards for tourism activities and guidelines for project tiger from time to time for tiger conservation in the buffer and core area of tiger reserves and ensure their due compliance
d. provide for management focus and measures for addressing conflicts of men and wild animal and to emphasize on co-existence in forest areas outside the National Parks, sanctuaries or tiger reserve, in the working plan code
e. provide information on protection measures including future conservation plan, estimation of population of tiger and its natural prey species, status of habitats, disease surveillance, mortality survey, patrolling, reports on untoward happenings and such other management aspects as it may deem fit including future plan conservation
f. approve, co-ordinate research and monitoring on tiger, co-predators, prey habitat, related ecological and socio-economic parameters and their evaluation
g. ensure that the tiger reserves and areas linking one protected area or tiger reserve with another protected area or tiger reserve are not diverted for ecologically unsustainable uses, except in public interest and with the approval of the National Board for Wild Life and on the advice of the Tiger Conservation Authority
h. facilitate and support the tiger reserve management in the State for biodiversity conservation initiatives through eco-development and people's participation as per approved management plans and to support similar initiatives in adjoining areas consistent with the Central and State laws
i. ensure critical support including scientific, information technology and legal support for better implementation of the tiger conservation plan
j. facilitate ongoing capacity building programme for skill development of officers and staff of tiger reserves, and
k. perform such other functions as may be necessary to carry out the purposes of this Act with regard to conservation of tigers and their habitat.

Government of India through National Tiger Conservation Authority has undertaken the following milestone initiatives for conservation and protection of tiger and other wild animals.

b. Important activities undertaken

1. In total, three new tiger reserves have been notified/approved under Section 38V of the Wildlife (Protection) Act, 1972—Guru Ghasidas–Tamo Pingla Tiger Reserve (Chhattisgarh), Ratapani Tiger Reserve (Madhya Pradesh), and Madhav National Park (Madhya Pradesh). With these additions, the

total number of tiger reserves in India has increased to 58. The expansion has further strengthened the country's protected area network, bringing the area under tiger reserves to nearly 2.5% of India's geographical area.



2. International Big Cat Alliance
   a. 19 countries have joined IBCA (India, Nicaragua, Eswatini, Somalia, Liberia, Cambodia, Guinea, Suriname, Bhutan, Ethiopia, Armenia, Rwanda, Nepal, Guatemala, Mongolia, Eritrea, Angola, Malaysia, South Sudan) and 3 countries as Observer Country (Thailand, Kazakhstan, Namibia), along with 9 partner organizations and 3 strategic partners.
   b. The Framework agreement on the establishment of IBCA officially came into force in January, 2025.



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विदेश मंत्रालय, नई दिल्ली
MINISTRY OF EXTERNAL AFFAIRS
NEW DELHI

No. AA/121/01/2022-184

3rd Feb 2025

Ministry of External Affairs presents its compliments to the Secretariat, International Big Cat Alliance, New Delhi and has the honour to refer to its letter No. IBCA/IC/01/2024 dated 15 January 2025, regarding the entry into force of the Framework Agreement on the Establishment of International Big Cat Alliance under Article XVIII of the Framework Agreement, and to state that as mentioned in Article XX) of the Framework Agreement, the Government of India (Ministry of External Affairs) will serve as the depository for instruments of ratification, acceptance, approval, or accession from countries seeking to become parties to the agreement under Article XII.

2. It may be noted that Article XVIII (1) of the Framework Agreement specifies that the Agreement will come into effect on the 15th day following the deposit of the 5th instrument of ratification, acceptance, or approval, which includes India's instrument. This provision ensures a coordinated and timely implementation of the Agreement among participating countries.

3. The Ministry would like to inform that, under Article XVIII(1) of the Framework Agreement, five countries - Nicaragua, Eswatini, India, Somalia, and Liberia - have submitted their ratifications/approvals on various dates. The ratification/approval of the agreement by Government of Liberia on January 8, 2025 marks the 5th instrument of ratification/approval upon which the framework agreement shall take effect on January 23, 2025.

4. In view of the above, the Ministry is delighted to confirm that the Framework Agreement on the Establishment of the International Big Cat Alliance (IBCA) will officially come into force on January 23, 2025. To ensure widespread awareness, the Ministry requests the IBCA Secretariat to publicize this milestone on relevant platforms.

5. The Ministry of External Affairs of the Republic of India avails itself of this opportunity to renew to the Secretariat, IBCA, New Delhi, the assurances of its highest consideration.

IBCA Secretariat
118, First floor,
Jorbagh,
New Delhi-110003

c. The Headquarters agreement was signed with the Government of India in April, 2025.



8.1 Climate Change

a. Introduction, objectives, and functions

The Ministry of Environment, Forest and Climate Change (MoEFCC) serve as the nodal agency for India's climate change policy. Within the Ministry, the Division is responsible for planning, coordinating, and overseeing the nation's comprehensive response to the climate crisis. The Division anchors India's engagement with the United Nations Framework Convention on Climate Change (UNFCCC), its Kyoto Protocol, and Paris Agreement, ensuring that national interests are effectively represented in global negotiations, including the annual session of Conference of the Parties (COP) to UNFCCC.

The Division coordinates with Adaptation Cell, National Communications (NATCOM) Cell, Climate Finance Vertical and the Cell for Carbon Market and Article 6 of the Paris Agreement to collectively manage diverse aspects of climate policy, ranging from international reporting (Transparency Framework) to the development of domestic carbon markets.

Domestically, the Division acts as the central coordinator for the National Action Plan on Climate Change (NAPCC), monitoring the progress of its nine sectoral missions. In addition, it oversees preparation of State Governments in formulating State Action Plans on Climate Change (SAPCCs) to ensure that climate objectives are mainstreamed into sub-national development strategies. The Division is instrumental in formulating India's key climate instruments, including the Nationally Determined Contributions (NDC) and the National Action Plan for Climate Change. By aligning the international commitments with national priorities, the Division ensures policy coherence. It functions as the strategic bridge between global climate obligations and India's developmental aspirations, ensuring that climate action remains participatory, transparent, and evidence-based.

b. Important Activities undertaken

- Preparation of India's Successive Nationally Determined Contribution
- India's participation at 30th Session of Conference of Parties to the UNFCCC (COP 30) held at Belem, Brazil in November, 2025
- India's participation in 62nd Session of Subsidiary Bodies to the UNFCCC held in June 2025 at Bonn Germany.
- Revision of National Action Plan on Climate Change
- Conceptualising the proposal to setup State Climate Forum
- Preparation of State Action Plan on Climate Change by the States of Maharashtra, Meghalaya and UT of Chandigarh

c. Progress / Innovation

- India's participation at 30th Session of Conference of Parties to the UNFCCC (COP 30) held at Belem, Brazil in November, 2025
- Preparation of State Action Plan on Climate Change by the States of Maharashtra, Meghalaya, and UT of Chandigarh

d. Important Cumulative Achievements

An inter-ministerial delegation led by the Hon'ble Minister of Environment, Forest and Climate Change and comprising of representatives from Ministry of External Affairs, Ministry of Power, Ministry of New and Renewable Energy, Ministry of Agriculture and Farmers Welfare, Ministry of Information and Broadcasting and the Ministry of Environment, Forest and Climate Change participated in COP 30.

The COP30 commenced amid expectations that it could mark a defining moment for advancing the priorities of the Global South. The Conference placed strong emphasis on translating commitments under the Paris Agreement into concrete action, with particular focus on climate finance, adaptation planning, and implementation frameworks. COP30 was significant as it was to take place when most of

the items of Paris Agreement rule were operational. The Brazilian Presidency underscored the need for meaningful and balanced outcomes across key agenda items, including enhanced climate finance, progress towards the Global Goal on Adaptation, advancement of the mitigation work programme, and operationalisation of the work programme on just transition. In addition, the Presidency has highlighted the importance of advancing the Gender Action Plan, strengthening technology implementation mechanisms, and addressing issues related to response measures, with a view to ensuring inclusive, equitable, and effective climate action.

India adopted a positive, constructive, and bridge-building approach during COP30 negotiations. This approach contributed to strengthening the collective voice of developing countries in shaping the conference outcomes. India worked closely with other developing country groups and negotiation platforms, including the G77 and China, the Like-Minded Developing Countries (LMDC), and BASIC, fostering strong convergence on key issues. India served as the spokesperson for the LMDC group and as Chair of BASIC, playing an active leadership role in advancing shared priorities. Throughout the negotiations, India effectively safeguarded its interests across major agenda items while maintaining broad engagement and avoiding isolation in the negotiation process.

The Hon'ble Minister of Environment, Forest and Climate Change delivered India's National Statement during the High-level Segment of COP 30. India called upon developed countries to demonstrate enhanced climate ambition and to honour their commitments under the Paris Agreement. It was emphasized that developed countries must achieve net-zero emissions significantly earlier than their current target timelines and deliver new, additional, and concessional climate finance at a scale commensurate with the challenge i.e. amounting to trillions rather than billions of dollars. India also highlighted the critical importance of ensuring that climate technologies are affordable and accessible, stressing that such technologies should not be constrained by restrictive intellectual property barriers. The progress on India's international commitments towards global climate actions was highlighted to demonstrate that economic development and environmental stewardship can progress simultaneously. In this context, it was conveyed that India's emission intensity has declined by over 36 percent since 2005, and that non-fossil fuel sources now constitute more than half of the country's total installed electricity capacity, which lead to achievement of one of India updated NDC (2021-30) targets five years ahead of the 2030 timeline.



To highlight our global leadership, India also referred to the initiatives such as the International Solar Alliance and the Global Biofuel Alliance, as well as the momentum generated by the Nuclear Mission and the Green Hydrogen Mission in advancing India's pathway towards net zero by 2070. It was informed that in alignment with the Paris Agreement's objectives on conserving and enhancing carbon sinks, more than two billion plants were planted through community-led efforts within a span of sixteen months.

On the climate finance; through the Like-Minded Developing Countries (LMDC) platform, India played a key role in securing a two-year work pro-

gramme on climate finance, including deliberations on Article 9 in the context of Article 9.1 of the Paris Agreement. This outcome reinforces the renewed emphasis on the climate finance obligations of developed countries as stipulated under Article 9.1, while also creating avenues for enhanced and additional financial support to developing countries. The Parties further agreed to pursue efforts to at least triple adaptation finance by 2035 relative to 2025 levels, with developed countries being urged to significantly scale up the trajectory of their collective provision of climate adaptation finance to developing countries.

The Parties agreed to adopt a set of 59 indicators to monitor progress towards the Global Goal on Adaptation (GGA). These indicators are voluntary, non-prescriptive, and non-intrusive in nature, and do not entail any additional reporting requirements for countries. This outcome is consistent with India's position and preserves national flexibility.

Overall, Indian delegation emphasized for recalibration of priorities, moving beyond a narrowly mitigation-focused framework to place stronger emphasis on adaptation and the means of implementation, particularly finance and technology. The net-zero narrative was appropriately corrected to reflect a global objective around 2050, rather than a one-size-fits-all deadline for countries, in recognition of differentiated national pathways such as India's commitment to achieve net-zero by 2070. Furthermore, attempts to advance top-down approaches through the Global Stocktake and synthesis reports were resisted, underscoring the importance of equity, national circumstances, and bottom-up action.

Hon'ble Minister of Environment, Forest and Climate Change anchored the negotiations on Technology Implementation Programme (TIP). Under the TIP, the Country Parties have agreed to focus on accelerating on-ground deployment of climate technologies in developing countries; address barriers across the technology cycle; support national innovation systems; build capacity for bankable project preparation; and enhance access to finance and partnerships, helping countries convert technology needs into implementable solutions aligned with NDCs and adaptation priorities.



During the closing plenary, India expressed its support and appreciation to Brazil and the international community for their cooperation. It called upon all Parties to make collective efforts to ensure that the path forward from Belém leads to a future grounded in fairness, solidarity, and shared prosperity for all. Reiterating India's principled approach to climate action, the India underscored the imperative that the burden of climate change mitigation must not be transferred to those who bear the least responsibility for its causes. It emphasized the urgent need for enhanced global support to vulnerable populations, predominantly located in the Global South, to enable them to effectively safeguard themselves against the intensifying impacts of climate change. India reaffirmed its steadfast commitment to science-based and equitable climate action, while reiterating its support for a rules-based global order that is fair, inclusive, and respectful of national sovereignty. The Statement further noted India's resolve to collaborate with all Parties to ensure that global climate ambition re-

mains inclusive, just, and equitable.

- **Thematic Installation at COP 30 in Belém.**
  At COP-30 in Belém, a thematic installation was setup to promote essence of community participation and a citizen-centric approach to collective climate action. The installation showcased initiatives such as **Ek Ped Maa Ke Naam and Mission LiFE (Lifestyle for Environment)**, which advocate sustainable living through everyday behavioral changes. It encouraged, inter-alia, actions such as conserving energy and water, adopting sustainable food choices, managing waste through Reduce, Reuse and Recycle, minimizing plastic use through cloth bags and reusable bottles, and opting for environmentally friendly products. Anchored in India's traditional wisdom of mindful consumption, it also highlighted the role of positive behavioural change and collective effort in environmental conservation.

- **India's participation in various side-events during CoP30 at Belem, Brazil**
  The Ministry, in collaboration with various international agencies, including the COP 30 Presidency, organized and participated in the following High-Level Ministerial side-events on the margins of COP 30:

  1. Honble Minister Bhupender Yadav and UK Secretary of State for Energy Security and Net Zero Edward Miliband held a "Fireside Chat"
  2. Lead-IT: Industry Leaders' Roundtable at COP30 "From ambition to investments: scaling industry transition this decade"
  3. Ministerial Event on International Big Cat Alliance
  4. 11th Japan Joint Credit Mechanism (JCM) Partner Countries' Meeting
  5. International Solar Alliance (ISA)- Small Island Developing Countries (SIDS) Platform Leadership Session

  These side events provided opportunity to showcase India's climate actions; exchange perspectives and draw attention to key initiatives aimed at addressing climate change. During these events, India effectively demonstrated its strong commitment to climate action, highlighting a range of measures being undertaken at both the domestic and international levels.

  Through these side events, the team India shared experiences, highlighted key our initiatives to deal with the climate change. The team effectively articulated and showcased India's commitment to climate action and various initiatives being taken at the international as well as domestic level.

- **Bilateral Meetings at COP 30**:
  During COP 30, HMEFCC held bilateral meetings with the EU, China, Germany, UK, Brazil, Nepal, Estonia, Cuba, Japan and a meeting with the UN Secretary General. Engagements were mainly focused on exchanging views and mobilizing convergence on COP agendas, however, issues of bilateral cooperation beyond the COP were also discussed. Notable discussions included; China proposed to establish a bilateral mechanism for Climate Cooperation, Japan proposed to expand existing JCM mechanism, Cuba requested for India's support on climate adaptation, India- Estonia agreed to explore cooperation in fields of technologies and renewable energy, Nepal requested for India's support on the mountain agenda within the UNFCCC. HMEFCC informed his Cuban, Japanese and Estonian counterparts that Indian Missions in these countries would take forward the discussions. Following this, messages to our Missions in Cuba, Japan and Estonia had already been sent by the UNES Division.

- **State Action Plan on Climate Change:**

State Action Plans on Climate Change (SAPCCs) serve as comprehensive strategic frameworks formulated by States and Union Territories to respond to climate-related challenges specific to their regional and administrative contexts. Developed in alignment with the National Action Plan on Climate Change (NAPCC), these plans form an essential component of India's broader climate mitigation and adaptation efforts. SAPCCs emphasize region-specific interventions aimed at lowering greenhouse gas emissions, strengthening climate resilience, and advancing sustainable development. In accordance with the Ministry's advisory, SAPCCs have been prepared by 34 States. During the year 2025, the Ministry reviewed and accorded approval to the revised SAPCCs of Maharashtra and Meghalaya and Union Territory, Chandigarh. So far, 16 States and UTs have revised their SAPCCs.

India submitted its 4th Biennial Update Report to the United Nations Framework Convention on Climate Change

India's 4th Biennial Update Report (BUR-4) to the United Nations Framework Convention on Climate Change was submitted on December 30, 2024. The BUR-4 updates the Third National Communication (TNC) and includes the National Greenhouse Gas (GHG) inventory for 2020. The report also embodies information on India's national circumstances, mitigation actions, an analysis of the constraints, gaps, related finance, technology, and capacity-building needs.

National Greenhouse Gas Inventory

In 2020, India's total GHG emissions, excluding Land Use Land-Use Change and Forestry (LULUCF), were 2,959 million tonnes of CO2e and net emissions of 2,437 million tonnes of CO2e with the inclusion of LULUCF. A summary of sector-wise and gas-wise national emissions and removals is presented in Table 1.

GHG sources and removals	CO_2 emission	CO_2 removal	CH_4	N_2O	HFC 23	CF_4	C_2F_6	SF_6	CO_2 equivalent
Energy	2181012	NO	1523	82	NO	NO	NO	NO	2238409
IPPU	201044	NO	232	8	2	1	0.27	0.004	238556
Agriculture	NO	NO	14290	342	NO	NO	NO	NO	405983
LULUCF	9369	-532357	41	1	NO	NO	NO	NO	-521933
Waste	NO	NO	2726	58	NO	NO	NO	NO	75641
Memo Items	802846	NO	0.09	0.11	NO	NO	NO	NO	802882
Total Emission	2382535	--	18771	489	2	1	0.27	0.004	2958589
Net Emission	2391904	532357	18811	490	2	1	0.27	0.004	2436656

Abbreviation: NO - Not Occurring.

The energy sector contributed the most to overall emissions (75.66 percent), followed by the agriculture (13.72 percent), Industrial Processes and Product Use (8.06 percent), and Waste (2.56 percent) (Figure 1). In 2020, India's forest and tree cover, along with other land use, sequestered approximately 522 million tonnes of CO2, equivalent to reducing 22% of the country's total carbon dioxide

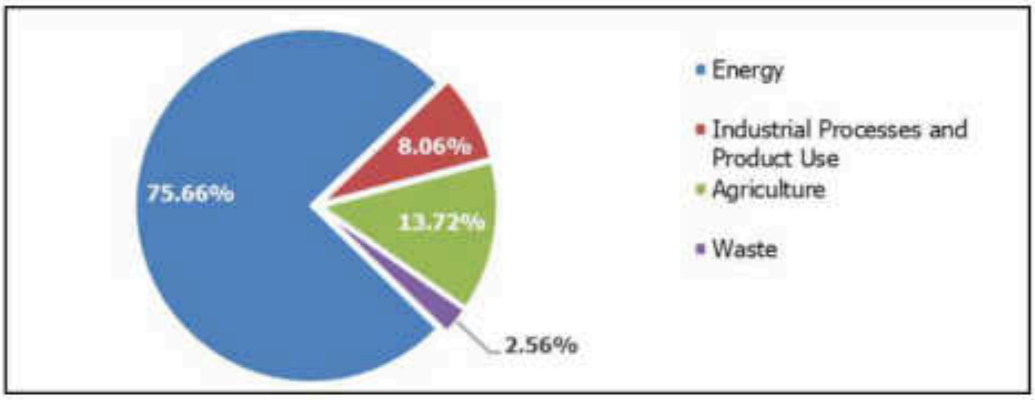


Figure 1: Distribution of GHG emissions (GgCO2e) by sector, 2020

Figure 2 shows the percentage share of the top 15 key emission categories in India for 2020, expressed as CO2e.

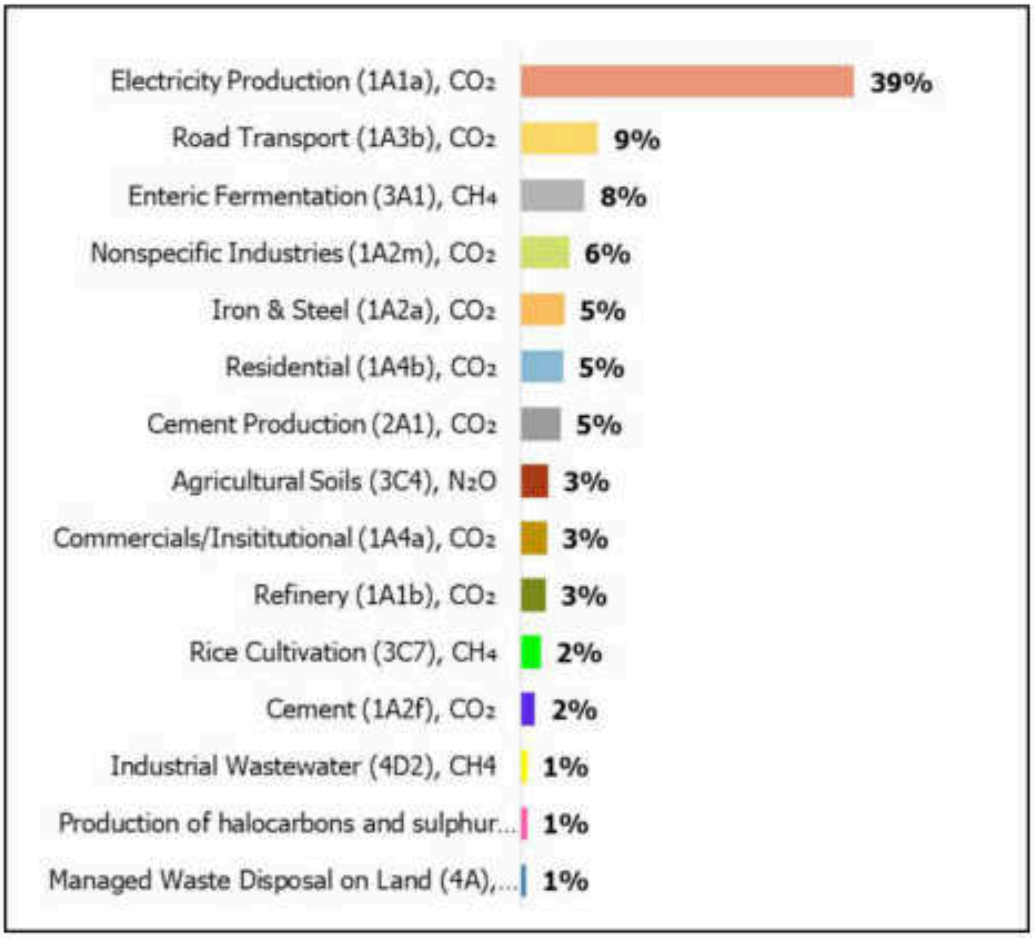


Figure 2: Percentage share of greenhouse gas emissions by category, 2020

emissions.

The CO2, CH4, and N2O emissions accounted for 80.53 per cent, 13.32 per cent, and 5.13 per cent, respectively, of the national GHG emissions excluding LULUCF (Figure 3). In the case of fluorinated gases, emissions of HFC, CF4, C2F6, and SF6 were 0.73 per cent, 0.20 per cent, 0.09 per cent, and 0.003 per cent, respectively. Total national GHG emissions (including LULUCF) have decreased by 7.93 per cent with respect to 2019 (Figure 4).

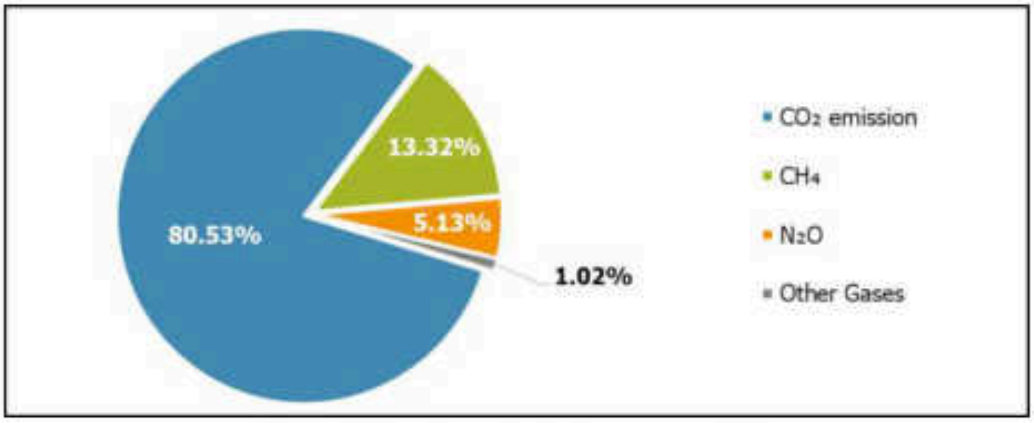


Figure 3: Gas-wise emission for the year 2020

Since 1994, India has undergone significant economic growth, resulting in an overall increase in greenhouse gas emissions. From 1994 to 2020, total CO2e emissions (excluding LULUCF) rose by 144 per cent. The waste sector experienced a growth of 226 per cent from 1994 to 2020, attributed to population growth and increased industrial activities. However, its contribution to overall emissions consistently hovers around 3 per cent, indicating a relatively low absolute contribution. The energy sector experienced a growth rate of 201 per cent from 1994 to 2020, primarily attributed to the ongoing rise in fossil fuel combustion. During the same period, the IPPU sector experienced a growth of 132 per cent, whereas the agriculture sector saw an increase of only 18 per cent (Figure 5). From 2000 to 2020, the LULUCF sector exhibited a 135 per cent increase in greenhouse gas removals.

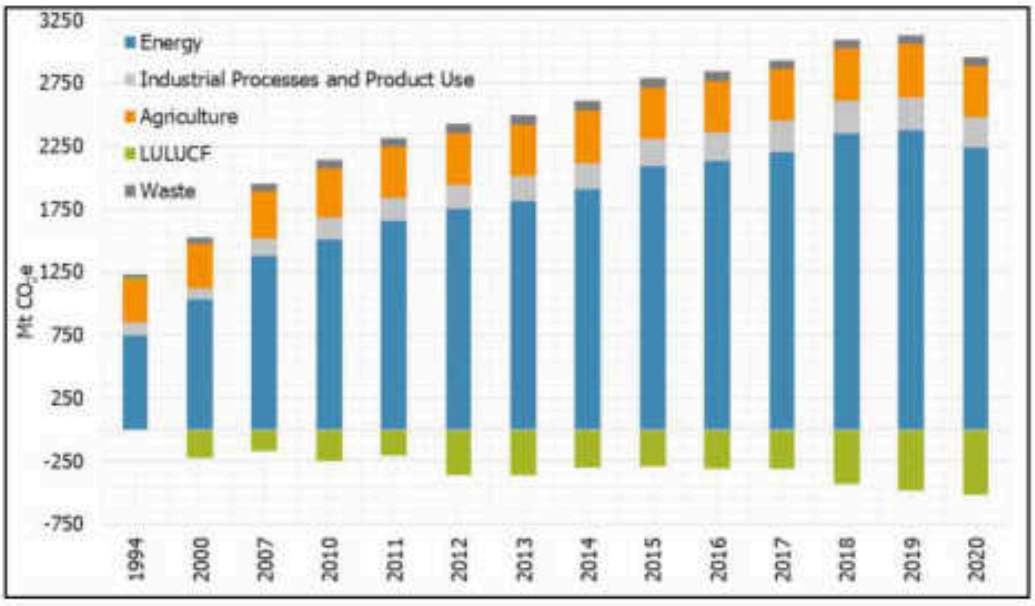


Figure 4: Sector-wise National GHG emission in Mt CO2e for 1994-2020.

Source: (MoEF, 2004); (MoEF, 2012); (MoEF, 2010); (MoEFCC, 2016); (MoEFCC, 2018); (MoEFCC, 2021); (MoEFCC, 2023).

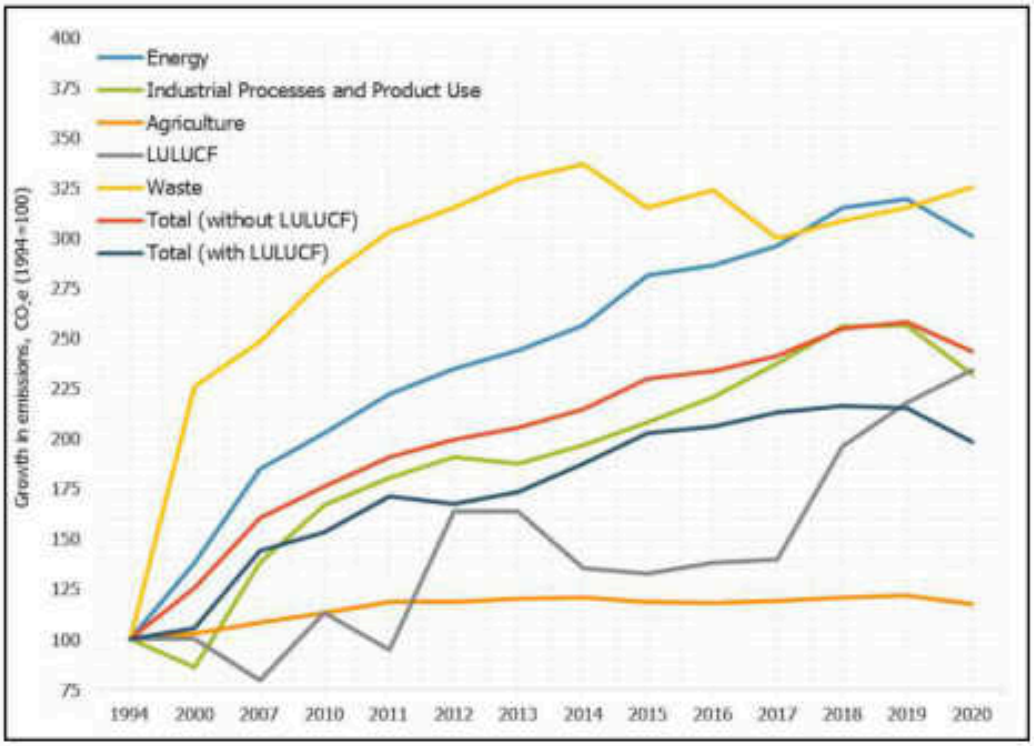


Figure 5: Growth in emissions of greenhouse gases, relative to 1994, illustrated by source categories, 1994-2020. Index 1994 = 100.

Source: (MoEF, 2004); (MoEF, 2012); (MoEF, 2010); (MoEFCC, 2016); (MoEFCC, 2018); (MoEFCC, 2021); (MoEFCC, 2023).

Mitigation Actions: India's achie respect of the NDC targets

Despite India's very low contribution to historical emissions and the current levels of global emissions, India has taken proactive actions to combat climate change in the context of sustainable development and it's developmental aspirations and in light of its national circumstances, reflecting principles of equity and common but differentiated responsibilities and respective capabilities (CBDR-RC) as enshrined in the UNFCCC and its Paris Agreement. The fundamental principle that informs India's climate policy is to pursue its development goals in line with national circumstances while staying within its fair share of the global carbon budget.

The Government of India has consistently undertaken initiatives to promote non-fossil-fuel-based energy sources. Owing to these, India stands 4th globally in renewable energy installed capacity, 4th in wind power capacity, and 3rd in solar power installed capacity. There has been a significant increase in non-fossil fuel-based sources in India,

primarily driven by a rapid increase in the installed capacity of solar photovoltaic (PV) based power supply and wind energy, alongside a steady rise in hydro and nuclear power capacity. As a proportion of total installed capacity in India, nonfossil fuel-based sources have consistently increased from ~30% in 2014 to 51.93% in Decemebr 2025 achieving India's NDC target five years ahead of the committed timeline of 2030.

Over the past decade, installed solar energy capacity has surged from approximately 2.63 GW in March 2014 to around 135.80 GW in December 2025, an increase of over 51% in more than a decade. These achievements align with India's updated Nationally Determined Contributions (NDCs), which include, inter alia, achieving about 50 per cent cumulative installed electric power capacity from non-fossil fuel-based energy resources by 2030. Table 2 summarises India's progress on its NDC targets.

Table 2: Progress of India's Nationally Determined Contributions (NDC)

Intended NDC (Oct 2015)	Updated NDC (Aug 2022)	Achievements	Remarks
NDC 1 - Reduction of the emissions intensity of GDP by **33% to 35%** from 2005 level by 2030	33-35% enhanced to **45%**	**36%** (between 2005-2020)	Original target achieved **11 years ahead** of schedule. On track to achieve revised target
NDC 2 - Achieve about **40%** cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030	40% enhanced to **50%**	**>51.93%** (December, 2025)	The original target was achieved **9 years ahead** of schedule. The revised target was also achieved **5 years ahead** of schedule.
NDC 3 - To create an additional carbon sink of **2.5 to 3** billion tonnes of CO_2 equivalent through additional forest and tree cover by 2030	No change	**2.29** billion tonnes of CO_2e by 2021	On track as per the Forest Survey of India estimates

e Multilateral Agreements Enviornmental (MEA)

- The 1st Indo Russia Joint Working Group (JWG) Meeting on Climate Change and Low Carbon Development was convened on 10 September 2025 in the Ministry. The sides noted the importance of expanding efforts to combat climate change and achieving the goals of the UNFCCC and its Paris Agreement. During the meeting, both sides held comprehensive discussions on key areas of mutual interest. These included the implementation of climate policies, with a focus on emissions reduction strategies and enhancement of energy efficiency across sectors. The JWG also exchanged views on adaptation to climate change, emphasizing the need for resilient infrastructure, climate-responsive planning, and knowledge sharing to address climate risks. Cooperation on the development of carbon markets was also the part of the discussions. In this connection, both the sides explored opportunities for collaboration under Article 6.2 and Article 6.4 of the Paris Agreement. In addition, discussions were held on the harmonization of approaches for measuring the carbon footprint of products, aimed at improving transparency, comparability, and trust in low-carbon value chains. The meeting reaffirmed the shared commitment of both countries to work closely in advancing climate ambition, promoting low-carbon development pathways, and strengthening bilateral relations in support of global climate goals.
- A Memorandum of cooperation on the Joint Credit Mechanism between the Government of the Republic of India and the Government of Japan was signed on 07.08.2025 under Article 6.2 of the Paris Agreement.

f. Progress / Achievements

- India has continued to successfully decouple economic growth from greenhouse gas emissions, with the emission intensity of GDP declining by 36 percent between 2005 and 2020.
- Further, India achieved a major milestone under its updated Nationally Determined Contribution (2022) by attaining one of the goals i.e. 50 percent of its installed electricity generation capacity from non-fossil fuel sources in June 2025, five years ahead of the 2030 target.
- With respect to the NDC target on creation of additional carbon sinks, the latest assessment under the India State of Forest Report (ISFR) 2023 indicates that India's total forest and tree carbon stock has reached 30.43 billion tonnes of CO_2 equivalent. This reflects the creation of an additional carbon sink of 2.29 billion tonnes of CO_2 equivalent compared to the 2005 baseline, demonstrating substantial progress towards the NDC goal of creating 2.5–3.0 billion tonnes of additional carbon sink by 2030.

g. Grant-in-aid released to Institutions / organisations including NGOs

The Division through NATCOM Cell; under the non-scheme establishment expenditure- Climate Change Action Programme, released the Grant-in-aid (General) to the following institutions in connection with ongoing National Carbonaceous Aerosols Programme (NCAP) Project.

Forest and Climate Change, Shri Bhupender Yadav, and Swedish Minister for Climate and Environment, Ms. Romina Pourmokhtari.

The co-chairs announced the selection of seven projects from the ITP calls for proposals, bringing together 18 industry and research partners from Indian and Swedish steel and cement sectors. The Summit also saw the participation of governments, industries and key stakeholders to reflect on the status of the industry transition globally and in the LeadIT countries, and emphasised on the role of collaboration to bring together the relevant stakeholders to accelerate the transition.



ITP Project Launch event, 08 Dec 2025, New Delhi, India

On 8 December 2025, the ITP project launch event was held at the Embassy of Sweden in New Delhi, bringing together the ITP Working Groups along with the companies and research institutes leading the selected projects under the ITP. The seven projects are structured as technology feasibility studies to explore the feasibility of Swedish innovations in the Indian industry with an aim to develop an outline for pilot projects. The partners in the seven projects and their objectives are -

1. Tata Steel, IIT-ISM Dhanbad, JK Cement, Cemvision - Steel Slag Reborn: Dual Value generation through high metallic recovery and sustainable supplementary cementitious material (SCM) production. Will create industrial symbiosis, where steel industry by-products become a resource for cement production
2. Tata Steel, IIT Hyderabad, GREEN14 - Microwave Plasma-Assisted CO2 Conversion of Blast Furnace Off-Gas. Explores microwave plasma-assisted conversion of steel blast furnace off-gas CO_2 into carbon monoxide (CO), which can be reinjected as a reducing agent
3. IIT Bombay, Ambuja Cements, EcoTech Solutions - feasibility study for establishing an integrated carbon capture and utilization unit at a cement industry facility
4. Prism Johnson, Datta Meghe College of Engineering, Ecometrix - Feasibility Study for AI-Based Platform ACORN to Optimize Concrete Mix Designs in the RMC Industry for reduced Climate Impact. Explores the potential of using AI for concrete recipe optimization.
5. Jindal Steel and Power, Kanthal, Swerim, - Electric heating methods for future CO2 neutral steel production. Assess concepts for electric gas heating, focusing on hydrogen and other process gases in ironmaking process, with the specific aim to reduce or eliminate CO2 emissions.
6. IIT Bhubaneswar,Metsol, Höganäs, Hoganas India - Electrified Green Ironmaking Pilot Using Rotary Kiln Technology. Aims to conduct a feasibility study for a pilot plant based on a novel fossil-free ironmaking process using hydrogen and an electrically heated rotary kiln.
7. My Home Industries (Maha Cement), Captimise AB (Andritz Group) - Decarbonisation of cement production with carbon capture. Screen three carbon capture technologies—amine absorption, hot potassium carbonate, and membranes.

2. National Adaptation Fund for Climate Change

The National Adaptation Fund for Climate Change (NAFCC) was established in 2015 to support adap-

tation activities in the States and Union Territories (UTs) of India that are vulnerable to the adverse effects of climate change. NAFCC is implemented in project mode and 30 projects have been sanctioned in 27 States and UTs.

The key achievements of the completed NAFCC Projects as per the Third Party Evaluation conducted in August 2025 are given below:

i. Community Outreach & Awareness: Conducted extensive awareness programmes and community mobilisation activities on climate adaptation, water security, and sustainable livelihoods.
ii. Climate-Resilient Agriculture: Promoted drought and saline-tolerant crops, integrated farming, and sustainable aquaculture to strengthen food security.
iii. Support for Vulnerable Communities: Enhanced livelihoods of tribal, women, and marginalised groups through skill building, resource access, and value-added production.
iv. Knowledge & Data Systems: Strengthened climate information access through GIS mapping, open data platforms, and integration with state climate plans.
v. Capacity Building for Adaptation: Delivered targeted training with implementing agencies and NGOs, improving adaptive skills, market linkages, and community resilience.

3. COP 30 Publication

In 2025, India released the flagship publication "One Earth, One Family, One Future: A Decade of Climate Action" on the sidelines of UNFCCC COP-30 in Belém, Brazil. The publication presents a comprehensive account of India's climate journey over the past decade, highlighting the country's people-centric, equity-based, and development-aligned approach to climate action. It documents India's progress across mitigation, adaptation, resilience-building, and sustainable lifestyles, while reaffirming the guiding philosophy of Vasudhaiva Kutumbakam—the World as one family. The publication also reflects India's achievement of several Paris Agreement commitments ahead of schedule.

The publication further serves as an important global reference by systematically showcasing India's adaptation strategies and resilience-building efforts, including climate-resilient agriculture, water security initiatives, disaster risk reduction, ecosystem restoration, and community-based actions. It highlights the role of national frameworks such as the National Action Plan on Climate Change (NAPCC), the ongoing development of the National Adaptation Plan (NAP), and large-scale initiatives such as Heat Action Plans, Mission LiFE, Jal Jeevan Mission, and ecosystem-based solutions. By consolidating policy frameworks, sectoral actions, state-level initiatives, and international engagement, the publication reinforces India's position as a leading voice of the Global South and a credible contributor to global climate adaptation and resilience efforts.

8.2 Ozone Layer Protection

1. Ozone Layer Protection

The stratospheric ozone layer is present from 10 to 50 kms above the Earth's surface. The stratospheric ozone layer acts as a shield and protects us from harmful ultraviolet-B (UV-B) radiation from the sun. Chlorofluorocarbons (CFCs), Hydrochlorofluorocarbons (HCFCs), Halon, Carbon Tetrachloride (CTC), Methyl Bromide, Bromofluoromethane (BFM) etc. are recognised as Ozone Depleting Substances (ODSs) that depletes the ozone layer. Depletion of Stratospheric Ozone

Layer has the potential to cause skin cancer, eye cataract, suppress body's immune system, decrease crop yield etc., which led to the adoption of the Vienna Convention for the Protection of the Ozone Layer in 1985 and the Montreal Protocol on Substances that Deplete the Ozone Layer in 1987. The Montreal Protocol is recognized as the most successful international environmental treaty enjoying universal ratification from all the 198 United Nations Member countries and has led to phase out of 99% of ODSs globally.

The MoEF&CC set up the Ozone Cell as a National Ozone Unit (NOU) to render necessary services for effective and timely implementation of the provisions of the Montreal Protocol and its ODSs phase-out program in India. The Ministry constituted an Empowered Steering Committee (ESC) under the Chairpersonship of the Secretary (EF&CC) to oversee the overall implementation of the provisions of the Montreal Protocol in the country.

In order to comply with the requirements of the Montreal Protocol, the Ministry has brought out comprehensive Ozone Depleting Substances (Regulation and Control) Rules, 2000 under the Environment (Protection) Act, 1986 for phasing out of various ODSs. The ODS Rules have been amended in 2001, 2003, 2004, 2005, 2007, 2014 and 2019 to implement ODS phase-out activities in the country aligned with the phase out schedule of Montreal Protocol.

India has played proactive role in the implementation of ozone depleting substances phase out activities in the country and adhered to all the phase-out schedules of the Montreal Protocol. India has been successfully implementing the phase-out program for ODSs in the country, wherein inter alia Chlorofluorocarbon (CFC), Carbon tetrachloride (CTC) and Halons have already been phased out for the controlled uses.

India is now phasing out Hydrochloro-fluorocarbons (HCFCs) as per the accelerated phase-out schedule of the Montreal Protocol i.e. 10% reduction as on 1st January, 2015, 35% reduction as on 1st January, 2020, 67.5% reduction as on 1st January, 2025 and complete phase-out of production and consumption of HCFCs will be by 2030 with a service tail of 2.5% annual average during the period 2030-2040.

2. Implementation of hydrochloro-fluorocarbons (HCFC) phase-out management plan (HPMP).

HPMP Stage-I

India successfully met 10% phase-out targets of HCFCs production and consumption for controlled application in 2015, as per the accelerated phase out schedule of the Montreal Protocol.

Through the HPMP Stage-I, a total of 341.77 ODP tonne of HCFCs have been phased out, in accordance with the agreement between the Government of India and the Executive Committee of the Multilateral Fund. The net direct emission reductions in terms of CO2 eq. tonne is 3,071,260.5.

HPMP Stage-II

India successfully met 35% phase-out targets of HCFCs production and consumption for controlled application in 2020, as per the accelerated phase out schedule of the Montreal Protocol.

In a significant first, amongst the Article 5 parties (developing countries) under the Montreal Protocol, India has been able to achieve the complete phase out of HCFC-141 b, one of the most potent chemicals involved in stratospheric ozone depletion, which was used in the foam manufacturing sector, as part of implementation of HPMP Stage II.

A separate notification S.O.No. 4724 (E) dated 31st December, 2019, amending the Ozone Depleting Substances (Regulation and Control) Rules, 2000, was published in the Gazette of India inter alia prohibiting the issuance of import license for HCFC-141b and its usage from 1st January, 2020.

Under implementation of HPMP Stage-II the following activities have been completed:

- Complete phase-out of HCFC-141b by the 158 Foam manufacturing participating enterprises.
- Complete phase-out of HCFC-22 by the 6 Air-Conditioning participating enterprises.
- Training of 17000 RAC service technicians on good servicing practices relating to installation and servicing, including safe handling of R-32 low GWP alternative during technology conversion in the air-conditioning manufacturing sector.

HPMP Stage-III

The HPMP Stage III for India was approved in the 91st meeting of the Ex-Com of the MLF to phase out 10,678.87 MT (i.e. 579.99 ODP tonne) of HCFCs consumption from the starting point, contributing to India's compliance with the 2025 and 2030 control targets for HCFCs under the Montreal Protocol.

The HPMP Stage-III is the last of the HPMPs for India, for assisting the country in achieving compliance with the Montreal Protocol 2025 and 2030 control targets. In pursuance of Ozone Depleting Substances (Regulation and Control) Rules, as amended in 2014, manufacture of all new equipment using HCFCs was phased out on 31st December, 2024. Use of HCFCs in the Refrigeration and air conditioning sector for servicing activities would continue till 2040, as per the accelerated phaseout schedule of the Montreal Protocol.

A total of 48 enterprises in the refrigeration manufacturing sector and 16 enterprises in the air-conditioning manufacturing sector have been participating in HPMP Stage-III.

In addition, HPMP Stage-III addresses conversion of technology from HCFC-22 to non-ODS and low-GWP in the RAC Manufacturing sector, Servicing sector, capacity building, project activities, policy and regulations, enforcement training, etc.

As part of the servicing sector activities under the HPMP Stage-III, equipment support is being provided to 120 ITIs across the country, in collaboration with the Directorate General of Training (DGT), Ministry of Skill Development and Entrepreneurship, considering regional and geographical balance. The equipment being provided are: (i) Basic refrigeration and electrical cycle unit (Frost-free Refrigerator), (ii) Split A C Training Unit (compatible to Inverter technology) Capacity: 1.5 Ton, (iii) Commercial Refrigeration Training Unit, (iv) Industrial Refrigeration Unit for Rack Systems, and (v) Split AC Training Unit Metal Frame.

In line with the accelerated phase out schedule of the Montreal Protocol for HCFCs, in 2024, India achieved reduction of 50% of HCFCs in consumption sector against 35% phase-out reduction target.

Enabling Activities under HPMPs

Ozone Cell, MoEF&CC, in association with the UN Environment (UNEP) have been implementing the enabling components (Capacity Building and Awareness creation) of HPMPs. The following activities have been undertaken as part of the enabling component under HPMPs.

- **Capacity Building of Enforcement Department/Agencies**

Activities relating to the capacity building of enforcement Departments are being imple-

mented in association with the National Academy of Customs Indirect Taxes and Narcotics (NACIN) for capacity building by imparting training of new Customs and Enforcement Officers in the field of import and export of ODSs, through a MoA. The following activities have been implemented:

i. A Two-Day Workshop for Supervisory Officers to Combat Illegal Trade of HCFCs under HPMP-Stage-II was conducted during 20th to 21st January, 2025 at Coorg, Karnataka.

- **Studies on key issues relating to Montreal Protocol implementation:**
  Following studies have been undertaken during 2025-26:

  i. Study report on Techno-economic feasibility Study of district cooling system in India. The study evaluates the potential for District Cooling Systems (DCS) across Indian cities supporting its adoption as a sustainable and energy-efficient cooling solution in building sector.
  ii. Study report on Low-GWP alternative Technologies for Chillers in Large Air Conditioning Building.
  iii. Study report on best Practices for Installation and Maintenance of Ammonia / Low-GWP Alternative Technology in Cold Chain Infrastructure to promote safe and efficient installation and maintenance of low-GWP technology systems within India's cold chain infrastructure.
  iv. Study report on Low-GWP alternative technologies in the Transport Refrigeration Sector.

- **Development and dissemination of knowledge products**
  Quarterly e-newsletter for RAC service technicians on the importance of good servicing practices including handling of different types of refrigerants, related safety issues and updates on refrigerant transition under the Montreal Protocol are being published in association with M/s TERI. During the year 2025, 4th and 5th edition of the e-newsletter were published.

3. Ratification of the Kigali amendment to the Montreal protocol for phase down of hydrofluorocarbons (HFCS) by India.

The Union Cabinet in its meeting held on 18 August 2021 approved the Ratification of Kigali Amendment to the Montreal Protocol for phase-down of HFCs by India. India ratified the Kigali Amendment to the Montreal Protocol for phase-down of HFCs on 27 September 2021.

As per the agreed phase-down of HFCs under the Kigali Amendment, India will have to complete its phase down of the production and consumption of HFCs in 4 steps from 2032 onwards with reduction of 10% in 2032, 20% in 2037, 30% in 2042 and 85% in 2047 respectively. The list of HFCs to be phase-down under the Kigali Amendment are as follows:

Sr. No.	Substance	100-Year Global Warming Potential
1.	HFC-134	1100
2.	HFC-134a	1 430
3.	HFC-143	353
4.	HFC-245fa	1030
5.	HFC-365mfc	794
6.	HFC-227ea	3220
7.	HFC-236cb	1340
8.	HFC-236ea	1370
9.	HFC-236fa	9810
10.	HFC-245ca	693
11.	HFC-43-10mee	1640
12.	HFC-32	675
13.	HFC-125	3500
14.	HFC-143a	4470
15.	HFC-41	92
16.	HFC-152	53
17.	HFC-152a	124
18.	HFC-23	14800

In accordance with the Montreal Proto provisions, the framework for licensing system for import and export of HFCs and reporting obligations has been put in place after the first national stakeholder consultation held on 20 January 2022. The DGFT, Ministry of Commerce and Industry, the nodal agency for licensing for import/export of HFCs is implementing the licensing system for HFCs, based on the recommendations of the Ozone Cell, MoEFCC. Data reporting for HFCs commenced from 2022 along with other ODS as per provisions of Article 7 of the Montreal Protocol.

4. India cooling action plan (ICAP)

MoEFCC developed and launched the India Cooling Action Plan (ICAP) during March 2019. India is one of the first countries in the World to have a comprehensive cooling action plan. ICAP aims to provide an integrated vision towards cooling across sectors encompassing inter alia reducing cooling demand, refrigerant transition, enhancing energy efficiency and better technology options with a 20-year time horizon, i.e. 2037-38.

ICAP recommends synergies with ongoing government programmes and schemes to maximize socio-economic co-benefits. The Government of India has taken steps to operationalize the recommendations given in ICAP after mapping of the recommendations for each thematic area, with the on-going government programmes/schemes and list out actions to be undertaken for achieving the said goals including identifying the required policy and regulatory interventions. Further, the following thematic groups during its meetings have identified a list of action points for operationalizing the recommendations of

HFC	Production (MT)	Export (MT)	Import (MT)	Consumption (MT)
HFC-43-10mee	0	0	3.948	3.948
R-404A* (HFC-125 = 44%, HFC-134a = 4%, HFC-143a = 52%)		348.793	2596.906	2248.113
R-407C* (HFC-32 = 23%, HFC-125 = 25%, HFC-134a = 52%)		384.448	82.56	505.852*
R-410A* (HFC-32 = 50%, HFC-125 = 50%)		6959.709	1089.02	4291.109*
R-438A* (HFC-32=8.5%, HFC-125= 45%, HFC-134a=44.2%, HC600=1.7%, HC-601a=0.6%)		0	120	279.70*
R-449A* (HFC-32=24.3%, HFC-125=24.7%, HFO-1234yf=25.3%, HFC-134a=25.7%)		0	7.042	7.042
R-449C* (HFC-32= 20%,HFC-125=20%,HFO-1234yf= 31%, HFC-134a= 29%)		0	1.298	1.298
R-454B* (HFC-32=68.9%, HFO-1234yf =31.1%)		0	8.596	8.596
R-507A* (HFC-125 = 50%, HFC-143a = 50%)		0	14.49	14.49
R-508B*(HFC-23 = 46%, PFC-116 = 54%)		0	0.84	0.84
R-513A* (HFO-1234yf=56%, HFC-134a= 44%)		0	0.938	0.938
HFC-365mfc/HFC-227ea* (87%/13%)		38.4	0	0
Total	46500.078	19821.517	22434.699	60280.898

*As per Article-7 and Country Programme Progress Report, no need to report the production of mixtures/blends. However, the consumption is calculated considering R-407C production= 807.740 MT, R-410A production= 10161.798 and R-438A production=159.70 MT during the year 2024.

8.3 United Nation Convention to Combat Desertification

Desertification Cell

a. Introduction

The Desertification Cell plans. coordinates and collaborates activities for restoration of degraded land and for the achievement of Land Degradation Neutrality ((LDN) in India.

- The focal point addressing the issues pertaining to the United Nations Convention to Combat Desertification (UNCCD), which was ratified by India on 17 December 1996.
- Co-ordination division for SDG 15.3 (combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation neutral world by 2030) in the Ministry.
- Coordination Division for National State of Environment Report for information pertaining to land degradation status.

b. Important Activities

i. **Aravalli Restoration**

The Ministry had announced its plan to revive the Aravallis, a major initiative to green the 5 km buffer area around the Aravalli hill range in the four states of Rajasthan, Gujarat, Haryana and parts of Delhi-NCR. The Detailed Action Plan (DAP) of the Aravalli Green Wall Initiative has been launched in June 2025 which shall be implemented through convergence with the identified line departments such as agriculture, watershed, etc. while being anchored in the State Forest Departments.

ii. **Desertification and Drought Day 2025**

On the occasion of the World Day to Combat Desertification and Drought 2025, the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India, organized a one-day national workshop at the Indian Council of Forestry Research and Education – Arid Forest Research Institute (ICFRE-AFRI), Jodhpur. The theme of the event was "Strategies for Combating Desertification and Drought", focusing on sustainable land management in arid and semi-arid ecosystems.

The event was graced by the Union Minister for Environment, Forest and Climate Change, Shri Bhupender Yadav, as Chief Guest in august presence of Union Minister for Tourism and Culture, Shri Gajendra Singh Shekhawat and Member of Parliament (Rajya Sabha), Shri Rajendra Gehlot. India's proactive measures to combat desertification and ecological restoration were highlighted. The importance of sustainable agricultural practices, community-driven initiatives, and the preservation of natural ecosystems were emphasized. Several new publications were launched at the event including:

- Information Booklet on Aravalli Districts
- Revised Mission Document of the Green India Mission
- Book on Sustainable Land Management (SLM)
- Launch of the National Afforestation Monitoring System (NAMS)
- Distribution of AFRI Shesham Clones to ten farmers by the Hon'ble Environment Minister

The workshop featured a series of technical sessions covering key themes in land restoration and desertification control. Discussions on Sustainable Land Management (SLM) highlighted MoEFCC and ICFRE institutes integrated, community-led restoration efforts. It also included presentations on Global and National case studies by development partners like UNDP, ADB, GIZ, KfW, AFD and World Bank. A dedicated session on the Aravalli Green Wall Project focused on inter-state collaboration to restore ecological balance in the Aravalli region. The final session addressed Land Degradation Neutrality (LDN) through multi stakeholder actions involving state governments, SAC, CAZRI, NGOs, and others. The event concluded with a wrap-up session affirming India's commitment to science-based, participatory, and policy-driven desertification mitigation.

iii. **India at CRIC 23, UNCCD**

An Indian delegation participated in CRIC-23 of the UNCCD (1–5 December 2025, Panama City), which served as the mid-term review of the UNCCD Strategic Framework (2018–2030) and a key milestone in the lead up to COP-17. Deliberations at CRIC-23 focused on progress and gaps in the implementation of Land Degradation Neutrality (LDN) targets, with emphasis on monitoring frameworks, indicators and reporting under PRAIS. Extensive discussions were held on drought resilience, including policy responses, early warning sys-

tems, financing mechanisms and the emerging global drought agenda. The session reviewed implementation aspects related to sand and dust storms, land tenure, gender-responsive approaches and capacity-building, highlighting challenges faced by developing countries. India actively participated in plenary sessions, contact groups and Asia-Pacific regional coordination meetings, and delivered interventions at both national and regional levels. India further engaged in thematic side events and technical sessions enabled exchange of experiences and good practices on land restoration, sustainable land management and integrated approaches. The outcomes of CRIC-23 provide important inputs for strengthening India's national implementation, refining reporting processes and shaping India's engagement and priorities for UNCCD COP-17.

iv. **Study Initiatives**

NAEB is implementing a pilot project "Enhanced Capacity building of Stakeholders and State Governments on Forest Landscape Restoration and Reporting Mechanism on Bonn Challenge" in consultation with IUCN is compiling the data for reporting the achievements under Bonn Challenge Commitments.

The project aims to develop and adapt best practices and monitoring protocols for the Indian States and build capacity within five pilot States (identified as Haryana, Madhya Pradesh, Maharashtra, Nagaland and Karnataka) on the same in the first phase of the project. This will be eventually rolled out across the country in all States in subsequent phases of the project.

36. का.आ. 5736(अ) तारीख 15 नवंबर, 2018 ;

37. का.आ. 5845(अ) तारीख 26 नवंबर, 2018 ;

38. का.आ. 345(अ) तारीख 17 जनवरी, 2019 ;

39. का.आ. 1960(अ) तारीख 13 जून, 2019 ; और

40. का.आ. 236(अ) तारीख 16 जनवरी, 2020 ।

MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE

NOTIFICATION

New Delhi, the 17th February, 2020

S.O. 751(E).—WHEREAS, the Central Government in the erstwhile Ministry of Environment and Forests, in exercise of its powers under sub-section (1) and clause (v) of sub-section (2) of section 3 of the Environment (Protection) Act, 1986 (29 of 1986) read with clause (d) of sub-rule (3) of rule 5 of the Environment (Protection) Rules, 1986 has published the Environment Impact Assessment Notification, 2006 (hereinafter referred to as the EIA Notification, 2006) *vide* number S.O.1533 (E), dated the 14th September, 2006, mandating Prior Environmental Clearance for certain category of projects;

AND WHEREAS, the Prior Environmental Clearance process involves four stages namely, screening; scoping; public consultation; and appraisal. The scoping is the process to determine detailed and comprehensive Terms of Reference (hereinafter referred to as ToR) addressing all relevant environmental concerns for the preparation of an Environmental Impact Assessment and Environment Management Report in respect of the project or activity for which Prior Environmental Clearance is sought;

AND WHEREAS, in order to streamline the process of scoping and bring the uniformity across the proposals, as a standard operating procedure, the Ministry has developed sector specific Standard Terms of References for all 39 class of projects or activities listed in the Schedule to the EIA Notification, 2006;

AND WHEREAS, the Expert Appraisal Committee constituted under the provisions of EIA Notification, 2006 can modify standard ToR and prescribe additional ToR based on examination of alternative sites proposed and the project specific requirements in respect of green field projects or activities;

AND WHEREAS, to expedite the process of granting standard Terms of Reference (ToR) in respect of expansion proposals and projects located within notified Industrial Estates, where there is no examination of alternative sites involved, the Ministry proposes to introduce the concept of issuance of an online Standard Terms of Reference (ToR) after acceptance of the proposal in Form-1 by the Regulatory Authority, automatically through the web portal developed by the Ministry to the Project Proponent;

AND WHEREAS, a draft notification further to amend the EIA Notification, 2006 was published in exercise of the powers conferred by sub-section (1) and clause (v) of sub-section (2) of section 3 of the Environment (Protection) Act, 1986 (29 of 1986), read with clause (d) of sub-rule (3) of rule 5 of the Environment (Protection) Rules, 1986 *vide* number S.O. 4085 (E), dated the 11th November, 2019, inviting objections and suggestions from all persons likely to be affected thereby, within a period of sixty days from the date on which copies of Gazette containing the said notification were made available to the public;

AND WHEREAS, copies of the said notification were made available to the public on 13th November, 2019;

AND WHEREAS, no objections or suggestions were received in response to the above-mentioned draft notification;

Now, therefore, in exercise of the powers conferred by sub-section (1) and clause (v) of sub-section (2) of section 3 of the Environment (Protection) Act, 1986 (29 of 1986), read with clause (d) of sub-rule (3) of rule 5 of the Environment (Protection) Rules, 1986, the Central Government hereby makes the following further amendments in the EIA Notification, 2006, namely:-

In the said notification, in paragraph 7, in sub-paragraph 7(i), for sub-heading II Stage (2)-Scoping and entries relating thereto, the following shall be substituted, namely:-

"II. Stage (2)-Scoping:

(i) "Scoping" refers to the process to determine detailed and comprehensive Terms of Reference (ToR) addressing all relevant environmental concerns for the preparation of an Environmental Impact Assessment and Environment Management Report in respect of the project or activity for which Prior Environmental Clearance is sought.

(ii) All projects or activities listed under Category "B2" of the schedule shall not require Scoping.

(iii) Sector specific Standard Terms of References developed by the Ministry of Environment, Forest and Climate Change, from time to time shall be displayed on its website.

(iv) The Standard Terms of References shall be issued to the following projects or activities through online mode, on acceptance of application within 7 working days, without referring to EAC or SEAC by the Ministry or SEIAA, as the case may be:

(a) All Highway projects in Border States covered under entry (i) and (ii) of column (3) and (4) against item 7(f) of the Schedule;

(b) All projects or activities proposed to be located in industrial estates or parks (item 7(c) of the Schedule) approved by the concerned authorities, and which are not disallowed in such approvals; and

(c) All expansion proposals of existing projects having earlier Prior Environmental Clearance:

Provided that EAC or SEAC may recommend additional specific Terms of Reference in addition to the Standard ToR, if found necessary, for a project or activity, within 30 days from the date of acceptance of application.

(v) All new projects or activities other than specified in sub-paragraph (iv) above, shall be referred to the EAC or SEAC by the Regulatory Authority, as the case may be, within 30 days from the date of application, for recommending the specific ToR in addition to the Standard ToR, deemed necessary. In case, the regulatory authority does not refer the matter to the EAC or SEAC, as the case may be, within 30 days of date of application in Form-I, sector specific Standard ToR shall be issued, online, on 30th day, by the Regulatory Authority.

(vi) Applications for Terms of Reference may be rejected by the regulatory authority concerned on the recommendation of the EAC or SEAC concerned. In case of such rejection, the decision together with reasons for the same after due personal hearing shall be communicated to the applicant in writing within sixty days of the receipt of the application.

(vii) The project proponent shall prepare the EIA report based on the sector specific Standard ToR as well as additional specific ToR, if any, stipulated by the EAC or SEAC.

(viii) The Terms of Reference for the projects or activities except for River valley and Hydro-electric projects, issued by the regulatory authority concerned, shall have the validity of four years from the date of issue. In case of the River valley and Hydro-electric projects, the validity will be for five years.

[F. No. 22-1/2019-IA.III]

GEETA MENON, Jt. Secy.

Note : The principal notification was published in the Gazette of India, Extraordinary, Part II, Section 3, Sub-section (ii) *vide* number S.O. 1533(E), dated the 14th September, 2006 and subsequently amended *vide* the following numbers: -

1. S.O. 1949 (E) dated the 13th November, 2006;
2. S.O. 1737 (E) dated the 11th October, 2007;
3. S.O. 3067 (E) dated the 1st December, 2009;
4. S.O. 695 (E) dated the 4th April, 2011;

Population and Vital Statistics

Highlights

- Population of the country is its most important demographic indicator. The population of elderly India had been increasing steadily since 1961. The growth in the elderly population became faster mainly due to decrease in the death rate because of various health interventions after the census 1981. The addition of the elderly population during 2001- 2011 was more than 27 million. The projected population for future years are required for preparation of perspective plan in various socio-economic sectors. According to the Report of the Technical Group on Population Projections for India and States 2011-2036, there are nearly 138 million elderly persons in India in 2021 comprising of 67 million males and 71 million females. (**Table 3.1**)

- It is interesting to note that upto Population Census 1991, the number of elderly females exceeded the number of elderly males. However, in the last two decades, the trend has been reversed and the elderly males outnumbered the elderly females. Further, it is projected that the number of elderly females will exceed the number of males in 2031. (**Table 3.1**)

- As per the Report of the Technical Group on Population Projections for India and States 2011-2036, an increase of nearly 34 million elderly persons was seen in 2021 over the Population Census 2011 and is further expected to increase by around 56 million elderly persons in 2031. (**Table 3.1**)

- The growth in elderly population may be due to the longevity of life achieved because of economic well-being, better healthcare and medical facilities and reduction in fertility rates. The general population has grown up by 12.4% during 2011-2021 in comparison to around 18% in earlier decade while the elderly population grow by 36% each in the last two decades (2001-2011 and 2011-2021). High growth rate in elderly population vis-à-vis of general population was observed earlier also in the two decades between 1961 and 1981. (**Table 3.2**)

- An increasing trend in percentage share of elderly persons to the total population of India is observed since 1961. In 1961, 5.6% of the population was in the age bracket of 60 years or more, the proportion has increased to 10.1% in 2021 and further likely to increase to 13.1% in 2031. A similar trend has also been observed in rural as well as in the urban areas. In rural areas, the proportion of

elderly persons has increased from 5.8% in 1961 to 8.8% in 2011, whereas in urban areas it has increased from 4.7% to 8.1% during 1961 to 2011. (**Table 3.3**)

- Population plays an important role for the development of a country as the potential for economic development of the country. As per population census data, the proportion of children in the age group 0-14 years increased till 1971 and gradually decreased afterwards. In 2021, the proportion of children in the age group 0-14 years is projected to be 25.5% whereas it was 30.8% in 2011. The proportion of elderly persons is ever increasing since 1951 and has projected to reach at 10.1% in 2021 and likely to further increase to 13.1% in 2031. The working age population has also an increasing trend since 1971 and has projected to reach 64.4% in 2021 and further increase to 65.2% in 2031as per the Report of the Technical Group on Population Projections for India and States 2011-2036. (**Table 3.4**)

- Age-wise distribution of population exhibits that there is an increasing trend in the proportion of population belonging to the age group 60 years and above from 7.4% in 2001 to the projection of 13.2% in 2031. (**Table 3.5**)

- As per the Report of the Technical Group on Population Projections for India and States 2011-2036, State-wise data on elderly population of 21 major states divulge that Kerala has the maximum proportion of elderly people in its population (16.5%) followed by Tamil Nadu (13.6%), Himachal Pradesh (13.1%), Punjab (12.6%) and Andhra Pradesh (12.4%) in 2021. On the contrary, proportion is the least in the States of Bihar (7.7%) followed by Uttar Pradesh (8.1%) and Assam (8.2%). (**Table 3.6**)

- Similarly, for the year 2031, maximum proportion of elderly people in its population is projected in the state of Kerala (20.9%) followed by Tamil Nadu (18.2%), Himachal Pradesh (17.1%), Andhra Pradesh (16.4%) and Punjab (16.2%). (**Table 3.6**)

- In comparison of the Population Census 2011 and Projected Population for the year 2021, it has been observed that the annual percentage growth rate of elderly population is 3.28%. Among the States (except UTs and North Eastern States), maximum average increase has been observed in Delhi (6.60%) followed by Gujarat (4.88%) and West Bengal (4.27%). However, average increase is least in state of Uttar Pradesh (2.16%) followed by Bihar (2.28%) and Madhya Pradesh (2.53%). (**Table 3.6**)

Economic Status

Highlights

- Old age dependency ratio gives an idea about the number of persons aged 60+ per 100 persons in the age group 15-59. An increasing trend is observed in old age dependency ratio. The ratio has increased from 10.9% in 1961 to 14.2% in 2011 and further projected to increase to 15.7% and 20.1% in 2021 and 2031 respectively. Also, the dependency ratio for females and males has an increasing trend and the projected dependency ratio for female and male is 14.8% and 16.7% respectively in 2021. However, over the periods, there are significant differences in old age dependency ratio for rural and urban areas and this may be due to relatively higher concentration of working age population in urban areas. As per Census 2011, the old-age dependency ratios were 15.1 and 12.4 for rural and urban areas respectively. (**Table 4.1**)

- Among the major states, the overall old-age dependency ratio, as per Census 2011, varied from 10.4% in Delhi to 19.6% in Kerala. However, for the projected figures for year 2021, it varies from 12.7% in Assam to 26.1% in Kerala. For the year 2031, the old age dependency ratio varies from 15.6% in Bihar to 34.3% in Kerala. (**Table 4.2**)

- The female old-age dependency ratio as compared to male is significantly high in the States like Kerala, Tamil Nadu, Himachal Pradesh and Punjab for the projected year 2021. (**Table 4.2**)

- As per NSS 75th Round: Social Consumption on Health in India conducted during 2017-18, about 70 per cent of the aged persons had to depend on others for their day-to-day maintenance. The situation of elderly females was not encouraging as only 10% and 11% were economically independent in rural and urban areas respectively whereas the elderly males were much better off as corresponding percentage for males were 48% and 57% in rural and urban areas respectively. (**Table 4.3**)

- The proportion of fully dependent elderly in rural as well as in urban areas have decreased from 52% in 2004 to 47% in 2017-18. (**Table 4.4(a) and 4.4(b)**)

- It is evident from State-wise (except UTs) data on economic independence that in rural areas, the proportion of elderly males who are fully dependent on others is highest in Assam (42%) followed by Sikkim (40%), Gujarat (38%) and Punjab (34%) and is lowest in Nagaland (4%) followed by Delhi (5%) and Himachal Pradesh (11%) whereas for females the figure is highest in Delhi (100%) followed by Meghalaya (81%), Punjab and West Bengal (80% each) and lowest in Manipur (26%) followed by Nagaland (35%). In urban areas, the proportion of fully dependent elderly males is highest in Sikkim (45%) followed by Goa and Telangana (43% each) and in Nagaland (38%) whereas for the proportion of fully dependent elderly females is highest in Jammu & Kashmir (85%) followed by Sikkim and Telangana (84% each) and is lowest in Arunachal Pradesh (16%). (**Table 4.4(b)**)

- Among economically dependent elderly men in 2017-18, it has been observed that in both rural and urban areas they were financially supported mainly by their own children followed by their spouses, grand-children and by others. In case of elderly women, more or less similar pattern has been observed. There is no significant change in this direction from that observed in 2004. (**Table 4.5**)

- It is evident from State-wise (except UTs) data on economic independence that in rural areas, the proportion of elderly females who are fully dependent on spouse is highest in Jammu & Kashmir (40%) followed by Haryana (35%), Mizoram (34%) and is lowest in Assam (4%) followed by 8% each in Delhi, Goa and Manipur whereas for males the figure is highest in Meghalaya (16%) followed by Andhra Pradesh (13%) and Manipur (12%). In urban areas, the proportion of fully dependent elderly females is highest in Rajasthan (35%) and lowest in Sikkim (2%). But for males the situation is much better in urban areas as proportion of fully dependent elderly males is highest in Chhattisgarh (19%) followed by Uttar Pradesh (8%). (**Table 4.6 (c) and 4.6(d)**)

- Of the economically independent elderly person in rural areas, the proportion of elderly person reported to be having with one or more dependents has decreased from 88% in 2004 to 83% in 2017-18 and for urban area, this proportion has decreased from 85% in 2004 to 79% in 2017-18. (**Table 4.7**)

- State-wise (except UTs) data on economic independence tell us that in rural areas, 100% aged persons were reported to be living with one or more dependents in the states of Delhi, Goa and Manipur whereas in urban areas, the proportion of elderly persons reporting living with one or more dependents is highest in Goa (98%) followed by Jharkhand (95%) and Telangana (94%). (**Table 4.8 (c) and 4.8(d)**)

- It is evident from the table that the proportion of participation in economic activity by the elderly person in the age-group 60-64 years has decreased from 49.3% in 2011-12 to 40.9% in 2018-19. Similar trend on participation in economic activity has also been noticed for the age group 65 years and above. (**Table 4.9**)

- As per the Periodic labour Force Survey (PLFS), 2018-19, about 65% of elderly men and 18% of elderly women in the age-group 60-64 years had participated in economic activity. However, there is wider difference in rural and urban areas. In rural areas, 72% of elderly men and 21% of elderly women participated in economic activities whereas in urban areas, it was only 51% among elderly men and 10% among elderly women. Similarly, in the age group 65 years and above, participation in economic activity by the elderly male and female is seen to be at a much reduced level. (**Table 4.9**)

Health and Social Status

Highlights

- Literacy levels among elderly males and females have improved over time in both rural and urban areas. However, huge gender gap has been observed in literacy rates. As per Population Census 2011, the literacy rate among elderly females (28%) is less than half of the literacy rate among elderly males (59%). The position in urban areas is expectedly better vis-à-vis that in rural areas. (**Table 5.1**)

- The general education level of aged persons in urban areas is significantly better than that observed in rural areas. Similarly, males are place at a better level than females in their general education level. Percentage of aged persons with education level secondary and above has increased in 2017-18 vis-à-vis 2011. (**Table 5.2**)

- It is evident from the table that the average number of years in formal education among persons aged 60 years & above is the highest in Chandigarh (12.7 years) followed by Delhi (10.7 years). This number is the lowest in Sikkim (6.2 years) followed by Mizoram (6.3 years) and Daman & Diu (6.5 years). (**Table 5.3**)

- The States/UTs of Bihar, Haryana, Himachal Pradesh, Jammu & Kashmir, Jharkhand, Manipur, Rajasthan, Uttarakhand, Uttar Pradesh, West Bengal and Chandigarh have exceeded all India average in number of years in formal education among persons aged 60 years and above in both rural and urban areas. (**Table 5.3**)

- Among the aged persons with illness, the proportion of aged persons with good or fair health, on their own perception, has shown increase in 2017-18 vis-à-vis 2004. (**Table 5.4**)

- Among the aged persons without illness, the proportion of aged persons against their current state of health as good or fair, based on their own perception, is same for men and women for the year 2017-18. (**Table 5.5(a), 5.5(b) and 5.5(c)**)

- Among the elderly persons it is observed that despite illness more men seemed to be feeling that they had a good or fair health condition as compared to the women in 2017-18. (**Table 5.5(a) and 5.5(b)**)

- The proportion of physically mobile elderly persons in the age-group 60 – 64 years is same in 2004 and 2017-18 in both rural and urban areas. With the increase of age, the proportion of physical mobility of elderly persons has shown decreasing trend. (**Table 5.6**)

- The percentage of female elderly persons (60 years and above) staying in the other's houses is more than double vis-à-vis male elderly persons. Also, the percentage of female elderly persons living alone not as an inmate of old age home is also much higher as compared to male elderly persons. (**Table 5.9(a) and 5.9(b)**)

- The proportion of persons that responded as ailing (PPRA) in a 15-day period among persons aged 60 years & above has been observed highest in Kerala (62.5%) followed by Andhra Pradesh (55.7%), Lakshadweep (54.6%) and West Bengal (47.9%). The PPRA is lowest in Meghalaya (1.2%) followed by Tripura (4.6%) and Arunachal Pradesh (5.1%). (**Table 5.10**)

- The percentage of female persons of age group 60 years and above participating in unpaid domestic services for household members is more than double as compared to males in rural as well as urban areas. Similarly, in the same age group, the proportion of females participating in employment & related activities is much less as compared to males in both rural and urban area. (**Table 5.13**)

- On an average, time spent by females of age group 60 years and above in unpaid domestic services for household members is 245 minutes as compared to males (112 minutes) on the same activity. (**Table 5.13**)

- Under the "self-care and maintenance activities", majority of time spent in a day per person by both males and females of age group 60 years and above is "sleep and related activities" followed by "eating and drinking activities". (**Table 5.15**)

- Under the "Traveling activities", majority of time spent in a day per person by both males and females of age group 60 years and above is for "travelling and commuting for employment". (**Table 5.17**)

- The most prevalent disability among elderly persons is locomotor disability followed by hearing disability and visual disability. (**Table 5.20**)

- Percentage of elderly disabled persons is slightly higher in rural areas as compared to urban areas in the broad type of disabilities of blindness, low vision, visual disability and hearing disability. (**Table 5.21**)

- More than 50% of elderly persons suffering from disability (other than mental illness) have taken the treatment from consulting doctors. (**Table 5.22**)

- The highest crime rate per lakh population against the senior citizens (60 years and above) has been found in the Delhi (93.8) followed by Gujarat (85.4), Chandigarh (74.5), Madhya Pradesh (73.2) and Chhattisgarh (67.3) whereas UTs

2

Maritime Security Imperatives and Influences

India's Maritime Outlook

Although land has been the primary and natural habitat of mankind, and is central to political, economic, military and social activities, the oceans have directly and indirectly influenced events on land. Over the past two centuries, in particular, technological and maritime developments have significantly altered the role and influence of the maritime environment. From a medium of transportation for trade, economy and the projection of power onto land, the oceans have become the primary conduits of international trade and are central to the global economy. The oceans and seabed are increasingly looked upon today as resource providers and critical contributors to national growth and prosperity. Maritime power is an important component of national power and is a key enabler for national growth and development. These aspects have prompted a steady, global shift of attention from land to the seas and an expanding maritime outlook, including for India.

The maritime outlook of a nation is shaped by the growth of population, industry, infrastructure and politico-economic power along the coast, and the ensuing dependence on the seas for national growth and prosperity. It is a central determinant of a nation's maritime interests and strategy. The key drivers of India's maritime outlook are its unique and advantageous geography, the need for sustained economic growth, a dynamic geo-strategic environment, the need to ensure safety and security of its SLOCs, and the security of Indian investments and other interests overseas, including Indian diaspora.

Maritime Geography

"The vital feature which differentiates the Indian Ocean from the Atlantic or the Pacific is the sub-continent of India, which juts out far into the sea for a thousand miles. It is the geographical position of India that changes the character of the Indian Ocean..."

– KM Panikkar[5]

Geography is a vital aspect, which can aid but also complicate maritime security, depending on the nation's geographic characteristics and the prevailing geo-strategic environment. India has a vast coastline extending to more than 7,500 km, with more than 1,200 islands, and a large Exclusive Economic Zone (EEZ) of about two million sq. km.[6] The anticipated addition of approximately 1.2 million sq. km of continental shelf would make India's total seabed area almost equal to the land mass.[7]

The Indian Ocean, through which much of the world's shipping transits, is distinguished by a land rim on three sides, with maritime access to the region possible only through certain 'choke points' leading to and from the Arabian Sea and the Bay of Bengal, and from the southern Indian Ocean. India flanks the first two regions and has a central position overseeing the third. Its peninsular feature provides a natural reach across wide sea spaces in all directions, extended by the islands in the Andaman & Nicobar and Lakshadweep Island groups.

India's central position in the IOR, astride the main International Shipping Lanes (ISLs), accords distinct advantages. It places the outer fringes of the IOR and most choke points almost equidistant from India, thereby facilitating reach, sustenance and mobility of its maritime forces across the region. India is, therefore, well positioned to influence the maritime space, and promote and safeguard its national maritime interests, across the IOR. At the same time, India's vast coastline and maritime zones require significant resources and investments to ensure their security.

Maritime power is an important component of national power and is a key enabler for national growth and development. These aspects have prompted a steady, global shift of attention from land to the seas and an increasing maritime outlook, including for India

Map 2.1: Important Choke Points and ISLs

Suez Canal

Strait of Hormuz

Persian Gulf

INDIA

Bab-el-Mandeb

Malacca and Singapore Straits

South China Sea

Ombai and Wetar Straits

Sunda Strait

Lombok Strait

Mozambique Channel

Cape of Good Hope

North Arabian Sea. Persian Gulf and Gulf of Oman to West Coast of India.

Central Arabian Sea. Red Sea/ Gulf of Aden and East Coast of Africa to West Coast of India.

South-West Indian Ocean. Cape of Good Hope and IOR Island States (Mauritius, Seychelles and Maldives) to India.

Bay of Bengal. East Coast of India to Arabian Sea, Bay of Bengal Eastern Rim, Andaman & Nicobar Islands and Malacca Strait.

South-East Indian Ocean. India to South-East Indian Ocean, including Indonesia and Australia.

Indo-Pacific. Indian Ocean to Pacific Ocean, through the various Indo-Pacific Straits and South/ East China and Philippines Seas.

Table 2.1: Choke Points in IOR



Cape of Good Hope

The Cape of Good Hope is not a conventional choke point, since adequate space and depth of water lies to its South and the passage of ships is not restricted by land. However, economic considerations and strong currents encourage ships to route closer to the coast. Ships that cannot pass through the Suez Canal follow this route. There was substantial increase in traffic in this area when the Suez Canal was shut down from 1967 – 1975.



Mozambique Channel

The Mozambique Channel is approximately 1,000 nautical miles (nm) long and 250 nm wide at its narrowest point. The usage of this channel reduced after the Suez Canal opened in 1869. However, with estimations of Mozambique holding over 100 trillion cubic feet of recoverable natural gas (one of the largest gas finds in the world) and huge coal reserves, the channel may regain prominence as an important sea route and choke point.



Bab-el-Mandeb

Bab-el-Mandeb is a 17 nm wide stretch of water that connects the Gulf of Aden with the Red Sea and, thence, the Suez Canal. Closure of the Bab-el-Mandeb, for any reason, would cut-off the Red Sea littoral from the Arabian Sea and Asia, and also prevent access to the Suez Canal. This would necessitate significant increase in shipping and tanker capacities, via other ISLs and choke points, to maintain the rate of oil and trade flow.



Suez Canal

The 105 nm long Suez Canal is the maritime gateway between Europe and Asia. Closure of the Suez Canal, for any reason, would cause traffic to be diverted around the Cape of Good Hope, thereby increasing the transit time and transportation costs. To illustrate, the distance between Mumbai to London is 6,200 nm via the Suez Canal, and 10,800 nm via the Cape of Good Hope.



Strait of Hormuz

The Strait of Hormuz, connecting the Persian Gulf to the Gulf of Oman and Arabian Sea, has a width of about 30 nm and primary navigable channel across a six nm wide zone. It has no maritime detour, rendering it as a critical choke point. About 17 million barrels of oil pass through this Strait each day, representing 35% of the world's oil trade by sea. Any closure of this strait would severely affect energy security of many nations, including India.



Malacca & Singapore Straits

The Straits of Malacca and Singapore link the Indian Ocean to the South China Sea and Pacific Ocean. Providing the shortest sea route from the Persian Gulf to East Asia/ West Pacific regions, it is a strategic choke point in the IOR. A dense shipping zone, more than 70,000 ships transit it each year. The narrowest point amongst the two straits is the 1.5 nm wide navigable stretch of the Phillip Channel in the Singapore Strait.



Sunda Strait

An alternative route to the Malacca and Singapore Straits is the Sunda Strait, which is 50 nm long and 15 nm wide at its North-East entrance. Large ships do not prefer passage through this strait due to navigational hazards, depth restrictions and strong currents.



Lombok Strait

At a minimum channel width of 11.5 nm, the Lombok Strait has sufficient width and depth, with lesser congestion, and provides an alternative passage between the Indian and the Pacific Oceans, especially for larger ships.



Ombai & Wetar Straits

The Ombai Strait lies between the islands of Alor and Timor, and the Wetar Strait lies between the islands of Timor and Wetar. Due to longer distance, routeing through this area is not normally preferred as an alternative to the Malacca and Singapore Straits. The route provides adequate depth for submarines to transit submerged between the Indian and Pacific Oceans.

Primary Areas

India's primary areas of maritime interest include the following:-

- India's coastal areas and maritime zones, including coastline, islands, internal sea waters, territorial waters, contiguous zone, EEZ and continental shelf.
- The Arabian Sea, Bay of Bengal, Andaman Sea, and their littoral regions.
- The Persian Gulf and its littoral, which is the source of majority of our oil supplies and gas imports, and is home to more than seven million expatriate Indians.
- The Gulf of Oman, Gulf of Aden, Red Sea, and their littoral regions.
- South-West Indian Ocean, including IOR island nations therein and East Coast of Africa littoral regions.
- The choke points leading to, from and across the Indian Ocean, including the Six-degree Channel; Eight/ Nine-degree Channels; Straits of Hormuz, Bab-el-Mandeb, Malacca, Singapore, Sunda and Lombok; the Mozambique Channel, and Cape of Good Hope and their littoral regions.
- Other areas encompassing our SLOCs, and vital energy and resource interests.

Secondary Areas

India's secondary areas of maritime interest include the following:-

- South-East Indian Ocean, including sea routes to the Pacific Ocean and littoral regions in vicinity.
- South and East China Seas, Western Pacific Ocean, and their littoral regions.
- Southern Indian Ocean Region, including Antarctica.
- Mediterranean Sea, West Coast of Africa, and their littoral regions.
- Other areas of national interest based on considerations of Indian diaspora, overseas investments and political relations.

The Indian Navy will continue to monitor all developments impacting the maritime domain, which may affect its roles and responsibilities, and incorporate the same into its strategic and operational plans



Maritime Threats

Threats and challenges to India's maritime interests emanate from traditional and non-traditional sources. The various sources and types of maritime threats must be catered for in the maritime security strategy. The strategic assessment of probable and possible sources of threats to India is a continuous process, and is carried out by various national agencies and the armed forces. The Indian Navy will continue to monitor all developments impacting the maritime domain, which may affect its roles and responsibilities, and incorporate the same into its strategic and operational plans.

Traditional Threats and Sources

The traditional sources refer to states with organised military capability and resources, which harbour adversarial posture and inimical intent towards India. Hostile actions by such states, in terms of scale, scope and intensity of force that may be applied, would potentially be of a higher order. Traditional sources, therefore, pose a higher level of threat to India's national security interests.

THE NATIONAL COMMISSION FOR WOMEN ACT, 1990

ACT NO. 20 OF 1990

[30*th August*, 1990.]

An Act to constitute a National Commission for Women and to provide for matters connected therewith or incidental thereto.

BE it enacted by Parliament in the Forty-first Year of the Republic of India as follows:—

CHAPTER I

PRELIMINARY

1. Short title, extent and commencement.—(*1*) This Act may be called the National Commission for Women Act, 1990.

(*2*) It extends to the whole of India except the State of Jammu and Kashmir.

(*3*) It shall come into force on such date[1] as the Central Government may, by notification in the Official Gazette, appoint.

2. Definitions.—In this Act, unless the context otherwise requires,—

(*a*) "Commission" means the National Commission for Women constituted under section 3;

(*b*) "Member" means a Member of the Commission and includes the Member-Secretary;

(*c*) "prescribed" means prescribed by rules made under this Act.

CHAPTER II

THE NATIONAL COMMISSION FOR WOMEN

3. Constitution of the National Commission for Women.—(*1*) The Central Government shall constitute a body to be known as the National Commission for Women to exercise the powers conferred on, and to perform the functions assigned to, it under this Act.

(*2*) The Commission shall consist of—

(*a*) a Chairperson, committed to the cause of women, to be nominated by the Central Government;

(*b*) five Members to be nominated by the Central Government from amongst persons of ability, integrity and standing who have had experience in law or legislation, trade unionism, management of an industry or organisation committed to increasing the employment potential of women, women's voluntary organisations (including women activists), administration, economic development, health, education or social welfare:

Provided that at least one Member each shall be from amongst persons belonging to the Scheduled Castes and Scheduled Tribes respectively;

(*c*) a Member-Secretary to be nominated by the Central Government, who shall be—

(*i*) an expert in the field of management, organisational structure or sociological movement, or

(*ii*) an officer who is a member of a civil service of the Union or of an all-India service or holds a civil post under the Union with appropriate experience.

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1. 31st January, 1992, *vide* notification No. S.O. 99(E), dated 31st January, 1992, *see* Gazette of India, Extraordinary, Part II, sec. 3(*ii*).

4. Term of office and conditions of service of Chairperson and Members.—(*1*) The Chairperson and every Member shall hold office for such period, not exceeding three years, as may be specified by the Central Government in this behalf.

(*2*) The Chairperson or a Member (other than the Member-Secretary who is a member of a civil service of the Union or of an all-India service or holds a civil post under the Union) may, by writing and addressed to the Central Government, resign from the office of Chairperson or, as the case may be, of the Member at any time.

(*3*) The Central Government shall remove a person from the office of Chairperson or a Member referred to in sub-section (*2*) if that person—

(*a*) becomes an undischarged insolvent;

(*b*) gets convicted and sentenced to imprisonment for an offence which in the opinion of the Central Government involves moral turpitude;

(*c*) becomes of unsound mind and stands so declared by a competent court;

(*d*) refuses to act or becomes incapable of acting;

(*e*) is, without obtaining leave of absence from the Commission, absent from three consecutive meetings of the Commission; or

(*f*) in the opinion of the Central Government has so abused the position of Chairperson or Member as to render that person's continuance in office detrimental to the public interest:

Provided that no person shall be removed under this clause until that person has been given a reasonable opportunity of being heard in the matter.

(*4*) A vacancy caused under sub-section (*2*) or otherwise shall be filled by fresh nomination.

(*5*) The salaries and allowances payable to, and the other terms and conditions of service of, the Chairperson and Members shall be such as may be prescribed.

5. Officers and other employees of the Commission.—(*1*) The Central Government shall provide the Commission with such officers and employees as may be necessary for the efficient performance of the functions of the Commission under this Act.

(*2*) The salaries and allowances payable to, and the other terms and conditions of service of, the officers and other employees appointed for the purpose of the Commission shall be such as may be prescribed.

6. Salaries and allowances to be paid out of grants.—The salaries and allowances payable to the Chairperson and Members and the administrative expenses, including salaries, allowances and pensions payable to the officers and other employees referred to in section 5, shall be paid out of the grants referred to in sub-section (*1*) of section 11.

7. Vacancies, etc., not to invalidate proceedings of the Commission.—No act or proceeding of the Commission shall be questioned or shall be invalid on the ground merely of the existence of any vacancy or defect in the constitution of the Commission.

8. Committees of the Commission.—(*1*) The Commission may appoint such committees as may be necessary for dealing with such special issues as may be taken up by the Commission from time to time.

(*2*) The Commission shall have the power to co-opt as members of any committee appointed under sub-section (*1*) such number of persons, who are not Members of the Commission, as it may think fit and the persons so co-opted shall have the right to attend the meetings of the committee and take part in its proceedings but shall not have the right to vote.

(*3*) The persons so co-opted shall be entitled to receive such allowances for attending the meetings of the committee as may be prescribed.

9. Procedure to be regulated by the Commission.—(*1*) The Commission or a committee thereof shall meet as and when necessary and shall meet at such time and place as the Chairperson may think fit.

(*2*) The Commission shall regulate its own procedure and the procedure of the committees thereof.

(*3*) All orders and decisions of the Commission shall be authenticated by the Member-Secretary or any other officer of the Commission duly authorised by the Member-Secretary in this behalf.

CHAPTER III

FUNCTIONS OF THE COMMISSION

10. Functions of the Commission.—(*1*) The Commission shall perform all or any of the following functions, namely:—

(*a*) investigate and examine all matters relating to the safeguards provided for women under the Constitution and other law;

(*b*) present to the Central Government, annually and at such other times as the Commission may deem fit, reports upon the working of those safeguards;

(*c*) make in such reports recommendations for the effective implementation of those safeguards for improving the conditions of women by the Union or any State;

(*d*) review, from time to time, the existing provisions of the Constitution and other laws affecting women and recommend amendments thereto so as to suggest remedial legislative measures to meet any lacunae, inadequacies or shortcomings in such legislations;

(*e*) take up the cases of violation of the provisions of the Constitution and of other laws relating to women with the appropriate authorities;

(*f*) look into complaints and take *suo moto* notice of matters relating to—

(*i*) deprivation of women's rights;

(*ii*) non-implementation of laws enacted to provide protection to women and also to achieve the objective of equality and development;

(*iii*) non-compliance of policy decisions, guidelines or instructions aimed at mitigating hardships and ensuring welfare and providing relief to women,

and take up the issues arising out of such matters with appropriate authorities;

(*g*) call for special studies or investigations into specific problems or situations arising out of discrimination and atrocities against women and identify the constraints so as to recommend strategies for their removal;

(*h*) undertake promotional and educational research so as to suggest ways of ensuring due representation of women in all spheres and identify factors responsible for impeding their advancement, such as, lack of access to housing and basic services, inadequate support services and technologies for reducing drudgery and occupational health hazards and for increasing their productivity;

(*i*) participate and advise on the planning process of socio-economic development of women;

(*j*) evaluate the progress of the development of women under the Union and any State;

(*k*) inspect or cause to be inspected a jail, remand home, women's institution or other place of custody where women are kept as prisoners or otherwise, and take up with the concerned authorities for remedial action, if found necessary;

(*l*) fund litigation involving issues affecting a large body of women;

(*m*) make periodical reports to the Government on any matter pertaining to women and in particular various difficulties under which women toil;

(*n*) any other matter which may be referred to it by the Central Government.

(*2*) The Central Government shall cause all the reports referred to in clause (*b*) of sub-section (*1*) to be laid before each House of Parliament along with a memorandum explaining the action taken or proposed to be taken on the recommendations relating to the Union and the reasons for the non-acceptance, if any of any of such recommendations.

(*3*) Where any such report or any part thereof relates to any matter with which any State Government is concerned, the Commission shall forward a copy of such report or part to such State Government who shall cause it to be laid before the Legislature of the State along with a memorandum explaining the action taken or proposed to be taken on the recommendations relating to the State and the reasons for the non-acceptance, if any, of any of such recommendations.

(*4*) The Commission shall, while investigating any matter referred to in clause (*a*) or sub-clause (*i*) of clause (*f*) of sub-section (*1*), have all the powers of a civil court trying a suit and, in particular, in respect of the following matters, namely:—

(*a*) summoning and enforcing the attendance of any person from any part of India and examining him on oath;

(*b*) requiring the discovery and production of any document;

(*c*) receiving evidence on affidavits;

(*d*) requisitioning any public record or copy thereof from any court or office;

(*e*) issuing commissions for the examination of witnesses and documents; and

(*f*) any other matter which may be prescribed.

CHAPTER IV

FINANCE, ACCOUNTS AND AUDIT

11. Grants by the Central Government.—(*1*) The Central Government shall, after due appropriation made by Parliament by law in this behalf, pay to the Commission by way of grants such sums of money as the Central Government may think fit for being utilised for the purposes of this Act.

(*2*) The Commission may spend such sums as it thinks fit for performing the functions under this Act, and such sums shall be treated as expenditure payable out of the grants referred to in sub-section (*1*).

12. Accounts and audit.—(*1*) The Commission shall maintain proper accounts and other relevant records and prepare an annual statement of accounts in such form as may be prescribed by the Central Government in consultation with the Comptroller and Auditor-General of India.

(*2*) The accounts of the Commission shall be audited by the Comptroller and Auditor-General at such intervals as may be specified by him and any expenditure incurred in connection with such audit shall be payable by the Commission to the Comptroller and Auditor-General.

(*3*) The Comptroller and Auditor-General and any person appointed by him in connection with the audit of the accounts of the Commission under this Act shall have the same rights and privileges and the authority in connection with such audit as the Comptroller and Auditor-General generally has in connection with the audit of Government accounts and, in particular, shall have the right to demand the production of books, accounts, connected vouchers and other documents and papers and to inspect any of the offices of the Commission.

(*4*) The accounts of the Commission, as certified by the Comptroller and Auditor-General or any other person appointed by him in this behalf, together with the audit report thereon shall be forwarded annually to the Central Government by the Commission.

13. Annual report.—The Commission shall prepare, in such form and at such time, for each financial year, as may be prescribed, its annual report, giving a full account of its activities during the previous financial year and forward a copy thereof to the Central Government.

(*18*) "Schedule" means a Schedule appended to this Act;

(*19*) "senior citizen" means a person defined as such under clause (*h*) of section 2 of the Maintenance and Welfare of Parents and Senior Citizens Act, 2007; 56 of 2007.

(*20*) "social audit" means the process in which people collectively monitor and evaluate the planning and implementation of a programme or scheme;

(*21*) "State Commission" means the State Food Commission constituted under section 16;

(*22*) "State Government", in relation to a Union territory, means the Administrator thereof appointed under article 239 of the Constitution;

(*23*) "Targeted Public Distribution System" means the system for distribution of essential commodities to the ration card holders through fair price shops;

(*24*) "Vigilance Committee" means a committee constituted under section 29 to supervise the implementation of all schemes under this Act;

(*25*) the words and expressions not defined here but defined in the Essential Commodities Act, 1955, or any other relevant Act shall have the meaning respectively assigned to them in those Acts. 10 of 1955.

CHAPTER II

PROVISIONS FOR FOOD SECURITY

Right to receive foodgrains at subsidised prices by persons belonging to eligible households under Targeted Public Distribution System.

3. (*1*) Every person belonging to priority households, identified under sub-section (*1*) of section 10, shall be entitled to receive five kilograms of foodgrains per person per month at subsidised prices specified in Schedule I from the State Government under the Targeted Public Distribution System:

Provided that the households covered under Antyodaya Anna Yojana shall, to such extent as may be specified by the Central Government for each State in the said scheme, be entitled to thirty-five kilograms of foodgrains per household per month at the prices specified in Schedule I:

Provided further that if annual allocation of foodgrains to any State under the Act is less than the average annual offtake of foodgrains for last three years under normal Targeted Public Distribution System, the same shall be protected at prices as may be determined by the Central Government and the State shall be allocated foodgrains as specified in Schedule IV.

Explanation.— For the purpose of this section, the "Antyodaya Anna Yojana" means, the scheme by the said name launched by the Central Government on the 25th day of December, 2000; and as modified from time to time.

(*2*) The entitlements of the persons belonging to the eligible households referred to in sub-section (*1*) at subsidised prices shall extend up to seventy-five per cent. of the rural population and up to fifty per cent. of the urban population.

(*3*) Subject to sub-section (*1*), the State Government may provide to the persons belonging to eligible households, wheat flour in lieu of the entitled quantity of foodgrains in accordance with such guidelines as may be specified by the Central Government.

Nutritional support to pregnant women and lactating mothers.

4. Subject to such schemes as may be framed by the Central Government, every pregnant woman and lactating mother shall be entitled to—

(*a*) meal, free of charge, during pregnancy and six months after the child birth, through the local *anganwadi*, so as to meet the nutritional standards specified in Schedule II; and

(*b*) maternity benefit of not less than rupees six thousand, in such instalments as may be prescribed by the Central Government:

Provided that all pregnant women and lactating mothers in regular employment with the Central Government or State Governments or Public Sector Undertakings or those who are in receipt of similar benefits under any law for the time being in force shall not be entitled to benefits specified in clause (*b*).

Nutritional support to children.

5. (*1*) Subject to the provisions contained in clause (*b*), every child up to the age of fourteen years shall have the following entitlements for his nutritional needs, namely:—

(*a*) in the case of children in the age group of six months to six years, age appropriate meal, free of charge, through the local *anganwadi* so as to meet the nutritional standards specified in Schedule II:

Provided that for children below the age of six months, exclusive breast feeding shall be promoted;

(*b*) in the case of children, up to class VIII or within the age group of six to fourteen years, whichever is applicable, one mid-day meal, free of charge, everyday, except on school holidays, in all schools run by local bodies, Government and Government aided schools, so as to meet the nutritional standards specified in Schedule II.

(*2*) Every school, referred to in clause (*b*) of sub-section (*1*), and *anganwadi* shall have facilities for cooking meals, drinking water and sanitation:

Provided that in urban areas facilities of centralised kitchens for cooking meals may be used, wherever required, as per the guidelines issued by the Central Government.

Prevention and management of child malnutrition.

6. The State Government shall, through the local *anganwadi*, identify and provide meals, free of charge, to children who suffer from malnutrition, so as to meet the nutritional standards specified in Schedule II.

Implementation of schemes for realisation of entitlements.

7. The State Governments shall implement schemes covering entitlements under sections 4, 5 and section 6 in accordance with the guidelines, including cost sharing, between the Central Government and the State Governments in such manner as may be prescribed by the Central Government.

CHAPTER III

FOOD SECURITY ALLOWANCE

Right to receive food security allowance in certain cases.

8. In case of non-supply of the entitled quantities of foodgrains or meals to entitled persons under Chapter II, such persons shall be entitled to receive such food security allowance from the concerned State Government to be paid to each person, within such time and manner as may be prescribed by the Central Government.

CHAPTER IV

IDENTIFICATION OF ELIGIBLE HOUSEHOLDS

Coverage of population under Targeted Public Distribution System.

9. The percentage coverage under the Targeted Public Distribution System in rural and urban areas for each State shall, subject to sub-section (*2*) of section 3, be determined by the Central Government and the total number of persons to be covered in such rural and urban areas of the State shall be calculated on the basis of the population estimates as per the census of which the relevant figures have been published.

State Government to prepare guidelines and to identify priority households.

10. (*1*) The State Government shall, within the number of persons determined under section 9 for the rural and urban areas, identify—

(*a*) the households to be covered under the Antyodaya Anna Yojana to the extent specified under sub-section (*1*) of section 3, in accordance with the guidelines applicable to the said scheme;

(*b*) the remaining households as priority households to be covered under the Targeted Public Distribution System, in accordance with such guidelines as the State Government may specify:

Provided that the State Government may, as soon as possible, but within such period not exceeding three hundred and sixty-five days, after the commencement of

the Act, identify the eligible households in accordance with the guidelines framed under this sub-section:

Provided further that the State Government shall continue to receive the allocation of foodgrains from the Central Government under the existing Targeted Public Distribution System, till the identification of such households is complete.

(*2*) The State Government shall update the list of eligible households, within the number of persons determined under section 9 for the rural and urban areas, in accordance with the guidelines framed under sub-section (*1*).

Publication and display of list of eligible households.

11. The State Government shall place the list of the identified eligible households in the public domain and display it prominently.

CHAPTER V

REFORMS IN TARGETED PUBLIC DISTRIBUTION SYSTEM

Reforms in Targeted Public Distribution System.

12. (*1*) The Central and State Governments shall endeavour to progressively undertake necessary reforms in the Targeted Public Distribution System in consonance with the role envisaged for them in this Act.

(*2*) The reforms shall, *inter alia*, include—

(*a*) doorstep delivery of foodgrains to the Targeted Public Distribution System outlets;

(*b*) application of information and communication technology tools including end-to-end computerisation in order to ensure transparent recording of transactions at all levels, and to prevent diversion;

(*c*) leveraging "aadhaar" for unique identification, with biometric information of entitled beneficiaries for proper targeting of benefits under this Act;

(*d*) full transparency of records;

(*e*) preference to public institutions or public bodies such as Panchayats, self-help groups, co-operatives, in licensing of fair price shops and management of fair price shops by women or their collectives;

(*f*) diversification of commodities distributed under the Public Distribution System over a period of time;

(*g*) support to local public distribution models and grains banks;

(*h*) introducing schemes, such as, cash transfer, food coupons, or other schemes, to the targeted beneficiaries in order to ensure their foodgrain entitlements specified in Chapter II, in such area and manner as may be prescribed by the Central Government.

CHAPTER VI

WOMEN EMPOWERMENT

Women of eighteen years of age or above to be head of household for purpose of issue of ration cards.

13. (*1*) The eldest woman who is not less than eighteen years of age, in every eligible household, shall be head of the household for the purpose of issue of ration cards.

(*2*) Where a household at any time does not have a woman or a woman of eighteen years of age or above, but has a female member below the age of eighteen years, then, the eldest male member of the household shall be the head of the household for the purpose of issue of ration card and the female member, on attaining the age of eighteen years, shall become the head of the household for such ration cards in place of such male member.

CHAPTER VII

GRIEVANCE REDRESSAL MECHANISM

Internal grievance redressal mechanism.

14. Every State Government shall put in place an internal grievance redressal mechanism which may include call centres, help lines, designation of nodal officers, or such other mechanism as may be prescribed.

District Grievance Redressal Officer.

15. (*1*) The State Government shall appoint or designate, for each district, an officer to be the District Grievance Redressal Officer for expeditious and effective redressal of grievances of the aggrieved persons in matters relating to distribution of entitled foodgrains or meals under Chapter II, and to enforce the entitlements under this Act.

(*2*) The qualifications for appointment as District Grievance Redressal Officer and its powers shall be such as may be prescribed by the State Government.

(*3*) The method and terms and conditions of appointment of the District Grievance Redressal Officer shall be such as may be prescribed by the State Government.

(*4*) The State Government shall provide for the salary and allowances of the District Grievance Redressal Officer and other staff and such other expenditure as may be considered necessary for their proper functioning.

(*5*) The officer referred to in sub-section (*1*) shall hear complaints regarding non-distribution of entitled foodgrains or meals, and matters relating thereto, and take necessary action for their redressal in such manner and within such time as may be prescribed by the State Government.

(*6*) Any complainant or the officer or authority against whom any order has been passed by officer referred to in sub-section (*1*), who is not satisfied with the redressal of grievance may file an appeal against such order before the State Commission.

(*7*) Every appeal under sub-section (*6*) shall be filed in such manner and within such time as may be prescribed by the State Government.

State Food Commission.

16. (*1*) Every State Government shall, by notification, constitute a State Food Commission for the purpose of monitoring and review of implementation of this Act.

(*2*) The State Commission shall consist of—

(*a*) a Chairperson;

(*b*) five other Members; and

(*c*) a Member-Secretary, who shall be an officer of the State Government not below the rank of Joint Secretary to that Government:

Provided that there shall be at least two women, whether Chairperson, Member or Member-Secretary:

Provided further that there shall be one person belonging to the Scheduled Castes and one person belonging to the Scheduled Tribes, whether Chairperson, Member or Member-Secretary.

(*3*) The Chairperson and other Members shall be appointed from amongst persons—

(*a*) who are or have been member of the All India Services or any other civil services of the Union or State or holding a civil post under the Union or State having knowledge and experience in matters relating to food security, policy making and administration in the field of agriculture, civil supplies, nutrition, health or any allied field; or

(*b*) of eminence in public life with wide knowledge and experience in agriculture, law, human rights, social service, management, nutrition, health, food policy or public administration; or

(*c*) who have a proven record of work relating to the improvement of the food and nutrition rights of the poor.

(*4*) The Chairperson and every other Member shall hold office for a term not exceeding five years from the date on which he enters upon his office and shall be eligible for reappointment:

Provided that no person shall hold office as the Chairperson or other Member after he has attained the age of sixty-five years.

(*5*) The method of appointment and other terms and conditions subject to which the Chairperson, other Members and Member-Secretary of the State Commission may be appointed, and time, place and procedure of meetings of the State Commission (including the quorum at such meetings) and its powers, shall be such as may be prescribed by the State Government.

(*6*) The State Commission shall undertake the following functions, namely:—

(*a*) monitor and evaluate the implementation of this Act, in relation to the State;

(*b*) either *suo motu* or on receipt of complaint inquire into violations of entitlements provided under Chapter II;

(*c*) give advice to the State Government on effective implementation of this Act;

(*d*) give advice to the State Government, their agencies, autonomous bodies as well as non-governmental organisations involved in delivery of relevant services, for the effective implementation of food and nutrition related schemes, to enable individuals to fully access their entitlements specified in this Act;

(*e*) hear appeals against orders of the District Grievance Redressal Officer;

(*f*) prepare annual reports which shall be laid before the State Legislature by the State Government.

(*7*) The State Government shall make available to the State Commission, such administrative and technical staff, as it may consider necessary for proper functioning of the State Commission.

(*8*) The method of appointment of the staff under sub-section (*7*), their salaries, allowances and conditions of service shall be such, as may be prescribed by the State Government.

(*9*) The State Government may remove from office the Chairperson or any Member who—

(*a*) is, or at any time has been, adjudged as an insolvent; or

(*b*) has become physically or mentally incapable of acting as a member; or

(*c*) has been convicted of an offence which, in the opinion of the State Government, involves moral turpitude; or

(*d*) has acquired such financial or other interest as is likely to affect prejudicially his functions as a member; or

(*e*) has so abused his position as to render his continuation in office detrimental to the public interest.

(*10*) No such Chairperson or Member shall be removed under clause (*d*) or clause (*e*) of sub-section (*9*) unless he has been given a reasonable opportunity of being heard in the matter.

17. The State Government shall provide for salary and allowances of Chairperson, other Members, Member-Secretary, support staff, and other administrative expenses required for proper functioning of the State Commission.

Salary and allowances of Chairperson, Member, Member-Secretary and other staff of State Commission.

Designation of any Commission or body to function as State Commission.

18. The State Government may, if considers it necessary, by notification, designate any statutory commission or a body to exercise the powers and perform the functions of the State Commission referred to in section 16.

Joint State Food Commission.

19. Notwithstanding anything contained in sub-section (*1*) of section 16, two or more States may have a Joint State Food Commission for the purposes of this Act with the approval of the Central Government.

Powers relating to inquiries.

20. (*1*) The State Commission shall, while inquiring into any matter referred to in clauses (*b*) and (*e*) of sub-section (*6*) of section 16, have all the powers of a civil court while trying a suit under the Code of Civil Procedure, 1908, and, in particular, in respect of the following matters, namely:— 5 of 1908.

(*a*) summoning and enforcing the attendance of any person and examining him on oath;

(*b*) discovery and production of any document;

(*c*) receiving evidence on affidavits;

(*d*) requisitioning any public record or copy thereof from any court or office; and

(*e*) issuing commissions for the examination of witnesses or documents.

(*2*) The State Commission shall have the power to forward any case to a Magistrate having jurisdiction to try the same and the Magistrate to whom any such case is forwarded shall proceed to hear the complaint against the accused as if the case has been forwarded to him under section 346 of the Code of Criminal Procedure, 1973. 2 of 1974.

Vacancies, etc., not to invalidate proceedings of State Commission.

21. No act or proceeding of the State Commission shall be invalid merely by reason of—

(*a*) any vacancy in, or any defect in the constitution of, the State Commission; or

(*b*) any defect in the appointment of a person as the Chairperson or a Member of the State Commission; or

(*c*) any irregularity in the procedure of the State Commission not affecting the merits of the case.

CHAPTER VIII

OBLIGATIONS OF CENTRAL GOVERNMENT FOR FOOD SECURITY

Central Government to allocate required quantity of foodgrains from central pool to State Governments.

22. (*1*) The Central Government shall, for ensuring the regular supply of foodgrains to persons belonging to eligible households, allocate from the central pool the required quantity of foodgrains to the State Governments under the Targeted Public Distribution System, as per the entitlements under section 3 and at prices specified in Schedule I.

(*2*) The Central Government shall allocate foodgrains in accordance with the number of persons belonging to the eligible households identified in each State under section 10.

(*3*) The Central Government shall provide foodgrains in respect of entitlements under sections 4, 5 and section 6, to the State Governments, at prices specified for the persons belonging to eligible households in Schedule I.

(*4*) Without prejudice to sub-section (*1*), the Central Government shall,—

(*a*) procure foodgrains for the central pool through its own agencies and the State Governments and their agencies;

(*b*) allocate foodgrains to the States;

SCHEDULE I

[*See* sections 3(*1*), 22(*1*), (*3*) and 24 (*2*), (*3*)]

SUBSIDISED PRICES UNDER TARGETED PUBLIC DISTRIBUTION SYSTEM

Eligible households shall be entitled to foodgrains under section 3 at the subsidised price not exceeding rupees 3 per kg for rice, rupees 2 per kg for wheat and rupee 1 per kg for coarse grains for a period of three years from the date of commencement of this Act; and thereafter, at such price, as may be fixed by the Central Government, from time to time, not exceeding,—

(*i*) the minimum support price for wheat and coarse grains; and

(*ii*) the derived minimum support price for rice,

as the case may be.

CABINET SECRETARIAT

RESOLUTION

New Delhi, the 1st January, 2015

No. 511/2/1/2015-Cab.—Mahatma Gandhi had said: "Constant development is the law of life, and a man who always tries to maintain his dogmas in order to appear consistent drives himself into a false position". Reflecting this spirit and the changed dynamics of the new India, the institutions of governance and policy have to adapt to new challenges and must be built on the founding principles of the Constitution of India, the wealth of knowledge from our civilizational history and the present day socio-cultural context.

2. The Planning Commission was set up on the 15th of March, 1950 through a Cabinet Resolution. Nearly 65 years later, the country has metamorphosed from an under-developed economy to an emergent global nation with one of the world's largest economies.

3. From being preoccupied with survival, our aspirations have soared and today we seek elimination, rather than alleviation, of poverty. The people of India have great expectations for progress and improvement in governance, through their participation. They require institutional reforms in governance and dynamic policy shifts that can seed and nurture large-scale change. Indeed, the 'destiny' of our country, from the time we achieved Independence, is now on a higher trajectory.

4. The past few decades have also witnessed a strengthening of Indian nationhood. India is a diverse country with distinct languages, faiths and cultural ecosystems. This diversity has enriched the totality of the Indian experience. Politically too, India has embraced a greater measure of pluralism which has reshaped the federal consensus. The States of the Union do not want to be mere appendages of the Centre. They seek a decisive say in determining the architecture of economic growth and development. The one-size-fits-all approach, often inherent in central planning, has the potential of creating needless tensions and undermining the harmony needed for national effort. Dr. Ambedkar had said with foresight that it is "unreasonable to centralise powers where central control and uniformity is not clearly essential or is impracticable".

5. At the heart of the dynamics of transforming India lies a technology revolution and increased access to and sharing of information. In the course of this transformation, while some changes are anticipated and planned, many are a consequence of market forces and larger global shifts. The evolution and maturing of our institutions and polity also entail a diminished role for centralised planning, which itself needs to be redefined.

6. The forces transforming India are many and include:

a. The industry and service sectors have developed and are operating on a global scale now. To build on this foundation, new India needs an administration paradigm in which the government is an "enabler" rather than a "provider of first and last resort". The role of the government as a "player" in the industrial and service sectors has to be reduced. Instead, government has to focus on enabling legislation, policy making and regulation.

b. India's traditional strength in agriculture has increased manifold on account of the efforts of our farmers and improvements in technology. We need to continue to improve, and move from pure food security to a focus on a mix of agricultural production as well as the actual returns that farmers get from their produce.

c. Today, we reside in a 'global village', connected by modern transport, communications and media, and networked international markets and institutions. As India 'contributes' to global endeavours, it is also influenced by happenings far removed from our borders. Global economics and geo-politics are getting increasingly integrated, and the private sector is growing in importance as a constituent within that. India needs to be an active player in the debates and deliberations on the global commons, especially in relatively uncharted areas.

d. India's middle class is unique in terms of its size and purchasing power. This formidable group is increasing with the entry of the neo-middle class. It has been an important driver of growth and has enormous potential on account of its high education levels, mobility and willingness to push for change in the country. Our continuing challenge is to ensure that this economically vibrant group remains engaged and its potential is fully realised.

e. India's pool of entrepreneurial, scientific and intellectual human capital is a source of strength waiting to be unleashed to help us attain unprecedented heights of success. In fact, the 'social capital' that is present in our people has been a major contributor to the development of the country thus far and, therefore, it needs to be leveraged through appropriate policy initiatives.

f. The Non-Resident Indian community, which is spread across more than 200 countries, is larger in number than the population of many countries of the world. This is a significant geo-economic and geo-political strength. Future national policies must incorporate this strength in order to broaden their participation in the new India beyond just their financial support. Technology and management expertise are self-evident areas where this community can contribute significantly.

g. Urbanisation is an irreversible trend. Rather than viewing it as an evil, we have to make it an integral part of our policy for development. Urbanisation has to be viewed as an opportunity to use modern technology to create a wholesome and secure habitat while reaping the economic benefits that it offers.

h. Transparency is now a sine qua non for good governance. We are in a digital age where the tools and modes of communication, like social media, are powerful instruments to share and explain the thoughts and actions of the government. This trend will only increase with time. Government and governance have to be conducted in an environment of total transparency – using technology to reduce opacity and thereby, the potential for misadventures in governing.

7. Technology and information access have accentuated the unity in diversity that defines us. They have helped integrate different capabilities of our regions, states and eco-systems towards an interlinked national economy. Indeed, Indian nationhood has been greatly strengthened on their account. To reap the benefits of the creative energy that emerges from the Indian kaleidoscope, our development model has to become more consensual and co-operative. It must embrace the specific demands of states, regions and localities. A shared vision of national development has to be worked out based on human dignity, national self-respect and an inclusive and sustainable development path.

8. The challenges we face as a country have also become more complex:

a. India's demographic dividend has to be leveraged fruitfully over the next few decades. The potential of our youth, men and women, has to be realized through education, skill development, elimination of gender bias, and employment. We have to strive to provide our youth productive opportunities to work on the frontiers of science, technology and knowledge economy.

b. Poverty elimination remains one of the most important metrics by which alone we should measure our success as a nation. Every Indian must be given an opportunity to live a life of dignity and self respect. The words of Tiruvalluvar, the sage-poet, when he wrote that "nothing is more dreadfully painful than poverty", and "gripping poverty robs a man of the lofty nobility of his descent", are as true today as they were when written more than two thousand years ago.

c. Economic development is incomplete if it does not provide every individual the right to enjoy the fruits of development. Pt. Deen Dayal Upadhyaya had enunciated this in his concept of Antyodaya, or uplift of the downtrodden, where the goal is to ensure that the poorest of the poor get the benefits of development. Inequalities based on gender biases as well as economic disparities have to be redressed. We need to create an environment and support system that encourages women to play their rightful role in nation-building. Equality of opportunity goes hand in hand with an inclusiveness agenda. Rather than pushing everyone on to a pre-determined path, we have to give every element of society – especially weaker segments like the Scheduled Castes and Scheduled Tribes - the ability to influence the choices the country and government make in setting the national agenda. In fact, inclusion has to be predicated on a belief in the ability of each member of society to contribute. As Sankar Dev wrote centuries ago in the Kirtan Ghosh: "To see every being as equivalent to one's own soul is the supreme means (of attaining deliverance)".

d. Villages (Gram) continue to be the bedrock of our ethos, culture and sustenance. They need to be fully integrated institutionally into the development process so that we draw on their vitality and energy.

e. India has more than 50 million small businesses, which are a major source of employment creation. These businesses are particularly important in creating opportunities for the backward and disadvantaged sections of the society. Policy making must focus on providing necessary support to this sector in terms of skill and knowledge upgrades and access to financial capital and relevant technology.

f. Responsible development implies environmentally sound development. India is one of the mega-diverse countries. Our environmental and ecological assets are eternal, and must be preserved and safeguarded. The country's legacy of respect for environment is reflected in our reverence for trees and animals. Our legacy to future generations must be sustainable progress. Each element of our environment (paryavaran) and resources, namely water, land and forest(Jal, Jameen evam Jungle) must be protected; and this must be done in a manner that takes into account their inter-linkages with climate (jal vayu) and people (jan). Our development agenda has to ensure that development does not sully the quality of life of the present and future generations.

9. The role of the government in achieving 'national objectives' may change with time, but will always remain significant. Government will continue to set policies that anticipate and reflect the country's requirements and execute them in a just manner for the benefit of the citizens. The continuing integration with the world – politically and economically - has to be incorporated into policy making as well as functioning of the government.

In essence, effective governance in India will rest on the following pillars:

a. Pro-people agenda that fulfils the aspirations of the society as well as individual,

b. Pro-active in anticipating and responding to their needs,

c. Participative, by involvement of citizens,

d. Empowering women in all aspects

e. Inclusion of all groups, with special attention to the economically weak (garib), the SC, ST and OBC communities, the rural sector and farmers (gaon and kisan), youth and all categories of minorities.

f. Equality of opportunity to our country's youth,

g. Transparency through the use of technology to make government visible and responsive.

10. Governance, across the public and private domains, is the concern of society as a whole. Everyone has a stake in ensuring good governance and effective delivery of services. Creating Jan Chetna, therefore, becomes crucial for people's initiative. In the past, governance may have been rather narrowly construed as public governance. In today's changed dynamics – with 'public' services often being delivered by 'private' entities, and the greater scope for 'participative citizenry', governance encompasses and involves everyone.

11. The institutional framework of government has developed and matured over the years. This has allowed the development of domain expertise which allows us the chance to increase the specificity of functions given to institutions. Specific to the planning process, there is a need to separate as well as energize the distinct 'process' of governance from the 'strategy' of governance.

In the context of governance structures, the changed requirements of our country, point to the need for setting up an institution that serves as a Think Tank of the government – a directional and policy dynamo. The proposed institution has to provide governments at the central and state levels with relevant strategic and technical advice across the spectrum of key elements of policy. This includes matters of national and international import on the economic front, dissemination of best practices from within the country as well as from other nations, the infusion of new policy ideas and specific issue-based support. The institution has to be able to respond to the changing and more integrated world that India is part of.

An important evolutionary change from the past will be replacing a centre-to-state one-way flow of policy by a genuine and continuing partnership with the states. The institution must have the necessary resources, knowledge, skills and, ability to act with speed to provide the strategic policy vision for the government as well as deal with contingent issues.

Perhaps most importantly, the institution must adhere to the tenet that while incorporating positive influences from the world, no single model can be transplanted from outside into the Indian scenario. We need to find our own strategy for growth. The new institution has to zero in on what will work in and for India. It will be a Bharatiya approach to development.

12. The institution to give life to these aspirations is the NITI Aayog (National Institution for Transforming India). This is being proposed after extensive consultation across the spectrum of stakeholders including inter alia state governments, domain experts and relevant institutions. The NITI Aayog will work towards the following objectives:

a. To evolve a shared vision of national development priorities, sectors and strategies with the active involvement of States in the light of national objectives. The vision of the NITI Aayog will then provide a framework 'national agenda' for the Prime Minister and the Chief Ministers to provide impetus to

b. To foster cooperative federalism through structured support initiatives and mechanisms with the States on a continuous basis, recognizing that strong States make a strong nation

c. To develop mechanisms to formulate credible plans at the village level and aggregate these progressively at higher levels of government

d. To ensure, on areas that are specifically referred to it, that the interests of national security are incorporated in economic strategy and policy

e. To pay special attention to the sections of our society that may be at risk of not benefitting adequately from economic progress

f. To design strategic and long term policy and programme frameworks and initiatives, and monitor their progress and their efficacy. The lessons learnt through monitoring and feedback will be used for making innovative improvements, including necessary mid-course corrections

g. To provide advice and encourage partnerships between key stakeholders and national and international like-minded Think Tanks, as well as educational and policy research institutions

h. To create a knowledge, innovation and entrepreneurial support system through a collaborative community of national and international experts, practitioners and other partners

i. To offer a platform for resolution of inter-sectoral and inter-departmental issues in order to accelerate the implementation of the development agenda

j. To maintain a state-of-the-art Resource Centre, be a repository of research on good governance and best practices in sustainable and equitable development as well as help their dissemination to stake-holders

k. To actively monitor and evaluate the implementation of programmes and initiatives, including the identification of the needed resources so as to strengthen the probability of success and scope of delivery

l. To focus on technology upgradation and capacity building for implementation of programmes and initiatives

m. To undertake other activities as may be necessary in order to further the execution of the national development agenda, and the objectives mentioned above

13. The NITI Aayog will comprise the following:

a. Prime Minister of India as the Chairperson

b. Governing Council comprising the Chief Ministers of all the States and Lt. Governors of Union Territories

c. Regional Councils will be formed to address specific issues and contingencies impacting more than one state or a region. These will be formed for a specified tenure. The Regional Councils will be convened by the Prime Minister and will comprise of the Chief Ministers of States and Lt. Governors of Union Territories in the region. These will be chaired by the Chairperson of the NITI Aayog or his nominee

d. Experts, specialists and practitioners with relevant domain knowledge as special invitees nominated by the Prime Minister

e. The full-time organizational framework will comprise of, in addition to the Prime Minister as the Chairperson:

i. Vice-Chairperson: To be appointed by the Prime Minister

ii. Members: Full-time

iii. Part-time members: Maximum of 2 from leading universities research organizations and other relevant institutions in an ex-officio capacity. Part time members will be on a rotational basis

iv. Ex Officio members: Maximum of 4 members of the Union Council of Ministers to be nominated by the Prime Minister

v. Chief Executive Officer: To be appointed by the Prime Minister for a fixed tenure, in the rank of Secretary to the Government of India

vi. Secretariat as deemed necessary

14. Swami Vivekananda said "Take up one idea. Make that one idea your life – think it, dream of it, live on that idea. Let the brain, muscles, nerves, every part of your body, be full of that idea and just leave every other idea alone. This is the way to success." Through its commitment to a cooperative federalism, promotion of citizen engagement, egalitarian access to opportunity, participative and adaptive governance and increasing use of technology, the NITI Aayog will seek to provide a critical directional and strategic input into the development process. This, along with being the incubator of ideas for development, will be the core mission of NITI Aayog.

15. This Resolution shall come into force and shall be deemed to have come into force with effect from the 1st day of January, 2015 and Resolution No. 1-P(C)/50 dated the 15th March 1950, shall stand superseded with effect from the date of coming into force of this Resolution.

SANJUKTA RAY, Director

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